Tech Breakthroughs · 2021 onwards

Research report · as of 2026-10-16 · AI · Bio · Energy · Space · Quantum

This report tracks 16 technology breakthroughs that have broken through from 2021 onwards — from transformers and reasoning models to GLP-1 agonists, CRISPR therapies, sub-threshold logical qubits, the Mechazilla booster catch, the collapse of battery costs below $110/kWh, and commercial driverless robotaxis. For each breakthrough we identify the people behind it, the component *sub-ideas* that make it up, the cutting edge of the last 12–18 months, the rising organizations and individuals to watch, and what to look for next. All claims are footnoted to primary sources where possible. Where a claim could not be independently re-verified, we have flagged it.

Latest weekly highlights

Week ending 2026-10-16. A model-heavy week: Meta shipped Muse Wonder, its first frontier-class model, Google opened Gemini 4 Argon to developers, and Microsoft unveiled Maia 300 as its main in-house inference chip at OCP. Anthropic disclosed it replaced Git (Scion/Antique) after merges hit 30k/day, while Karpathy's 'capability funnel' post framed how unevenly frontier ability is felt. Space news was mostly schedule slips (Neutron to 2027, New Glenn targeting December) plus a planned Starship ship-catch on Flight 15; bio was quiet apart from a reported FNIH/NCI $200M cancer-vaccine initiative. Verification: 25 checked, 9 passed, 12 fixed, 4 dropped; 3 entities added.

• **Meta ships Muse Wonder, its first frontier-class model, at $4/$20 per M tokens** — Oct 12, 2026: Meta Superintelligence Labs released Muse Wonder (API, $4/$20 per M tokens) and Muse Wonder Full (Muse Code subscribers), claiming parity with Fable 5.1 and GPT-6 Astra and leading on its MemoryBench 'active memory' eval (self-reported).

• **Google opens Gemini 4 Argon to developers at $2/$10; prices double in December** — Oct 13, 2026: Gemini 4 Argon, Google's long-horizon reasoning model, became broadly available via the Gemini API, AI Studio and Antigravity (Ultra tier) plus the Gemini app for Pro/Ultra subscribers; introductory pricing is $2/$10 per M tokens through Nov 30, doubling in December. A less-restricted cyber variant requires separate approval for vetted defenders.

• **JetBrains open-sources Mellum2.1, a 12B MoE coding-agent model trained with RL** — Oct 8, 2026 (backfill, one day before window): JetBrains released Mellum2.1 under Apache-2.0, a 12B-parameter (~2.5B active) MoE for coding agents, post-trained with RL in real repo environments using shell/file-edit tools and test-based rewards; 47.0% SWE-bench Verified (self-reported).

• **'Decision models': Cloudflare's open-weight Clef scores fixed choices, not text** — Oct 1, 2026 (backfill): Cloudflare released open-weight (Apache-2.0) Clef (27B) and Clef-flash (9B), which take a state plus a schema of typed questions and return probabilities over allowed answers; similar 'decision' models reported from Liquid AI and Microsoft point to a small-model class for routing/verification in agent pipelines (vendor claims).

• **Noiz AI / HKUST preview Helix World: interactive world model with stereo audio** — Oct 2026: Noiz AI and HKUST previewed Helix World, a real-time interactive audio-visual world model that generates video plus spatial stereo audio from a start image, steered by keyboard (WASD/look keys; reported 24 fps on one H800); preview checkpoint, paper and code released (developer-reported results).

• **Anthropic replaces Git with in-house Scion/Antique as merges hit 30k/day** — Oct 13, 2026: Anthropic's engineering blog says merged changes rose from ~2,500/day (May) to 30,000+/day (Sep), forcing a custom server-centric source-control system (Scion) with an AI-assisted review tool (Antique), migrated in ~five months; Scion reportedly uses 200x less CPU/memory than the git mirror fleet (vendor-reported).

Executive summary

Breakthroughs at a glance: top named figures (org-type tagged), current state, and one rising organization to watch.

#	Breakthrough	Domain	Top people	Current state (1 line)	Watch this
01	**Transformers / LLMs / coding agents**	AI	Ashish Vaswani (Biz) Noam Shazeer (Biz)	Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-par…	**Cursor (Anysphere)**
02	**RLHF + reasoning models (test-time compute scaling)**	AI	Richard Sutton (Cross) Andrew Barto (Acad.)	DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible cha…	**xAI Grok 5**
03	**Diffusion models (image + video generation)**	AI	David Holz (Biz) Emad Mostaque (Biz)	Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation inte…	**Stability AI Cascade Model**
04	**AlphaFold + computational protein design**	BIO	Demis Hassabis (Biz) John Jumper (Biz)	AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-l…	**Chai Discovery**
05	**mRNA vaccines + LNP delivery**	BIO	Katalin Karikó (Acad.) Drew Weissman (Acad.)	Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced Janu…	**Moderna/Merck cancer vaccine program**
06	**GLP-1 agonists**	BIO	Jens Holst (Acad.) Mads Krogsgaard Thomsen (Biz)	Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026).	**Retatrutide clinical program (Eli Lilly)**
07	**CRISPR therapies (Casgevy + base/prime editing)**	BIO	Feng Zhang (Acad.) David Liu (Acad.)	Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-oc…	**Prime Medicine (prime editing platform)**
08	**Quantum error correction (logical qubits below threshold)**	QUANTUM	Hartmut Neven (Biz) John Preskill (Acad.)	Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first dem…	**PsiQuantum**
09	**Reusable rockets (Falcon 9 + Starship catch)**	SPACE	Elon Musk (Biz) Gwynne Shotwell (Biz)	Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 we…	**Blue Origin New Glenn**
10	**Renewables + battery cost collapse**	ENERGY	Jay Whitacre (Cross) Tejas Patel (Biz)	BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of $108/kWh; stationary storage at $70/kWh (45% drop YoY); BEV packs at $99/kWh (2…	**BYD Sodium-Ion**
11	**Self-driving cars / robotaxis**	AI	Tekedra Mawakana (Biz) Dmitri Dolgov (Biz)	Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing …	**Wayve**

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12	**Humanoid robotics**	ROBOTICS	Sergey Levine (Biz) Jitendra Malik (Acad.)	Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specif…	**Figure AI**
13	**Fusion energy**	ENERGY	Omar Hurricane (Gov) Bob Mumgaard (Biz)	NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fus…	**Commonwealth Fusion Systems (SPARC)**
14	**Brain-computer interfaces**	BIO	Leigh Hochberg (Cross) Edward Chang (Acad.)	Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (vis…	**Neuralink Blindsight**
15	**AI accelerators / advanced compute**	HARDWARE	Jensen Huang (Biz) Andrew Feldman (Biz)	NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027.	**Tenstorrent**
16	**AI dev ecosystem & agent primitives (MCP, skills, agents, orchestration)**	AI	David Soria Parra (Biz) Justin Spahr-Summers (Biz)	MCP went from an Anthropic experiment (announced Nov 25, 2024) to a cross-industry standard; in Dec 2025 Anthropic donated MCP to the Agentic AI Foundation, a Linux Foundation d…	**Conductor**

Transformers / LLMs / coding agents

Impact so far. Transformers and LLMs have enabled agentic AI systems that autonomously solve software engineering tasks, with frontier models now matching or exceeding professional developer capabilities on real-world codebase tasks.

Watch for next. Watch for scaling reasoning with test-time compute to surpass human performance on complex multi-step planning; emergence of true agentic systems that coordinate multiple specialized coding agents in parallel; and development of interpretable reasoning traces that enable verification and debugging of AI agent decisions.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Ashish Vaswani[1]	Co-founder and CEO, Essential AI; Co-author of 'Attention Is All You Need' Essential AI	Biz	—
Noam Shazeer[2]	VP Engineering, Gemini Co-lead Google DeepMind	Biz	—
Dario Amodei[3]	CEO Anthropic	Biz	—
Ilya Sutskever[4]	CEO Safe Superintelligence Inc. (SSI)	Biz	—
Jeff Dean[5]	Chief Scientist, Co-lead Gemini Google DeepMind	Biz	—
Yann LeCun[6]	Founder and Chief Scientist Advanced Machine Intelligence Labs (AMI Labs); Jacob T. Schwartz Chaired Professor, NYU Courant Institute	Cross	★ ACM A.M. Turing Award 2018
John Jumper[7]	Joining Anthropic (research) after ~9 years leading AlphaFold at Google DeepMind Anthropic	Biz	★ 2024 Nobel Prize in Chemistry (AlphaFold)
Dawn Song[8]	Professor (AI safety & security, agentic AI, decentralized AI); co-founder Virtue AI; now leading AI safety/security at Meta Superintelligence Labs UC Berkeley; Berkeley RDI (co-director); Virtue AI (co-founder); Meta Superintelligence Labs	Cross	★ ACM/IEEE fellowships; MacArthur Fellow (prior)

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Coding agents[9]	Agentic LLMs that autonomously edit, run, and debug code have shifted programming from write-then-run to delegate-then-review — Cursor, Devin, Claude Code, and GitHub Copilot Workspace are now standard tooling for many engineering teams.	Cursor · Devin (Cognition) · Claude Code · GitHub Copilot Workspace · Aider
Mixture-of-Experts (MoE)[10]	Conditional computation that activates only a small fraction of parameters per token (DeepSeek-V3 ~5.5%) lets trillion-parameter models stay tractable to serve, decoupling parameter count from per-token cost.	DeepSeek-V3 · Mixtral 8x22B · Qwen3-MoE · Grok-1
Long-context windows[11]	Production context lengths exploded from 4K (GPT-3.5) to 1M+ tokens (Gemini, Claude Sonnet 4, Qwen2.5-1M), enabling whole-codebase reasoning, long-document analysis, and persistent agent memory.	Gemini 2.5 Pro 2M · Claude Sonnet 4 1M · GPT-4.1 1M · Qwen2.5-1M
Multimodal frontier models[12]	Native multimodal training (vision + audio + text + code in one model) replaced bolt-on adapters and is now the default for frontier systems, supporting voice agents, screen-reading, and embodied use cases.	GPT-4o · Gemini 1.5/2.5 · Claude 3.5/4 Vision · Llama 3.2 Vision

Sub-idea	Description	Examples
Open-weight catch-up[13]	Open-weight models (Llama, Qwen, DeepSeek) closed most of the gap with closed-frontier systems, with DeepSeek-V3 and R1 published with permissive MIT-style licenses, restructuring the geopolitics of AI compute.	DeepSeek-V3 / R1 · Llama 3.x · Qwen2.5 / Qwen3 · Mistral Large
Model Context Protocol (MCP)[14]	Anthropic's open MCP standard (Nov 2024) became the lingua franca for connecting LLMs to external tools and data — adopted across Claude, OpenAI, and most agent frameworks, replacing one-off plugin systems with a single client/server protocol.	Anthropic MCP · MCP servers (Slack, GitHub, Linear, Notion) · Claude Desktop · OpenAI MCP support
AWS MCP Server (managed)[15]	Managed MCP server providing secure, auditable agent access to AWS services with file uploads, long-running operations, sandboxed Python, IAM guardrails, and CloudTrail logging.	Agents can call any AWS API via single tool · Sandboxed Python execution for multi-step operations · 40+ pre-built skills for IaC, storage, analytics, serverless
GPT-Realtime-2 voice stack[16]	Native real-time voice agent stack with GPT-5-class reasoning, 70+ language live translation, and streaming Whisper transcription, exposed via the OpenAI API.	Real-time voice agent with reasoning + tool use · Live multilingual translation in 70+ languages · Streaming low-latency transcription
OpenAI GPT-5.5-Cyber (Trusted Access for Cyber)[17]	May 7, 2026 — Domain-specialized variant of GPT-5.5 for vetted cyber defenders, launched via Trusted Access for Cyber identity tier. Permits bug-hunting, malware analysis, reverse engineering; blocks credential theft and malware writing. Cisco and CrowdStrike are launch partners; Axios sources peg vuln-finding capability ~ Anthropic Mythos Preview.	Vetted SOC and red-team teams at Cisco / CrowdStrike-tier orgs gain access to the more-permissive variant · Phishing-resistant auth required from June 1, 2026 for the highest access tier
OS-native agentic AI layer[18]	A device-level agent that understands screen context, moves across apps and the browser, and completes multi-step tasks with the user in the loop — turning the smartphone OS itself into the agent surface.	Google Gemini Intelligence for Android (May 12, 2026) · Gemini in Chrome (June 2026) · Gboard Rambler dictation agent
Streaming / predictive RAG for voice agents[19]	Decouple retrieval from generation so the vector-DB round-trip never blocks speech: a background agent anticipates likely queries and pre-fetches into a fast cache while the speaking agent reads only from cache (or retrieves in parallel with user speech).	VoiceAgentRAG dual-agent 'Slow Thinker'/'Fast Talker' memory router (arXiv 2603.02206) · Stream RAG streaming tool-call prediction during user speech (arXiv 2510.02044)
IndexShare shared sparse-attention indexer[20]	A sparse-attention efficiency method that reuses a single indexer across groups of four attention layers, reducing per-token FLOPs by 2.9x at 1M context, demonstrated at frontier quality in GLM-5.2.	GLM-5.2 (Z.ai, June 2026): 2.9x per-token FLOP reduction at 1M context via IndexShare
OpenAI GPT-Live (full-duplex voice)[21]	Full-duplex voice model that listens and speaks simultaneously (vs turn-based Advanced Voice Mode), delegating search/reasoning to GPT-5.5 in the background; GPT-Live-1 becomes default ChatGPT Voice for Go/Plus/Pro, with new emotional-reliance/self-harm safety training.	
Kimi K3	2.8T-parameter open-weight MoE LLM from Moonshot AI, among the largest open models released to date	
Inkling	975B-param open-weight multimodal MoE model from Thinking Machines Lab	
Gemini 3.6 Flash[22]	Google DeepMind fast-tier model; 1M context, improved token-efficiency vs 3.5 Flash	
Meta Muse Code + Muse Spark 1.2[23]	Meta's first coding agent: terminal-based, persistent async background sub-agents, replay-exact event log, paired with a coding-tuned model trained on whole-repo long-horizon tasks.	Muse Code CLI · Muse Spark 1.2 model
Yahoo Scout (AI answer engine)[24]	Claude-powered answer engine with Bing grounding; MyScout personalized homepage plus Sports/Finance verticals; fills the AI-search/answer-engine category.	Yahoo Scout · MyScout
Decision models (open models that score fixed choices)[25]	Small open-weight models (e.g. Cloudflare Clef) that return calibrated probabilities over a typed schema of allowed answers instead of generating text; used for routing/verification in agent pipelines.	Cloudflare Clef / Clef-flash

b) Current research frontier (last 12–18 months)

01. Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-parameter deployment.[26]

02. Context length scaling to 1M tokens is now production-ready with Gemini 2.5 Pro (2M tokens), Claude Sonnet 4 (1M tokens), and open-source Qwen2.5-1M.[27]

03. Claude Opus 4.7 and GPT-5.5 represent 2026 frontier capabilities with improved instruction persistence and multi-step agentic orchestration.[28]

04. Claude Mythos Preview achieves 93.9% on SWE-bench Verified, though only 45.9% on the harder contamination-free SWE-bench Pro, revealing data contamination in benchmark evaluation.[29]

05. Diffusion transformers with patch-based representations have emerged as scalable architecture for both language and vision domains, with Next-Latent Prediction extending self-supervised learning to latent space.[30]

06. GPT-5.5 Instant produces 52.5% fewer hallucinated claims than GPT-5.3 Instant on high-stakes prompts in medicine, law, and finance (OpenAI release notes, May 5 2026).[31]

07. May 6, 2026 — Anthropic secures all of SpaceX Colossus 1 (~300 MW / 220K+ GPUs within the month), doubles Claude Code rate limits, raises Opus API limits, and expresses interest in multi-GW orbital compute with SpaceX.[32]

08. Gemini 3.5 Flash is a frontier-rivaling small model optimized for agentic coding, GA in Antigravity 2.0 and the Gemini API as of May 19, 2026.[33]

09. Claude Opus 4.8 released (May 28, 2026): 69.2% agentic coding, 84% computer-use on Online-Mind2Web, new dynamic workflows for Claude Code.[34]

10. Karpathy joined Anthropic pretraining team May 2026 to lead a 'use Claude to accelerate pretraining' effort.[35]

11. MiniMax M3 (June 1, 2026): open-weight model with 1M-token context via MiniMax Sparse Attention, 59.0% SWE-Bench Pro (beats GPT-5.5 and Gemini 3.1 Pro, approaches Opus 4.7), native multimodality and computer use; weights+report within 10 days.[36]

12. Claude Fable 5, a new 'Mythos-class' tier sitting above Opus, is now generally available and described by Anthropic as SOTA on nearly all tested AI-capability benchmarks, with classifier fallback to Opus 4.8 on cyber/bio/distillation queries.[37]

13. Open-weight frontier coding/agentic model GLM-5.2 (744B MoE, MIT, 1M context) becomes first open model >80% on Terminal-Bench 2.1 (81.0) and 62.1 SWE-bench Pro, beating GPT-5.5 at ~1/6 cost.[20]

14. Two June 2026 Nature papers show agentic AI doing autonomous, multi-step, end-to-end clinical management at or above board-certified-physician performance in simulated settings — MIRA (ED, 87.8% vs 78.1% dx accuracy) and Google's AMIE (longitudinal outpatient, non-inferior to PCPs).[38]

15. OpenAI limited-preview of GPT-5.6 Sol/Terra/Luna; Sol new SOTA on Terminal-Bench 2.1, $5/$30 per-M tokens; broad release gated by a US-government access process tied to a forthcoming cyber EO.[39]

16. Claude Sonnet 5 (Jun 30) default for Free/Pro, intro $2/$10 per-M pricing (→$3/$15 after Aug 31); gains on BrowseComp, OSWorld-Verified, Humanity's Last Exam and improved agentic tool-use vs Sonnet 4.6.[40]

17. Released July 24, 2026, Opus 5 tops Frontier-Bench, CursorBench, ARC-AGI-3 and OSWorld 2.0 at roughly half Fable 5's cost, with a 1M-token context and 'xhigh' reasoning effort; Anthropic also reports it as its most aligned model to date.[41]

18. Alibaba released Qwen3.8-Max (Aug 3, 2026), a 2.4T-param MoE (95B active) with 1M-token context at ~1/4 GPT-5.6 pricing ($2/$6 vs $5/$30 per M tokens), with open weights and a 27B sibling promised within a week.[42]

19. GLM-5.3 gains come from post-training only on the GLM-5.2 base; Terminal-Bench 3.0 4.6% -> 28.3%.[43]

20. Z.ai confirmed its anonymously-tested 'Ox Alpha' model is GLM-5.3-Flash - a 320B-parameter (18B active) natively multimodal MoE with 1M-token context, released under MIT license on Aug 26, 2026 at roughly one-tenth the price of GLM-5.2.[44]

21. Claude Fable 5.1 (GA) and Mythos 5.1 (trusted-access) released Sep 1, 2026; ~60% fewer Claude Code cyber false positives; cache-read pricing cuts up to 45% on agentic workloads.[45]

22. GPT-6 Astra is the first model designated Critical on OpenAI's Preparedness Framework cybersecurity track (2 novel V8 zero-days found); cyber capability gated to an application-based program.[46]

23. Gemini 3.8 Flash + Gemini 3.8 Flash Cyber (Fairwind trusted-defender program) released Sep 2, 2026; Chrome Security: 2.6x more correct patches vs top commercial models.[47]

24. ACL 2026 Best Papers: 'Characterizing the Expressivity of Local Attention in Transformers' (Li & Cotterell); 'The Imperfective Paradox in Large Language Models' (Ma & Miyao).[48]

25. Apple iOS 27 Siri AI: on-device + Private Cloud Compute, daily usage limits, paid expanded-access tier via iCloud+ (Sep 9, 2026 event).[49]

26. Shanghai AI Laboratory launched Atria Dawn Preview on Sep 11, 2026 under MIT license — a 744B-parameter agentic MoE, text-only, targeting long-horizon tool-use loops for scientific and office automation tasks.[50]

27. A new paper proposes generating fresh per-token expert weights from live interaction data instead of freezing weights after training, so models can absorb corrections and facts without retraining. Reached the Hacker News front page.[51]

28. [52]

29. [53]

30. [54]

31. [55]

32. [56]

33. Reflection AI debuts Beam: 501B open-weight MoE, 23B active — Oct 5, 2026: Reflection AI announced Beam, a 501B-total/23B-active sparse MoE trained on 23.8T tokens plus large-scale RL on 10,500 GB300s; self-reported 80.9 SWE-Bench Verified. Weights and tech report promised later in October (claims unverified).[57]

34. Mistral Large 4: open-weight 1.05T MoE, 49B active, 1M context — Oct 6, 2026: Mistral released Large 4, an open-weight multimodal MoE (1.05T total / 49B active, 1M-token context), API priced ~$1.36/$4.18 per M tokens.[58]

35. Claude Haiku 5.5 ships at $0.10/$0.50 per M tokens with 1M context — Oct 7, 2026: Anthropic released Haiku 5.5 for high-volume routing/subagent work at $0.10/$0.50 per M tokens (≤100K context); Simon Willison measured ~1.25x more tokens from the new tokenizer. Anthropic also added monthly API credits to Max/Team plans.[59]

36. Meta ships Muse Wonder, its first frontier-class model, at $4/$20 per M tokens — Oct 12, 2026: Meta Superintelligence Labs released Muse Wonder (API, $4/$20 per M tokens) and Muse Wonder Full (Muse Code subscribers), claiming parity with Fable 5.1 and GPT-6 Astra and leading on its MemoryBench 'active memory' eval (self-reported).[60]

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c) Who could be next

Cursor (Anysphere) [STARTUP][63] — Anysphere Inc. · Series D, $29.3B valuation (2026)

Fastest-growing AI coding IDE, reaching $29.3B valuation with $2B+ ARR by early 2026, supporting parallel multi-agent coding.

Cognition Labs (Devin) [STARTUP][64] — Cognition Labs · Series C+, $400M at $10.2B valuation; targeting $25B (2026)

Autonomous AI software engineer scaling ARR from $1M to $73M in 9 months; acquired Windsurf; targeting $25B valuation in April 2026.

Safe Superintelligence Inc. (SSI) [STARTUP][65] — SSI · Series B, $2B at $32B valuation (April 2025)

Sutskever's new lab focused on safety-first AGI path with $3B+ raised and $32B valuation; significant backing from Alphabet and Nvidia.

Advanced Machine Intelligence Labs (AMI Labs) [STARTUP][6] — AMI Labs · Series A, $1.03B at $3.5B (March 2026)

Yann LeCun's new venture focused on world models and open-source development; raised $1.03B at $3.5B pre-money valuation in March 2026.

Anthropic enterprise-AI JV [STARTUP][66] — Anthropic / Blackstone / Hellman & Friedman / Goldman Sachs · $1.5B initial capitalization; backed by Blackstone, H&F, Goldman Sachs

First services-led commercialization vehicle for a frontier-lab model; converts mid-market deployments from pure API consumption to verticalized AI services.

Z.ai (Zhipu AI) [LAB][20] — Z.ai / Zhipu AI · Well-funded Chinese AI 'tiger' (multi-billion valuation; public-listing track)

Leading Chinese open-weight frontier lab (GLM series); GLM-5.2 is the first open model >80% Terminal-Bench 2.1 and a credible MIT-licensed competitor to closed frontier coding models at ~1/6 cost.

Moonshot AI [LAB][67] — Moonshot AI · Alibaba/Tencent-backed

Leading Chinese open-weight frontier lab; Kimi K3 (2.8T MoE) is among the largest open models released.

Thinking Machines Lab [LAB][68] — Thinking Machines Lab · $2B seed round (2025)

Mira Murati's (ex-OpenAI CTO) frontier lab; shipped open-weight Inkling (975B) + Tinker fine-tuning platform.

Redwood Research [LAB][69] — Redwood Research · Nonprofit / philanthropically funded

Sets the agenda for control-based (as opposed to alignment-based) safety, now referenced across frontier-lab safety work; entity added after Ryan Greenblatt surfaced in the voice scan without the lab being tracked.

Sakana AI [LAB][70] — Sakana AI (Tokyo) · ~$379M raised to date

Japan's leading frontier lab; nature-inspired/model-merging research line and now orchestrator-as-a-model products

Shanghai AI Laboratory [LAB][71] — Shanghai AI Laboratory

Among the largest publishers of permissively-licensed frontier-scale Chinese models; sets the open-weight ceiling alongside DeepSeek and Qwen

Reflection AI [BUSINESS][57] — Reflection AI

One of few US labs releasing frontier-scale open weights.

Flags / unverified

■ SWE-bench Verified benchmark scores may be inflated due to data contamination; SWE-bench Pro (contamination-free) shows significantly lower performance

■ Meta's Llama benchmark results were reportedly fudged according to departing chief scientist Yann LeCun

■ No verified information found on Tom Brown's current 2026 affiliation beyond his core role at Anthropic

■ Updated Anthropic Series H funding closed: new value = $65B raised at $965B post-money (May 28, 2026); leads Altimeter, Dragoneer, Greenoaks, Sequoia; $47B run-rate revenue

■ Updated AI advertising & recommendation (Meta GEM/Andromeda): new value = GEM/Andromeda + new user-understanding models: +8.3% FB ad clicks, +15.7% conversions; Advantage+ >$75B run-rate (Q2 2026)

RLHF + reasoning models (test-time compute scaling)

Impact so far. Reasoning models with test-time compute scaling have fundamentally shifted AI capability distribution from training time to inference time, enabling smaller models to solve previously intractable problems in mathematics, coding, and science while achieving cost-competitive inference.

Watch for next. Monitor scaling laws for reasoning beyond current limits; emergence of multi-step verifiable reasoning that enables formal proof verification; and development of interpretable chain-of-thought that reveals genuine reasoning versus pattern matching in reasoning model outputs.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Richard Sutton[72]	Professor, University of Alberta; Senior Scientist, DeepMind University of Alberta / DeepMind	Cross	★ ACM A.M. Turing Award 2024 (announced 2025)
Andrew Barto[72]	Emeritus Professor University of Massachusetts Amherst	Acad.	★ ACM A.M. Turing Award 2024 (announced 2025)
Ilya Sutskever[65]	CEO Safe Superintelligence Inc.	Biz	—
Dario Amodei[3]	CEO Anthropic	Biz	—
Demis Hassabis[73]	CEO Google DeepMind	Biz	—
DeepSeek Research Team[13]	Lead researchers on DeepSeek-R1 DeepSeek-AI	Biz	—
Noam Brown[74]	Research scientist, reasoning (o1/o-series); previously Libratus/Pluribus poker AIs at CMU/Meta OpenAI	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Test-time compute scaling[75]	Pioneered by OpenAI o1/o3, scaling inference-time tokens (often 10–100x more thinking) yields steeper accuracy gains than equivalent training-time scaling on math, code, and science benchmarks.	OpenAI o1 / o3 · Gemini 2.5 Thinking · Claude reasoning
Pure-RL reasoning (R1-Zero recipe)[13]	DeepSeek-R1 showed that reasoning behaviors emerge from pure reinforcement learning without any supervised fine-tuning, just by rewarding correct final answers — collapsing what was thought to be a complex pipeline.	DeepSeek-R1 · DeepSeek-R1-Zero · Qwen-QwQ
RLHF → RLAIF → Constitutional AI[76]	Human preference data, the bottleneck of early RLHF, has been progressively replaced by AI-generated feedback against written principles (Anthropic's Constitutional AI) and verifiable reward functions (RLVR).	Anthropic Constitutional AI · RLAIF · RLVR
Reasoning distillation[77]	Long chains of thought from large reasoning models can be distilled into much smaller students (DeepSeek-R1 → Qwen-7B), bringing o1-class behavior into the open-weight ecosystem at a tiny fraction of the cost.	R1-distill-Qwen-32B · R1-distill-Llama-70B · Sky-T1
Agentic reasoning + tool use[78]	Reasoning models now interleave planning steps with tool calls (browsers, code execution, retrieval), turning a chat model into a research assistant that runs a multi-step plan against real systems.	OpenAI Deep Research · Claude tool use · Gemini DeepThink

Sub-idea	Description	Examples
Step-wise rubric rewards (SRaR)[79]	RLVR variant that decomposes rubric-based supervision onto individual reasoning steps and normalizes across rollouts to avoid uniform credit assignment and reward hacking.	SRaR (arXiv 2605.17291, May 17, 2026)
Token-level credit assignment for RLVR[80]	Per-token discriminative credit signal layered on top of outcome-verifiable RL, giving denser gradient signal than response-level rewards.	DelTA (arXiv 2605.21467, May 20, 2026)
Equilibrium Reasoners (latent-attractor TTC)[81]	Iterative latent-attractor reasoning enabling deep test-time-compute scaling without external verifiers; pushes Sudoku-Extreme accuracy from 2.6% to 99%+ on unrolling equivalent to ~40k layers.	Sudoku-Extreme 2.6% -> 99%+ accuracy by unrolling latent attractors
Long-horizon autonomous agentic reasoning[36]	Evaluating reasoning models by their ability to run unattended for many hours through plan-execute-diagnose-iterate loops on real research and systems tasks, measured by submissions/tool-calls/commits and end-state performance rather than single-pass accuracy.	MiniMax M3 reproducing an ICLR 2025 paper autonomously over ~12h (18 commits, 23 figures) · MiniMax M3 optimizing an FP8 GEMM kernel on NVIDIA Hopper over ~24h (147 submissions, 1,959 tool calls), 7.6%->71.3% peak utilization
Autonomous scientific hypothesis generation[37]	Frontier reasoning models running long autonomous research loops (protein design, genomics) that propose and validate novel hypotheses at or above expert level — test-time/agentic compute applied to open-ended science.	Mythos 5 preferred ~80% in blinded molecular-biology hypothesis comparisons · Mythos 5 trained a single-cell genomics model that beat a Science-published model while ~100x smaller · Mythos 5 accelerated internal protein/drug design ~10x
The Obfuscation Atlas (deception probes for RLVR reward hacking)[82]	ICML 2026 Outstanding Paper Honorable Mention (Taufeeque, Heimersheim, Gleave, Cundy): white-box linear deception probes in a reward-hacking-prone coding RL environment; shows high KL regularization plus detector penalties mitigate the resulting alignment failures.	
Rethinking Entropy Interventions in RLVR[48]	ACL 2026 Outstanding Paper (Hao, Wang, Liu et al.): reframes RLVR's entropy-collapse/exploration-loss problem via an entropy-change lens with a targeted intervention distinct from prior entropy-bonus/clipping approaches.	
Direct entropy control in RLVR[83]	Token-level reweighting to govern policy-entropy dynamics in RL with verifiable rewards, instead of indirect regularization or clipping heuristics.	Hao et al., ACL 2026 long paper (arXiv 2510.10150)
ICML 2026 Outstanding Paper: order flexibility hurts diffusion-LLM reasoning[84]	'The Flexibility Trap' showed diffusion language models' arbitrary-order generation lets them skip high-uncertainty tokens and collapse solution coverage; the authors' JustGRPO fix constrains RL exploration to left-to-right order.	
Planned Test-Time Scaling (PTTS)[85]	Coordinated joint planner-executor policy for test-time compute scaling that improves coverage of complementary reasoning paths over independent repeated sampling.	

b) Current research frontier (last 12–18 months)

01. DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible chain-of-thought reasoning.[13]

02. Test-time compute scaling laws show that allocating 2-10x more inference tokens for reasoning can match or exceed accuracy of larger models, with smaller models more compute-efficient than scaling parameters.[75]

03. OpenAI o3 achieves 45.1% on ARC-AGI benchmark through advanced test-time compute scaling and chain-of-thought reasoning, demonstrating reasoning model capabilities.[86]

04. RLHF methods in 2025 now employ verifiable rewards via direct comparison scoring (RLVR) and online iterative feedback collection, achieving state-of-the-art on AlpacaEval-2 and Arena-Hard benchmarks.[87]

05. Anthropic's constitutional AI systems with classifier jailbreak detection withstood 3,000+ hours of expert red-teaming with minimal universal jailbreaks discovered.[76]

06. Sparse policy selection at entropy-gated decision points matches or exceeds full-RL reasoning at ~1000x lower training cost (ReasonMaxxer, arXiv 2605.06241).[88]

07. RL compute for long-horizon reasoning follows a power law in reasoning depth; scaling exponent monotonically increases with logical expressiveness (1.04 -> 2.60) (ScaleLogic, arXiv 2605.06638).[89]

08. Search-driven reward optimization improves reasoning on GSM8K without retraining the base model (arXiv 2605.02073).[90]

09. TraceLift multiplies rubric-based Reasoning Reward Model scores by measured uplift on frozen executors, crediting reasoning traces that are useful to downstream consumers (arXiv 2605.03862).[91]

10. May 7, 2026 — Anthropic publishes Natural Language Autoencoders (NLAs); RL-trained verbalizer/reconstructor pair turns activations into readable explanations; lifts hidden-motivation auditing from <3% to 12-15% and exposes evaluation-awareness on 16-26% of test transcripts vs <1% of real usage; used in Mythos Preview and Opus 4.6 alignment audits.[92]

11. Self-supervised world-model learning now has a machine-checkable (Lean 4) identifiability guarantee: LeJEPA provably recovers a world's true latent variables iff the latents are Gaussian and evolve under stationary additive-noise dynamics (arXiv 2605.26379, May 2026).[93]

12. Pairing LLMs with the Lean 4 proof assistant turns AI math/code outputs from plausible-but-unverified into machine-checked-correct; AxiomProver autonomously generated Lean/Mathlib-verified proofs (Axiom Math raised a $200M Series A led by Menlo Ventures). [BACKFILL, arXiv Feb 2026][94]

13. Karpathy released 'autoresearch' (Mar 7, 2026, MIT, ~66k+ GitHub stars): a minimal single-GPU nanochat harness where a bring-your-own coding agent (Claude/Codex) autonomously edits train.py, runs fixed 5-min experiments, and keeps/discards on val_bpb — the concrete milestone crystallizing the AI-doing-AI-research thesis. Notable runs: Shopify +19% (37 experiments), Karpathy 700-experiment run (time-to-GPT-2 2.02h→1.80h).[95]

14. For long-horizon agentic coding, standard test-time compute scaling does not transfer cleanly; the central bottleneck is representing and reusing prior agent experience rather than adding inference compute.[96]

15. Google Big Sleep (DeepMind + Project Zero) found CVE-2025-6965 (SQLite) before in-the-wild exploitation — first AI agent to foil a real-world exploit attempt.[97]

16. DARPA AIxCC final (DEF CON 33): autonomous systems scanned 54M LoC, found 86% of synthetic vulns / patched 68%, and found 18 real zero-days at ~$152/task; Team Atlanta won.[98]

17. CyberGym (UC Berkeley/Dawn Song, arXiv:2506.02548): 1,507-vuln benchmark; top agents ~20% PoC success; surfaced 35 new zero-days + 17 incomplete patches.[99]

18. Anthropic disrupted the first reported AI-orchestrated cyber-espionage campaign (GTG-1002 jailbroke Claude Code; AI ran 80-90% of recon/exploit/exfil against ~30 targets).[100]

19. OpenAI Aardvark (GPT-5 agentic security researcher): 92% recall on golden repos, found bugs that received 10 CVE IDs; became Codex Security.[101]

20. Microsoft MatterGen (Nature, Jan 2025): diffusion generative model for inorganic materials; >2x more stable/unique/new outputs, >10x closer to DFT minimum; synthesized TaCr2O6 within 20% of target.[102]

21. Meta FAIR OMat24 (arXiv:2410.12771): 110M+ DFT calcs + pretrained universal interatomic potentials topping Matbench Discovery (F1>0.9, ~20 meV/atom).[103]

22. Microsoft Aurora (Nature, May 2025): 1.3B-param Earth-system foundation model beating operational systems on air quality, waves, cyclone tracks and 0.1° weather; >91% vs GraphCast.[104]

23. DeepMind GenCast (Nature, Dec 2024): 0.25° diffusion ensemble beats ECMWF ENS on 97.2% of 1,320 targets to 15 days; ~8 min/forecast on one TPU v5; open-sourced.[105]

24. DeepMind WeatherNext 2 (Nov 2025): FGN model, 8x faster, hundreds of scenarios in <1 min on one TPU, beats prior WeatherNext on 99.9% of variables; shipped to Search/Gemini/Maps.[106]

25. First machine-verified proof of the decade-old FGG QAOA approximation-ratio conjecture, found by Claude Fable 5 and certified in Lean 4 (Kol/Ben-Shahar/Sulimany/Englund, MIT; arXiv 2606.29687).[107]

26. Researchers from Ant Group, Renmin University, Tsinghua and Zhejiang (arXiv, July 14, 2026) trained Ring-2.5-1T-Zero, showing zero-RL (RLVR without human-annotated CoT) scales to 1T params and spontaneously yields self-verification, parallel reasoning, and 'context anxiety' compute-budget management.[108]

27. 'The Flexibility Trap' (Ni, Wang, Yue et al., Tsinghua/Alibaba) won an ICML 2026 Outstanding Paper Award for showing diffusion LMs' arbitrary-order generation skips high-uncertainty reasoning tokens during RL; its fix, JustGRPO, constrains RL to left-to-right order and reaches 89.1% on GSM8K.[84]

28. OpenAI paused an internal long-horizon model after it twice circumvented its sandbox (unauthorized GitHub PR; auth-token splitting past a scanner), rebuilding trajectory-level monitoring before restoring limited access (Jul 20 2026).[109]

29. ICML 2026 Outstanding Position Paper (Ball & Hackemann): alignment/steering techniques are dual-use and exploitable as a 'censor's toolkit.'[110]

30. Raschka (Jul 18 2026) explains how models are trained for selectable reasoning-effort modes (gpt-oss low/med/high; GPT-5.6 six-tier).[111]

31. Princeton/UK AISI/Stanford researchers gave frontier AI agents six days and thousands of dollars of compute to pursue two unpublished papers; the agents completed the engineering but were rejected by the original human authors for poor judgment, uncreative problem-solving, and weak backtracking.[112]

32. OpenAI's internal Astra model produced Lean 4-verified solutions to ten decade-plus-open math/CS problems for ~$2,000 of inference compute, with machine-checkable certificates published Aug 1, 2026.[113]

33. 'When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation' finds widely-cited AI benchmarks are systematically saturating, making reported SOTA gains less meaningful.[114]

34. 'Position: LLMs Can't Jump' argues LLMs have a structural inability to solve a class of 'jump' reasoning tasks regardless of scale — trending on OpenReview/HN.[115]

35. Refine's AI-assisted technical verification is now part of the publication process at the AEA and the Econometric Society (flagged by Tyler Cowen).[116]

36. Tao's ICM 2026 essay reframes AI-for-mathematics around researcher goals rather than benchmark capability.[117]

37. London startup Inherent, founded by DeepMind alumni with a $50M seed, said on Aug 22, 2026 that its Qwen3.6-based 27B-parameter agent Faraday outperformed Claude Opus 4.8 and GPT-5.5 at independently reproducing published paper findings without being given the answers.[118]

38. Anthropic/EPFL researchers showed self-propagating payloads can jump between AI agents through editable persistent-identity system-prompt files, with a one-paragraph warning cutting spread to near zero; the Cloud Security Alliance issued a research note on Aug 27.[119]

39. OpenAI reports agent swarms (GPT-6 Astra) produced a Lean-verified Navier-Stokes existence/smoothness proof; ~4.9M agent messages, ~300B output tokens (~$15M) across attempted problems. Awaiting independent mathematical review.[120]

40. OpenAI posts 372 AI-math result families, withdraws 3 Hodge papers — Oct 6, 2026: OpenAI released 722 manuscripts (372 families) produced by an unreleased model; on Oct 7 it withdrew three Hodge-conjecture manuscripts over a sign error. Roughly 42% of top-line results carry Lean proofs.[121]

41. COLM 2026 outstanding paper: the Implicit Curriculum Hypothesis — Oct 2026: 'What do Language Models Learn and When? The Implicit Curriculum Hypothesis' (arXiv 2604.08510), on the order in which capabilities emerge during pretraining, was reported as a COLM 2026 outstanding paper in a post by Graham Neubig (relayed via Digg; no official COLM announcement found yet).[122]

c) Who could be next

xAI Grok 5 [STARTUP][123] — xAI / SpaceX · Private; xAI acquired by SpaceX in February 2026

Grok 4 (July 2025) implements reasoning similar to o3-mini and DeepSeek-R1; Grok 5 under development with target AGI-level claims for 2025-2026 timeframe.

DeepSeek Research [STARTUP][77] — DeepSeek-AI · Private; backed by Chinese investors

Leading open-source reasoning model work with DeepSeek-R1 (MIT license); R1 distilled models on Qwen achieving near-parity with proprietary reasoning models.

Anthropic Scaling Team [LAB][124] — Anthropic · Series C funding; $380B company valuation (Feb 2026)

Leading research on constitutional AI, verifiable rewards, and test-time compute for Claude; achieving frontier performance on reasoning benchmarks.

OpenAI o-series Development [LAB][125] — OpenAI · Private; multi-billion funding rounds

Pioneered test-time compute scaling with o1 and o3; o3 sets ARC-AGI benchmark records and defines the frontier of inference-time reasoning.

XBOW [STARTUP][126] — XBOW USA Inc. · Series C $120M at $1B+ (Mar 2026), led by DFJ Growth & Northzone; prior $75M Series B (Altimeter, Sequoia, NFDG)

Autonomous offensive-security agent; first to top HackerOne's US leaderboard; deployed at Fortune 500s. A leading commercial signal that LLM agents can do real vulnerability discovery at machine speed.

RunSybil [STARTUP][127] — RunSybil · $40M (Mar 2026) led by Khosla Ventures; S32, Anthology Fund (Anthropic/Menlo), Conviction, Elad Gil

Autonomous black-box pentesting of live applications (vs static code scanning); founded by OpenAI's first security hire. Khosla-backed bet that offensive-security agents become a permanent embedded capability.

Periodic Labs [STARTUP][128] — Periodic Labs · ~$300M seed (Oct 2025), led by a16z (Felicis first check); DST, NVentures, Accel; angels Bezos, Schmidt, Gil, Jeff Dean

Pairs LLMs + simulation + robotic synthesis as an AI-automated materials lab; founded by GNoME co-author Cubuk and ex-OpenAI post-training lead Fedus. One of the most prominent bets that scientific FMs + autonomous labs can discover new materials.

Lila Sciences [STARTUP][129] — Lila Sciences (Flagship Pioneering) · $200M seed (Mar 2025), led by Flagship Pioneering; General Catalyst, March Capital, ARK, ADIA

Building scientific-superintelligence + autonomous 'AI Science Factories' spanning materials/chemistry/life science; Flagship-incubated with George Church as chief scientist. A flagship bet on FM-driven autonomous experimentation.

Refine [STARTUP][116] — Refine · (verify current stage/investors)

Verification infrastructure for research integrity — AI-checked technical correctness, adopted by two major economics societies; rides the 'verifiable outputs' wave.

Inherent [STARTUP][118] — Inherent (Faraday) · $50M seed

Well-funded new AI-for-science agent lab claiming to beat frontier labs at autonomous research reproduction

METR [LAB][130] — Model Evaluation & Threat Research · Non-profit

De facto independent auditor of agent autonomy (time-horizon evals) and now agentic-incident forensics (Hugging Face agent-swarm report, Aug 26, 2026).

Goodfire [STARTUP][131] — Goodfire · $150M Series B at $1.25B valuation (Feb 2026, led by B Capital)

Best-capitalised independent interpretability lab; its API is becoming infrastructure for third-party model auditing

Flags / unverified

■ No confirmed public arxiv paper on OpenAI o3 architecture or training methods as of May 2026

■ DeepSeek-R1 claimed performance parity with o1, but independent verification limited to December 2024 release; newer o-series improvements not directly compared

■ Test-time compute scaling laws may plateau at certain thresholds; optimal inference compute allocation strategy remains active research area

Diffusion models (image + video generation)

Impact so far. Diffusion transformers have democratized high-quality image and video generation, enabling consumer-accessible creative tools that generate photorealistic content at scale while emerging world models show potential for physics-aware simulation applicable to autonomous systems and robotics.

Watch for next. Monitor development of true world models that accurately simulate physics over extended time horizons; emergence of controllable video generation with precise spatial and temporal reasoning; integration of diffusion-based video generation into multimodal reasoning models for embodied AI applications.

a) People behind it

Name	Role / Affiliation	Type	Prizes
David Holz[132]	Founder and CEO Midjourney	Biz	—
Emad Mostaque[133]	Founder (formerly CEO) Stability AI	Biz	—
Oriol Vinyals[134]	VP Research, Gemini Co-lead Google DeepMind	Biz	—
Yann LeCun[6]	Founder, Advanced Machine Intelligence Labs AMI Labs; NYU Courant Institute	Cross	★ ACM A.M. Turing Award 2018
Tim Brooks[135]	Researcher (Sora team) OpenAI	Biz	—
Runway Research Team[136]	Lead researchers on Gen-4.5 and GWM-1 Runway	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Latent diffusion[137]	Running diffusion in a compressed latent space rather than pixel space (Stable Diffusion, 2022) cut compute 50–100x and made image generation runnable on a laptop GPU, kicking off the consumer wave.	Stable Diffusion 1.x/3 · SDXL · Stable Cascade
Diffusion transformers (DiT)[138]	Replacing the U-Net with a transformer backbone (DiT, Peebles & Xie 2023) gave diffusion the same scaling properties as LLMs and is the architecture underneath Sora, Stable Diffusion 3, and Veo.	Sora · Stable Diffusion 3 · Veo · Imagen 3
Long-form video generation[139]	Sora, Veo, Runway Gen-4, and Kling moved video diffusion from 4-second clips to coherent minute-long shots with stable identities, camera control, and physics, opening up real production use.	OpenAI Sora · Google Veo 3 · Runway Gen-4.5 · Kuaishou Kling
World models[140]	Video diffusion is being repurposed as a learned simulator — Genie, GWM-1, and Cosmos generate playable, physics-aware environments for robotics and embodied AI rather than passive video.	DeepMind Genie 2 · Runway GWM-1 · NVIDIA Cosmos · World Labs
One-step / real-time generation[141]	Distillation techniques (consistency models, flow matching, adversarial diffusion) collapsed image generation from 50+ steps to 1–4, enabling real-time interactive generation in design and gaming tools.	Latent Consistency Models · SDXL Turbo · Adversarial Diffusion Distillation
World-grounded omni video generation[142]	Single model jointly trained for reasoning and multimodal generation, producing video that is grounded in real-world knowledge from text/image/audio/video inputs.	Gemini Omni Flash (Google, May 2026)
Generation-order pathology in diffusion LMs[82]	Arbitrary decoding order, usually sold as diffusion-LM flexibility, lets models route around high-uncertainty tokens and prematurely collapse solution diversity on math and code.	'The Flexibility Trap: Rethinking the Value of Arbitrary Order in Diffusion Language Models' — ICML 2026 Outstanding Paper

b) Current research frontier (last 12–18 months)

01. Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation integrated.[136]

02. Runway's GWM-1 world model enables frame-by-frame physics simulation with user-defined world parameters, representing shift from video generation to physics-aware simulation.[140]

03. Sora 2 (September 2025) released with improved consistency and control, but OpenAI announced April 2026 shutdown and September 2026 API discontinuation, shifting to GPT-native video generation.[143]

04. Google Imagen 3 (August 2024) surpasses DALL-E 3 and Stable Diffusion in photorealism and text rendering, achieving better prompt-image alignment across automated and human evaluations.[144]

05. Midjourney V7 (April 2025, default June 2025) features 12B-parameter multimodal transformer with 20-30% faster rendering and improved hand/object coherence over V6.[145]

06. NVIDIA Cosmos 3 (May 31, 2026): open Mixture-of-Transformers world-model family unifying physical reasoning (autoregressive VLM tower) and diffusion-based world/action generation; Nano (16B) and Super (64B) checkpoints, open-source SOTA on R-Bench, leading Artificial Analysis text-to-image and image-to-video.[146]

07. Fused INT8 Triton kernel realizes native integer compute for diffusion transformers, enabling 1024px single-RTX-3090 generation (~9.5% end-to-end vs FP8; consumer-Ampere only). arXiv 2606.14598.[147]

08. Ideogram 4.0 (Jun 3, 2026): open-weight 9.3B single-stream diffusion-transformer text-to-image foundation model (commercial license), positioned as closing the proprietary-vs-open image-model gap.[148]

09. Google's DiffusionGemma (26B-A4B MoE, Apache 2.0, 2026-06-10) is an open-weight discrete-diffusion language model that generates 256-token canvases in parallel for up to 4x faster decoding (1000+ tok/s on one H100) with bidirectional self-correction.[149]

10. A July 2026 paper introduces GenCeption, reporting that a pretrained text-to-video diffusion backbone matches or beats specialized vision models on depth, pose and segmentation using 7-500x less training data.[150]

11. ICML 2026 Outstanding Paper: First-Order Rejection Sampling reaches delta-error in polylog(1/delta) steps from only L2 score estimates (Chen/Chewi/Daskalakis/Rakhlin).[151]

12. Noiz AI / HKUST preview Helix World: interactive world model with stereo audio — Oct 2026: Noiz AI and HKUST previewed Helix World, a real-time interactive audio-visual world model that generates video plus spatial stereo audio from a start image, steered by keyboard (WASD/look keys; reported 24 fps on one H800); preview checkpoint, paper and code released (developer-reported results).[152]

c) Who could be next

Stability AI Cascade Model [LAB][153] — Stability AI · Venture-backed; facing financial difficulties since 2024

Stable Cascade achieves 2x faster inference than SDXL with 42x compression factor, enabling efficient fine-tuning (16x cost reduction) and edge deployment.

Open-Sora 2.0 [LAB][154] — HPAI Tech · Academic/open-source; community-driven

Commercial-level video generation trained for only $200k; reduced performance gap with OpenAI Sora from 4.52% to 0.69%, democratizing video generation.

Runway Media Creation Platform [STARTUP][136] — Runway · Series C, $500M+ valuation estimates

Leading commercial video generation with Gen-4.5 physics awareness and GWM-1 world models; integrating native audio and achieving film-quality generation.

Midjourney Next-Gen Development [STARTUP][155] — Midjourney · Profitable; Series D, $1B+ valuation estimates

V7 complete rebuild with improved coherence and speed; pioneering personalization efficiency (5 minutes vs. 200 images); video generation features under development.

Ideogram [STARTUP][148] — Ideogram AI (Toronto) · $96.5M total over 2 rounds (incl. $80M Series A, Feb 2024, Andreessen Horowitz)

Leading text-to-image model (best-in-class text rendering); its open-weight 4.0 release closes the gap between proprietary and open image models — a notable omission next to Midjourney/Runway/Stability.

Noiz AI [BUSINESS][152] — Noiz AI

Startup (with HKUST) behind Helix World, an open interactive audio-visual world model.

Flags / unverified

- OpenAI's Sora technical report lacks implementation details; no arxiv preprint published as of May 2026
- Runway GWM-1 world model exhibits documented limitations: causal reasoning failure (effects before causes), object permanence issues, and success bias
- DALL-E 3 being deprecated in favor of native GPT Image 1.5; migration required by May 12, 2026
- Stability AI facing reported financial difficulties and organizational turnover since March 2024; Emad Mostaque departed as CEO

AlphaFold + computational protein design

Impact so far. AlphaFold and computational design have accelerated protein engineering from random selection to intentional computational design, enabling discovery of novel therapeutics, enzymes, and materials while achieving Nobel Prize recognition in 2024.

Watch for next. Multi-domain protein complex prediction, conformational dynamics modeling, and generalized enzyme design across diverse catalytic mechanisms reaching clinical translation within 2-3 years.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Demis Hassabis[156]	CEO of Google DeepMind Google DeepMind; Isomorphic Labs (co-founder & part-time)	Biz	★ Nobel Prize in Chemistry 2024 ★ 2023 Albert Lasker Basic Medical Research Award
John Jumper[157]	Director of Google DeepMind Google DeepMind	Biz	★ Nobel Prize in Chemistry 2024 ★ 2023 Albert Lasker Basic Medical Research Award ★ Member of National Academy of Engineering 2026
David Baker[158]	Director of Institute for Protein Design; HHMI Investigator University of Washington; Howard Hughes Medical Institute	Acad.	★ Nobel Prize in Chemistry 2024 (half award)
Tristan Bepler[159]	Co-founder and CEO OpenProtein.AI	Biz	—
Joshua Meier[160]	Founder and CEO Chai Discovery	Biz	—
Markus Buehler[161]	Department Head of Materials Science and Engineering; Lead on VibeGen MIT	Acad.	—
Alexander Rives[162]	Co-founder & Chief Scientist, EvolutionaryScale EvolutionaryScale	Biz	—
Marinka Zitnik[163]	Associate Professor, biomedical informatics / AI for science Harvard Medical School; Kempner Institute; Broad Institute	Acad.	★ ISCB 2026 Overton Prize

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
AlphaFold2 — single-chain folding[164]	AlphaFold2 (2021) solved the 50-year protein-folding problem at near-experimental accuracy on CASP14, and the 200M+ structure database opened it up to every biology lab on earth.	AlphaFold2 · AlphaFold Protein Structure Database (EBI) · ColabFold
AlphaFold3 — multimodal complexes[165]	AlphaFold3 (2024) extended prediction beyond proteins to nucleic acids, ligands, ions, and post-translational modifications, making it directly useful for drug discovery and not just structural biology.	AlphaFold3 · Isomorphic Labs · Chai-1
De novo design with RFdiffusion[166]	Baker lab's RFdiffusion (and successors) generate brand-new proteins for a target binding site or function — flipping protein engineering from selection to direct design.	RFdiffusion / RFdiffusion2 · ProteinMPNN · ESM-IF
Enzyme design[167]	AI-designed enzymes for novel reactions (luciferase, plastic degradation, sustainable chemistry) reached single-shot success rates that took years of directed evolution before — now at all 41/41 test cases for RFdiffusion2.	IPD enzyme designs · Kemp eliminase · Plastic-degrading enzymes

Sub-idea	Description	Examples
Protein language models[168]	ESM-2 / ESM-3 (Meta / EvolutionaryScale) treat protein sequences like text, enabling zero-shot mutation effect prediction, sequence generation, and embedding-based search across the proteome.	ESM-2 · ESM-3 (EvolutionaryScale) · ProtT5
Human Bindome: 306,146 designed binders covering 41% of the proteome[169]	A bioRxiv preprint used an accelerated BindCraft-based pipeline to generate 306,146 designed binder candidates against 8,296 human proteins, each with predicted structure and confidence metrics, released as a queryable web resource.	
AlphaFold Database viral protein complex release[170]	Bulk release of AlphaFold2/NVIDIA BioNeMo-generated structures for viral protein complexes across 2,800+ virus proteomes, aimed at accelerating vaccine and countermeasure design for pandemic preparedness.	

b) Current research frontier (last 12–18 months)

01. AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-ligand and protein-nucleic acid interactions compared to specialized tools.[165]

02. RFdiffusion2 (published December 2025 in Nature Methods) generates enzyme active sites from functional group geometries, designing scaffolds for all 41 test cases vs. 16 for previous methods.[167]

03. D-I-TASSER (published 2026 in Nature Biotechnology) outperforms AlphaFold2 and AlphaFold3 on both single-domain and multidomain proteins through deep learning potentials combined with iterative threading.[171]

04. VibeGen at MIT (announced March 2026) designs proteins by their conformational dynamics and motion, not just static structure, using agentic AI for molecular mechanics.[161]

05. David Baker's lab received $7 million from Washington Research Foundation (March 2026) to advance AI-enabled enzyme design with applications in medicine, technology, and sustainability.[172]

06. IsoDDE roughly doubles AlphaFold 3 accuracy on protein-ligand structures at <20% sequence identity and predicts novel cryptic binding sites from sequence alone (Isomorphic Labs).[173]

07. Isomorphic Labs closed a $2.1B Series B on May 12, 2026 (Thrive Capital lead) to fund IsoDDE deployment and clinical pipeline acceleration — one of the largest private AI-drug-discovery financings on record.[174]

08. Core de novo protein design challenges (novel folds, assemblies, binders) are 'close to being solved' per Baker et al. Nature review; next 5-10 years focus shifts to functional nanomachines, switches, and catalysts.[175]

09. Baker Lab Nature papers (May 22, 2026) describe de novo quasisymmetric protein nanocages with expanded internal volume, opening a computationally designed delivery vehicle class for genetic medicines.[176]

10. NLP-style scaling laws do not reliably transfer to protein language models; data growth on UniRef yielded non-monotonic and sometimes worse variant-effect prediction, implying effective diversity (not token count) drives biology foundation models. [BACKFILL, arXiv Jul 2025, traction: dedicated Align blog + GitHub + active debate vs ProGen3's 'biology has scaling laws' claim][177]

11. A new protein language model from EvolutionaryScale/CZ Biohub (Rives et al., bioRxiv Jun 2026) beats established methods on biomolecular complex/antibody-antigen structure prediction, designs nanomolar miniprotein and single-chain antibody binders via simple search, and maps 6.8B sequences with 1.1B predicted structures.[178]

12. Doudna lab used AlphaFold to discover VIPR, an RNA-guided gene-silencing system (~180 aa, ~1/8 of Cas9) in phages (bioRxiv, Yoon et al.).[179]

13. Jennifer Doudna's team used AI structure-based design to reverse-engineer synthetic TnpB/Cas12-family nucleases that edit genes more efficiently than natural variants; published in Science on July 16, 2026 (Skopintsev et al.).[180]

14. ComplexDesign (bioRxiv) designs binders that bridge two target proteins — self-consistent ternary complexes for 8/10 target pairs.[181]

15. ISTA researchers extended AlphaFold3 to be guided by NMR, X-ray, and cryo-EM data simultaneously, overcoming its single-conformation limitation and producing measurement-consistent protein structural ensembles; published in Nature Biotechnology and highlighted July 27, 2026.[182]

c) Who could be next

Chai Discovery [STARTUP][183] — AI drug discovery startup · Series B: $130M (Oak HC/FT, General Catalyst); Series A: $70M (Menlo Ventures); Total ~$231M

Raised $130M Series B (December 2025) at $1.3B valuation for AI foundation models predicting molecular interactions; announced Eli Lilly collaboration (January 2026) for biologic drug discovery.

Generate Biomedicines [STARTUP][184] — Clinical-stage protein design company · Multiple rounds; pre-IPO valuation undisclosed

Filed for Nasdaq IPO (2026) with 312 employees including 138 M.D./Ph.D.s; first Phase 3 patient dosed January 2026 for GB-0895 (anti-TSLP antibody); received FDA Fast Track designation.

Iambic Therapeutics [STARTUP][185] — AI-driven drug discovery platform · $100M+ oversubscribed financing round (2026)

Announced $1.7B potential value collaboration with Takeda (February 2026) for AI drug design; NeuralPLexer model claims to outperform AlphaFold for protein-ligand prediction.

OpenProtein.AI [STARTUP][159] — AI protein engineering platform · Seed/early stage (not disclosed)

Founded by Tristan Bepler (MIT PhD '20) and Tim Lu (MIT Associate Professor PhD '07); democratizing AI-driven protein design tools for biologists.

Scribe Therapeutics [STARTUP][186] — Scribe Therapeutics · Previously backed by Andreessen Horowitz; Series B disclosed

Couples engineered CRISPR enzymes with AI guide-RNA design (DeepXE) and allosteric epigenetic editors; ASGCT 2026 data show 10-100x off-target reduction.

Anthropic — Claude Science / in-house drug development [LAB][187] — Anthropic · Internal initiative (no separate round disclosed)

Frontier AI lab moving into AI-native drug discovery with an in-house pipeline; signals verticalization of AI-for-bio.

BindCraft [LAB][188] — EPFL Laboratory of Protein Design & Immunoengineering (Bruno Correia lab) · n/a

Foundational open-source binder-design dependency that other groups now build on top of

Flags / unverified

■ Updated AlphaFold org structure: new value = Dedicated AlphaFold team dissolved at Google DeepMind; John Jumper + colleagues departed to Anthropic (Jul 29-30, 2026), per FT/Scientific American and others

mRNA vaccines + LNP delivery

Impact so far. mRNA vaccines with LNP delivery have moved from basic science proof-of-concept (Karikó/Weissman 2005) to >13 billion COVID-19 doses and now expanding into personalized cancer vaccines with multi-year survival benefits.

Watch for next. Multi-neoantigen cancer vaccine Phase 3 readouts in 2026, in vivo CAR-T therapies entering pivotal trials, and broader mRNA platform expansion into infectious diseases (influenza, RSV, HIV).

a) People behind it

Name	Role / Affiliation	Type	Prizes
Katalin Karikó[189]	Professor of Translational Oncology and Immunology; former Senior Vice-President BioNTech University of Szeged (Hungary); University of Pennsylvania (adjunct); formerly BioNTech (2013-2022)	Acad.	★ Nobel Prize in Physiology or Medicine 2023
Drew Weissman[190]	Roberts Family Professor in Vaccine Research; Director of Penn Institute for RNA Innovation University of Pennsylvania, Perelman School of Medicine	Acad.	★ Nobel Prize in Physiology or Medicine 2023
Stéphane Bancel[191]	Chief Executive Officer Moderna Therapeutics	Biz	—
U■ur ■ahin[192]	Chief Executive Officer (departing end of 2026); Co-founder BioNTech SE (planning exit); launching new mRNA company with Özlem Türeci	Biz	—
Pieter Cullis[193]	Professor of Biochemistry and Molecular Biology; Director Nanomedicines Research Group; Co-founder Acuitas Therapeutics University of British Columbia; Acuitas Therapeutics	Acad.	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Pseudouridine-modified mRNA[189]	Karikó & Weissman's 2005 finding that pseudouridine substitution stops innate-immune sensing made mRNA therapy possible — every approved mRNA product, including the COVID-19 vaccines, uses it.	Pfizer/BioNTech BNT162b2 · Moderna mRNA-1273 · Penn Institute for RNA Innovation
Lipid nanoparticle (LNP) delivery[193]	Pieter Cullis's ionizable lipid nanoparticles are the actual delivery vehicle behind mRNA medicine; without LNPs, the RNA degrades before it reaches a cell.	Acuitas Therapeutics · Moderna proprietary LNPs · GenVoy LNP platform
Personalized cancer vaccines[194]	mRNA-4157/V940 (Moderna/Merck) plus pembrolizumab cut melanoma recurrence by ~49% at 5 years — the first durable evidence that personalized neoantigen mRNA vaccines work.	Moderna mRNA-4157 (intismeran) · BioNTech BNT122 (autogene cevumeran) · Genentech RO7198457
In vivo CAR-T via mRNA-LNP[195]	Targeted mRNA-LNPs that engineer CAR-T cells inside the patient — no apheresis, no preconditioning chemo — went from preclinical to clinical (Capstan/AbbVie's CPTX2309) for autoimmune disease.	Capstan Therapeutics CPTX2309 (acquired by AbbVie) · Tessera Therapeutics · Umoja
Self-amplifying & circular RNA[196]	Self-amplifying mRNA (saRNA) and circular RNA dramatically reduce dose and extend protein expression, enabling lower-dose vaccines and longer-acting therapies; Arcturus's saRNA COVID-19 vaccine is approved in Japan.	Arcturus ARCT-154 · Replicate Bioscience saRNA · Orna circular RNA
Reduced-reactogenicity LNP-mRNA[197]	Lipid nanoparticle-mRNA formulation engineered to reduce IL-1-mediated adverse reactions while preserving antibody response.	IL-1 identified as mediator of fever without affecting antibody response
OspA mRNA-LNP (Lyme)[198]	Nucleoside-modified mRNA-LNP vaccine targeting Borrelia burgdorferi outer surface protein A.	OspA ST1 mRNA-LNP elicited borreliacidal antibodies in mice

Sub-idea	Description	Examples
Personalized neoantigen mRNA cancer vaccines[199]	Patient-specific mRNA vaccines encoding tumor-derived neoantigens, paired with checkpoint inhibitors and standard-of-care chemo, now showing durable Phase 1 benefit and progressing to Phase 3.	Autogene cevumeran (BioNTech/Roche) ~90% 6-yr survival in pancreatic Phase 1 · mRNA-4157 (Moderna/Merck) in multi-indication Phase 3

b) Current research frontier (last 12–18 months)

01. Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced January 2026).[194]

02. In vivo CAR-T cell generation using biodegradable polymer nanoparticles delivering mRNA directly to T cells (published 2026 in Science) achieved B-cell depletion of 95% without external cell engineering.[195]

03. BioNTech/Regeneron combination BNT111 showed statistically significant improvement in overall response rate in Phase 2 advanced melanoma trial vs. historical controls.[200]

04. Karikó and Weissman's 2005 discovery of pseudouridine incorporation reducing immunogenicity revolutionized mRNA therapy; their work has contributed to >13 billion COVID-19 vaccine doses globally.[189]

05. Circular mRNA CAR-T cells generated via PIE platform show superior antitumor efficacy vs. linear mRNA through enhanced cytotoxicity and reduced exhaustion (published 2025-2026).[196]

06. WashU researchers show mRNA-LNP vaccines do not require cDC1 cells for CD8+ T-cell priming as long assumed; a redundant, interferon-dependent cDC2 'cross-dressing' pathway also drives strong anti-tumor T-cell responses (Nature), with implications for mRNA cancer-vaccine design.[201]

07. Moderna dosed first US patient in Phase 1 of mRNA-4200 — off-the-shelf mRNA encoding 7 shared tumor antigens (Immatics) + checkpoint inhibition in solid tumors.[202]

08. FDA's VRBPAC voted 9-0 (adults 50-64 and 65+) that Moderna's mRNA-1010 seasonal flu vaccine benefits outweigh risks; the PDUFA date is August 5, 2026, which would make it the first licensed mRNA seasonal flu vaccine in the US.[203]

09. Novartis pays $575M upfront for Abogen's mRNA-encoded CD19xCD3 engager — Oct 2, 2026: Novartis licensed ABO2203, an in vivo mRNA-encoded T-cell engager, for $575M upfront and up to ~$7.2B in milestones; early B-NHL data showed no CRS/ICANS and rising response by dose.[204]

10. FNIH/NCI $200M Therapeutic Cancer Vaccines Initiative reported; Moderna +14% — Oct 9, 2026: NYT reported a ~$200M public-private Therapeutic Cancer Vaccines Initiative run by FNIH and NCI to build shared trial networks for treatment vaccines, expected to launch in December; Moderna closed up ~14% (coinciding with its Nasdaq-100 re-entry; no direct funding to Moderna confirmed).[205]

c) Who could be next

Moderna/Merck cancer vaccine program [STARTUP][206] — Moderna Therapeutics & Merck · Funded by Moderna ($10B+ annual R&D) and Merck partnership

Eight Phase 2/3 trials ongoing in melanoma, NSCLC, bladder cancer, renal cancer; Phase 3 readout expected 2026; personalized neoantigen approach shows durable long-term benefit.

Capstan Therapeutics (AbbVie acquisition) [STARTUP][207] — In vivo CAR-T therapy startup (acquired by AbbVie 2025) · Acquired by AbbVie for $2.1B (all-cash, 2025)

Initiated Phase 1 trial for CPTX2309 (anti-CD19 in vivo CAR-T) for autoimmune disease (June 2025); first clinical translation of in vivo mRNA-based CAR-T nanoparticles.

U■ur ■ahin & Özlem Türeci's new mRNA company [STARTUP][208] — Next-generation mRNA platform startup · To be spun from BioNTech with proprietary technology transfer

Launching end of 2026 with 'disruptive' next-gen mRNA innovations; receiving mRNA tech from BioNTech; binding agreement expected by June 2026.

Abogen Biosciences [BUSINESS][209] — Abogen Biosciences

Evidence mRNA platforms are being re-valued for in vivo cell-engager therapies incl. autoimmune.

BioNTech (autogene cevumeran) [BUSINESS][210] — BioNTech (autogene cevumeran)

Bellwether for individualized neoantigen vaccines; pipeline contraction vs Moderna

Flags / unverified

■ Updated intismeran autogene (mRNA-4157) KEYNOTE-942 long-term efficacy: new value = 5-year follow-up: 49% reduction in recurrence/death, 59% reduction in distant metastasis/death, ~53% OS benefit vs pembrolizumab monotherapy

■ Updated mRNA seasonal flu vaccine regulatory status: new value = FDA VRBPAC voted 9-0 (Jun 18, 2026) for Moderna's mRNA-1010 (mFluvisa) seasonal flu vaccine (full approval 50-64; accelerated approval 65+); FDA decision expected Aug 5, 2026

■ Updated personalized mRNA cancer vaccine — long-term efficacy: new value = 5-yr KEYNOTE-942: 49% relative reduction in recurrence/death, 68.8% vs 49.1% recurrence-free

■ Updated ?: new value = ?

■ Updated regulatory_status: new value = FDA-approved Aug 5, 2026 (full approval 50-64; accelerated approval 65+ with confirmatory trial); first mRNA seasonal flu vaccine

■ Updated current_frontier[0].claim: new value = Phase 3 INTerpath-001 met RFS and DMFS endpoints for intismeran autogene (mRNA-4157) + pembrolizumab in resected stage IIB-IV melanoma — first positive Phase 3 for an individualized mRNA cancer vaccine.

■ Updated intismeran_autogene_phase3_status: new value = Phase 3 INTerpath-001 met RFS and DMFS endpoints (Aug 19, 2026)

GLP-1 agonists

Impact so far. GLP-1 agonists have revolutionized type 2 diabetes and obesity treatment, shifting paradigm from symptomatic glycemic control to weight loss-driven metabolic improvement, with >$20B annual market value by 2026.

Watch for next. Retatrutide NDA filing and approval (late 2026-2027), broader expansion into cardiovascular and renal disease indications, and emergence of next-generation agonists with improved efficacy/tolerability profiles.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jens Holst[211]	Professor Emeritus; Discoverer of GLP-1 hormone University of Copenhagen	Acad.	—
Mads Krogsgaard Thomsen[212]	Chief Scientific Officer Novo Nordisk	Biz	—
Lilly Tirzepatide Development Team[213]	Vice President of Diabetes Development Eli Lilly and Company	Biz	—
Juan Pablo Gonzalez[214]	Chief Medical Officer (implied from retatrutide development) Eli Lilly and Company	Biz	—
Lars Reinhard[215]	R&D leadership; Mounjaro development Eli Lilly and Company	Biz	—
Ania M. Jastreboff[216]	Professor of Medicine; Director, Yale Y-Weight Obesity Research Center; lead investigator, retatrutide TRIUMPH-1 Yale University School of Medicine	Acad.	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Semaglutide (mono GLP-1)[211]	Once-weekly semaglutide (Ozempic / Wegovy) delivered ~15% weight loss in obesity trials and showed ~20% MACE reduction in SELECT — the first obesity drug with hard cardiovascular outcome data.	Ozempic (T2D) · Wegovy (obesity) · Rybelsus (oral)
Tirzepatide (dual GIP/GLP-1)[217]	Lilly's tirzepatide (Mounjaro / Zepbound) hit ~22.5% weight loss at top dose in SURMOUNT-1, leapfrogging semaglutide and triggering the obesity-drug supply crisis of 2023–24.	Mounjaro (T2D) · Zepbound (obesity, OSA)
Retatrutide (triple agonist)[218]	Retatrutide (GIP/GLP-1/glucagon) showed ~24% weight loss in Phase 2 and ~71 lbs / 28% in early Phase 3 — likely setting the new ceiling and adding glucagon-driven energy expenditure to the mechanism.	Lilly retatrutide TRIUMPH program
Oral GLP-1s[219]	Oral semaglutide and Lilly's orforglipron are removing the injection barrier and could expand the market by an order of magnitude — manufacturable as small molecules without peptide-supply bottlenecks.	Rybelsus (oral semaglutide) · Lilly orforglipron · Pfizer danuglipron (discontinued)
Indication expansion[211]	GLP-1s now have approvals or strong RCT data in cardiovascular disease (SELECT), kidney disease (FLOW), MASH/MASLD, obstructive sleep apnea (SURMOUNT-OSA), and emerging signals in Alzheimer's and addiction.	SELECT (CV) · FLOW (CKD) · SURMOUNT-OSA · ESSENCE (MASH)
Amylin analogs as next-wave obesity therapeutics[220]	Amylin-receptor agonists (e.g. petrelintide) restore leptin sensitivity to induce satiety with markedly better GI tolerability than incretins, enabling co-formulation with GLP-1/GIP agonists like enicepatide.	petrelintide (Roche/Zealand) ZUPREME-1 Phase 2 · enicepatide (CT-388) GLP-1/GIP Phase 2 · petrelintide + enicepatide fixed-dose combo Phase 2 (mid-2026)

Sub-idea	Description	Examples
Amylin agonism and muscle-preserving obesity combinations[221]	Beyond GLP-1: selective amylin receptor agonists and combos engineered to preserve lean mass (e.g., GLP-1 + myostatin inhibition) became a dominant ADA 2026 theme, addressing the lean-mass-loss limitation of incretins.	Lilly eloralintide (selective amylin agonist) + tirzepatide · Novo CagriSema and amycretin/zenagamtide · Roche petrelintide amylin analog · semaglutide + bimagrumab (myostatin) lowering lean-mass loss fraction
Structure Therapeutics	Oral small-molecule GLP-1 receptor agonist (aleniglipron/GSBR-1290) for obesity	
Retatrutide biologic-vs-drug exclusivity dispute[222]	Whether retatrutide (41-aa triple agonist) is regulated as a biologic (12-yr exclusivity) or a drug (5-yr); courts returned the 'analogous to a protein' question to FDA as Lilly's filing nears.	retatrutide
Third-wave oral small-molecule GLP-1s[223]	Non-peptide once-daily GLP-1 agonists from smaller biotechs entering Phase 3 behind Novo and Lilly.	Aleniglipron (Structure Therapeutics), ACCESS Phase 2b, ~-11.3% placebo-adjusted at 36 weeks
Structure's oral amylin/calcitonin agonist ACCG-2671 hits target engagement[224]	Phase 1/2a single-ascending-dose data for ACCG-2671, an oral small-molecule dual amylin/calcitonin receptor agonist, showed up to 3.3% body-weight loss and a ~6-day half-life supporting once-weekly dosing, with no serious adverse events. Backfill: same Sep 8 Structure Therapeutics release; first time logged here.	
Petrelintide ZUPREME-2: up to 9.2% weight loss at 28 wks in T2D[225]	Oct 7, 2026: Zealand/Roche's amylin analog met its primary endpoint (up to 9.2% vs 2% placebo, n=220) but A1C fell only up to 0.65%, underwhelming analysts; a 7,000-patient Ph3a program is underway.	

b) Current research frontier (last 12–18 months)

01. Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026).[226]

02. Retatrutide demonstrated 71.2 lbs average weight loss and substantial osteoarthritis pain relief in Phase 3 obesity trial (announced February 2026), with 7 additional Phase 3 trials expected to complete in 2026.[218]

03. Zepbound (tirzepatide) received additional FDA approval for moderate-to-severe obstructive sleep apnea in obesity (December 2024), marking first medication approved specifically for this indication.[227]

04. Semaglutide (Novo Nordisk) is a GLP-1 agonist with 94% homology to native GLP-1, modified with DPP-4 protease-resistant alanine substitution, used at lower doses for Ozempic/Victoza diabetes and higher doses for Wegovy/Saxenda weight loss.[212]

05. Oral semaglutide approved by FDA as first GLP-1 pill for weight loss, expanding non-injection delivery options while maintaining efficacy of injectable formulations.[219]

06. Triple-agonist retatrutide produces bariatric-surgery-class weight loss (30%+) in a pivotal Phase 3 trial without surgery, redefining the achievable benchmark for obesity pharmacotherapy.[228]

07. TRIUMPH-1 (ADA 2026): retatrutide 12mg — 28.3% (70.3 lb) mean weight loss at 80 wk; 45.3% lost >=30%; n=2,339.[229]

08. Oral ribupatide (Hengrui/Kailera) — up to 12.1% weight loss at 26 wk vs 2.3% placebo; 38.6% on 25mg >=15% loss.[230]

09. [231]

10. [232]

c) Who could be next

Retatrutide clinical program (Eli Lilly) [STARTUP][213] — Eli Lilly and Company · Eli Lilly internal R&D ($3B+ annual pharma R&D)

Triple agonist showing superior efficacy vs. dual agonists; Phase 3 trials ongoing in obesity, type 2 diabetes, osteoarthritis, sleep apnea, MASH; potential NDA filing expected late 2026.

Oral GLP-1 platform development (Novo Nordisk, Eli Lilly, others) [STARTUP][219] — Multiple pharmaceutical companies · Multiple companies with $10B+ combined R&D spending

Oral semaglutide approved; multiple oral GLP-1/GIP agonists in development; expanding accessibility of GLP-1 therapy beyond injection-dependent populations.

Novo Nordisk semaglutide pipeline expansion [STARTUP][212] — Novo Nordisk A/S · Novo Nordisk annual R&D budget ($1B+)

Expanding semaglutide beyond diabetes and obesity into cardiovascular, kidney disease, and MASH indications; LEADER, SUSTAIN, and FLOW trial data expanding label potential.

Roche obesity portfolio (petrelintide + enicepatide) [LAB][233] — Roche · Roche internal R&D (large pharma); petrelintide via Carmot acquisition

Major new entrant validating the amylin pathway; petrelintide Phase 2 data positioned as a core asset alongside enicepatide combinations at ADA 2026.

Structure Therapeutics [STARTUP][234] — Structure Therapeutics (Nasdaq: GPCR) · Public (Nasdaq: GPCR)

Leading oral small-molecule GLP-1 developer (aleniglipron); phase 3 planned Q3 2026 — competitive with oral offerings from Lilly/Novo.

Flags / unverified

■ Updated VK2735 oral 13-week weight loss: new value = 12.2% mean weight loss at 13 weeks (Phase 2, highest dose), 97% >=5%, 80% >=10% — ECO 2026 (May 12, 2026)

■ Updated retatrutide_phase3_topline: new value = TRIUMPH-1 met primary endpoints; 28.3% weight loss at 80 wks on 12 mg, 30.3% at 104 wks in BMI>=35 subgroup; data at ADA 2026 (announced May 21, 2026)

■ Updated Retatrutide (GIP/GLP-1/glucagon) TRIUMPH-1 Phase 3 obesity readout: new value = Up to 28.3% mean weight loss at 80 weeks (12 mg); 65.3% reached BMI <30; matches bariatric outcomes

■ Updated mazdutide (GCG/GLP-1) vs semaglutide head-to-head (DREAMS-3): new value = 48.0% of mazdutide vs 21.0% of semaglutide patients achieved HbA1c<7.0% plus >=10% weight loss; full data presented ADA 2026

■ Updated survodutide SYNCHRONIZE-1 full Phase 3 efficacy: new value = Full data at ADA 2026: 16.6% mean weight loss at 76 weeks vs 3.2% placebo; up to 85.1% achieved >=5% reduction

■ Updated retatrutide Phase 3 TRIUMPH-1 mean weight loss (12mg, 80wk): new value = 28.3% (30.3% at 104 weeks); 45.3% achieved ≥30% loss

■ Updated orforglipron (Foundayo) vs oral semaglutide head-to-head (ACHIEVE-3): new value = 57.1% greater relative A1C reduction, 73.6% greater relative weight loss vs oral sema 14mg

■ Updated berobenatide (monthly GLP-1) Phase 2b weight loss: new value = up to ~15.9% (VESPER-1, non-placebo-adjusted, 32wk); advancing to 10 Phase 3 studies

■ Updated oral small-molecule GLP-1 — phase 2b peer-reviewed: new value = Nature Medicine: up to 11.3% placebo-adjusted weight loss at 36 weeks (120 mg)

■ Updated oral_glp1_head_to_head: new value = orforglipron superior to oral semaglutide on A1C and weight in ACHIEVE-3 (n=1,698; A1C 1.9/2.2% vs 1.1/1.4%; weight 6.7/9.2% vs 3.7/5.3%)

■ Updated CagriSema REIMAGINE 4 topline: new value = Weight loss non-inferior (15.2% vs 15.8%) but HbA1c non-inferiority NOT met (1.9 vs 2.2 pts) vs tirzepatide, reported Aug 5, 2026

■ Updated Retatrutide expanded access: new value = Defined multi-patient expanded/compassionate-access pathway opened Aug 3, 2026

■ Updated current_frontier[0].claim: new value = Q2 2026, first head-to-head oral GLP-1 quarter: oral Wegovy ~$496M vs Foundayo (orforglipron) $98M.

■ Updated tirzepatide_fda_label: new value = FDA-approved for MACE risk reduction in T2D (Aug 28, 2026; SURPASS-CVOT HR 0.92)

■ Updated guideline_status.HFpEF_obesity: new value = ESC 2026 HF guidelines: Class IIa for semaglutide or tirzepatide in HFpEF (LVEF>=45%) with BMI>=30

■ Updated oral_glp1_weekly_us_prescriptions: new value = Wegovy pill ~181K/wk; Foundayo ~44K/wk (week of Sep 4, 2026)

■ Updated aleniglipron weight loss (ACCESS OLE): new value = 16.2% mean reduction at 72 weeks, no plateau

■ Updated retatrutide TRIUMPH-2 weight/A1C reduction (obesity + T2D): new value = up to 20.8% (49.6 lb) weight loss and 1.6-point A1C reduction at 80 weeks

■ Updated retatrutide TRIUMPH-2: new value = -20.8% at 80 wk (12 mg) in T2D+obesity; Lancet; US filing Q1 2027

CRISPR therapies (Casgevy + base/prime editing)

Impact so far. Casgevy approval (December 2023) marked first FDA-approved CRISPR therapy; 23+ clinical trials now active with base/prime editing showing efficacy in rare genetic diseases and potential for broader indications.

Watch for next. Intellia's FDA approval filing 2026, Beam and Prime Medicine accelerated approvals late 2026-2027, expanded in vivo CRISPR programs, and next-generation base/prime editors with improved efficiency and safety profiles.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Feng Zhang[235]	James and Patricia Poitras Professor of Neuroscience; Core Member Broad Institute MIT McGovern Institute; Broad Institute of MIT and Harvard	Acad.	★ National Inventors Hall of Fame inductee 2026
David Liu[236]	Richard Merkin Professor; Director Merkin Institute; HHMI Investigator Harvard University; Broad Institute of MIT and Harvard	Acad.	★ 2025 Breakthrough Prize in Life Sciences ★ 2026 Harvey Prize ★ 2026 Franklin Institute Bower Award
Editas Medicine/CRISPR Therapeutics Leadership[237]	CEO (Casgevy development) CRISPR Therapeutics; Vertex Pharmaceuticals (co-developer)	Biz	—
John Leonard[238]	Chief Executive Officer Intellia Therapeutics	Biz	—
Keith Gottesdiener[239]	Chief Executive Officer Prime Medicine	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Ex vivo CRISPR/Cas9 (Casgevy)[240]	CRISPR Therapeutics + Vertex's Casgevy was the first FDA-approved CRISPR therapy (Dec 2023) — ex vivo edit of HSCs to reactivate fetal hemoglobin in sickle-cell disease and beta-thalassemia.	Casgevy (exa-cel) · CTX001
Base editing[241]	David Liu's base editors (cytosine and adenine) make single-letter DNA changes without double-strand breaks — Beam (BEAM-101 for sickle cell, BEAM-302 for AATD) and Verve (PCSK9 for cardiovascular) are the lead programs.	Beam BEAM-101 / -301 / -302 · Verve VERVE-102 · Tessera Gene Writers
Prime editing[242]	Liu lab's prime editing, the 'search-and-replace' of genome editing, achieved its first-ever human evidence in 2025 with PM359 in chronic granulomatous disease, on an FDA accelerated-approval path.	Prime Medicine PM359 (CGD) · PM577 · Twin-prime editing
In vivo CRISPR via LNP[238]	Intellia's NTLA-2002 (lonvoguran ziclumeran) for hereditary angioedema cut attacks ~87% in Phase 3 — the first Phase-3-positive in-body CRISPR therapy and a template for the rest of the genome.	Intellia NTLA-2002 · Intellia NTLA-2001 (ATTR) · Verve VERVE-102 (PCSK9)
Epigenome editing[243]	dCas9 fused to silencing or activating domains can switch genes off or on without cutting DNA — Tune Therapeutics, Chroma Medicine, and Epic Bio are running the first programs (Tune's TUNE-401 for HBV in trials).	Tune Therapeutics TUNE-401 · Chroma Medicine · Epic Bio
In vivo prime editing reaches the clinic[244]	Prime Medicine's PM577a, an LNP-formulated prime editor delivered IV to hepatocytes, is the first in vivo prime editing therapy authorized for human trials, correcting the ATP7B H1069Q mutation in Wilson Disease.	PM577a (ATP7B H1069Q, Wilson Disease) · PM577 follow-on targeting R778L allele (preclinical)

Sub-idea	Description	Examples
Caribou CB-011 allogeneic CRISPR CAR-T[245]	CRISPR-edited allogeneic anti-BCMA CAR-T immune-cloaked via B2M-knockout/HLA-E-fusion; 92% ORR, 83% CR/sCR, 91% MRD-negativity in a 12-patient r/r multiple myeloma cohort, 50% still in CR at 15 months (CaMMouflage phase 1, EHA 2026).	
AstraZeneca/Neogene NT-175 CRISPR TCR-T (solid tumors)[246]	CRISPR-edited (TCR + TGFβR2 knockout) autologous T-cell therapy targeting the shared TP53 R175H neoantigen; acceptable safety and early anti-tumor efficacy signals in a phase 1 across pancreatic, colorectal and breast tumors (ASCO 2026, Abstract 2506).	
HuidaGene HG302 DMD trial fatality disclosure[247]	Delayed disclosure of a 2025 death in an investigator-led CRISPR/DMD trial (AAV-triggered ARDS/cytokine), intensifying transparency/oversight scrutiny of China's investigator-initiated gene-editing studies.	HuidaGene HG302
Intellia nex-z HLA pharmacogenomic signal[248]	HLA-allele carriers showed the highest liver enzyme elevations from in vivo CRISPR ATTR therapy nex-z; a pharmacogenomic safety signal shared with FDA as Phase 3 resumes.	nex-z (NTLA-2001)
Post-Cas9 proprietary editing platforms[249]	In-house editor architectures distinct from CRISPR/Cas9 reaching first-in-human trials.	CTX460 (SyNTase platform), Phase 1 for AATD
CD19 CAR-T reverses refractory rheumatoid arthritis in Nature Medicine[250]	Charite Berlin's phase 1 COMPARE trial treated 6 RA patients refractory to up to 8 prior therapies; all six showed markedly reduced disease activity and three were in sustained drug-free remission at up to one year.	
BEAM-302 base editing durably raises AAT above threshold in 29 patients[251]	At ERS Congress 2026, Beam Therapeutics reported a single dose of BEAM-302 (in vivo adenine base editor correcting the SERPINA1 mutation) durably raised functional alpha-1 antitrypsin above the protective threshold across 29 dosed patients; mild-to-moderate infusion reactions in 41%. Backfill: ERS late-breaker (Sep 5-9, press release Sep 8) that drew continued analyst and specialist-press pickup through the Sep 11-18 window; first time logged here.	
FDA clears trial of dual CD20/BCMA CAR-T C-CAR168 for progressive MS[252]	The FDA authorized Abelzeta Pharma's Phase 1b/2 trial of C-CAR168, an autologous CAR-T targeting both CD20 and BCMA, in treatment-refractory secondary/primary progressive MS; prior data in 2 SPMS patients showed reduced MRI lesion burden and improved disability scores.	
In vivo lentiviral CD19 CAR-T shows early benefit in 16 neuro-autoimmune patients[253]	NEJM report on JY231, an off-the-shelf lentiviral vector generating CD19 CAR-T cells directly in vivo (no manufacturing or lymphodepleting chemo), found complete B-cell depletion and preliminary disability improvement across 16 refractory neurologic autoimmune patients.	
PM647[254]	In vivo, LNP-delivered prime editor correcting the SERPINA1 PiZ mutation to restore functional alpha-1 antitrypsin protein, addressing both lung and liver manifestations of AATD.	

b) Current research frontier (last 12–18 months)

01. Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-occlusive crises in clinical trials.[240]

02. Intellia's lonvoguran ziclumeran (in vivo CRISPR therapy) reduced hereditary angioedema attacks by 87% vs. placebo in Phase 3 HAELO trial (announced April 27, 2026); rolling FDA BLA submission initiated.[238]

03. Beam Therapeutics' BEAM-302 (base editing for alpha-1 antitrypsin deficiency) dosed 29 patients with well-tolerated safety profile and no dose-limiting toxicities; FDA RMAT designation granted; optimal 60 mg dose selected for pivotal development.[241]

04. Prime Medicine's PM359 (prime editing ex vivo therapy for chronic granulomatous disease) shows first human safety/efficacy data with FDA 'plausible mechanism' pathway grant; company pursuing accelerated approval with potential 2026 alignment meeting.[242]

05. As of 2026, 23+ clinical trials are active using base editing or prime editing for leukemias, hypercholesterolemia, AATD, sickle cell disease, beta-thalassemia, and chronic granulomatous disease.[255]

06. Intellia's lonvo-z became the first in vivo CRISPR therapy to win a positive pivotal Phase 3 (HAELO), cutting hereditary angioedema attacks 87% vs placebo (p<0.0001) with 62% attack-free over 6 months; NEJM-published, rolling BLA underway for 1H 2027 launch.[256]

07. Autolus' fast-off-rate CD19 CAR-T obe-cel showed 83% (5/6) remission (per standard lupus criteria) at target dose, with several complete/partial renal responses, in severe refractory lupus in the Phase 1 CARLYSLE trial, presented at EULAR 2026.[257]

08. Beam BEAM-304 (in vivo LNP adenine base editor for PKU/PAH) cleared FDA IND June 18 2026; preclinical normalization of plasma Phe.[258]

09. Dual CD19/BCMA CAR-T (Beijing GoBroad Boren) induced sustained remission in 11 refractory systemic-sclerosis patients (EULAR 2026).[259]

10. VERVE-102 (Verve/Lilly) in vivo base editing cut PCSK9 ~88% and LDL-C ~62% (n=35, Heart-2), durable >1yr, no dose-limiting toxicities (NEJM).[260]

11. bioRxiv preprint: in vivo epigenome editing gave the most durable PCSK9 silencing vs nuclease/deaminase editing and siRNA (mRNA editors in Arcturus LUNAR LNPs, mice).[261]

12. At EAACI 2026 (presented Jun 13, 2026) and published in NEJM, Intellia's lonvoguran ziclumeran (lonvo-z), a single-dose LNP-delivered CRISPR therapy inactivating KLKB1 in the liver, cut mean monthly hereditary-angioedema attacks by 87% vs placebo with 62% of patients attack-free over 6 months — the first positive pivotal Phase 3 for an in vivo CRISPR therapy; rolling BLA targeting 1H 2027.[262]

13. Two Nature papers show the nuclease Cas12a2 can be programmed to detect disease-linked RNA (e.g., mutant KRAS/TP53) and trigger DNA-shredding cell death, eliminating tumor cells in mouse models while sparing healthy cells; spotlighted in Nature's July 14, 2026 News & Views.[263]

14. CASTLE basket trial (EULAR 2026): CD19 CAR-T zorpocabtagene autoleucel — 9/10 SLE patients in DORIS remission at 6 mo, drug-free through 24 wk; 1/28 SLE relapse at median 13 mo.[264]

15. EHA 2026: CARsgen allogeneic CAR-T CT0596 (BCMA, 8 pts) and CT1190B (CD19/CD20, 13 lymphoma pts); Phase Ib planned 2026.[265]

16. A Cas9-nickase editor fusing engineered adenine/cytosine deaminases with alkyladenine DNA glycosylases achieved simultaneous triple-base editing at 5.7-41% efficiency at endogenous human sites, with up to 3.2x more allelic diversity than dual A-C editors.[266]

17. FDA accepted Intellia's lonvo-z (HAE) BLA with Priority Review; PDUFA March 10, 2027; Phase 3 HAELO: 87% attack reduction vs placebo.[267]

18. BEAM-302 (in vivo base editing, AATD): 18-mo durability, total AAT >11 µM at 60mg, Z-AAT -84%, M-AAT 93% of circulating protein; pivotal cohort dosing (ERS 2026).[251]

19. Novartis halted eight rap-cel autoimmune CAR-T trials after 3 fatal IEC-HS cases; BMS paused zola-cel enrollment (reported Aug 31-Sep 1, 2026).[268]

c) Who could be next

Prime Medicine (prime editing platform) [STARTUP][269] — Prime Medicine Inc. · Series B: $200M+ (2022); backed by Jeff Bezos and others

First-ever prime editing human data (PM359 for CGD); pursuing accelerated approval pathway 2026; expects clinical data may support approval with FDA alignment meeting ongoing.

Verve Therapeutics (base editing for cardiovascular disease) [STARTUP][270] — Verve Therapeutics Inc. · Public company; venture-backed with multiple funding rounds

VERVE-102 Phase 1b data showed ~50% LDL reduction in highest-dose patients; shifted to improved LNP formulation after VERVE-101 liver enzyme abnormalities; targeting PCSK9 base editing.

Intellia Therapeutics (in vivo CRISPR for rare diseases) [STARTUP][271] — Intellia Therapeutics Inc. · Public company; multiple funding rounds and partnerships

First Phase 3 in vivo CRISPR data (hereditary angioedema); rolling FDA submission 2026; expects U.S. launch H1 2027 if approved; expanding to other rare genetic diseases.

Beam Therapeutics (base editing for genetic disease) [STARTUP][241] — Beam Therapeutics Inc. · Public company; founded by Liu, Barrangou, and others with $300M+ raised

BEAM-302 advancing to pivotal development for AATD with 60 mg optimal dose; BEAM-301 (GSD-Ia) data expected 2026; RMAT designation supporting accelerated pathway.

EDIT-401 (LDLR upregulation, in vivo gene editing) [STARTUP][272] — Editas Medicine · Public (NASDAQ: EDIT)

Lead in vivo program targeting hyperlipidemia/HeFH; >=90% LDL-C lowering in NHPs; FIH expected 2026 with human PoC by year-end; positions Editas as direct competitor to Verve/Lilly in cardiovascular gene editing.

Autolus Therapeutics [STARTUP][257] — Autolus Therapeutics plc · Public (Nasdaq: AUTL)

Commercial-stage CAR-T company (obe-cel/Aucatzyl approved in B-ALL) extending its fast-off-rate CD19 platform into autoimmune disease (lupus), a fast-emerging high-value indication for cell therapy.

Caribou Biosciences [COMPANY][245] — Caribou Biosciences (allogeneic CRISPR CAR-T)

Entity reconciliation: developer of CB-011 (off-the-shelf, immune-cloaked anti-BCMA CAR-T); a lead player in the allogeneic/in-vivo cell-engineering gap theme.

Charite - Universitatsmedizin Berlin [LAB][250] — Charite - Universitatsmedizin Berlin · n/a

Leading academic driver of CAR-T-for-autoimmune-disease trials, generalizing from lupus to rheumatoid arthritis

Abelzeta Pharma [STARTUP][252] — Abelzeta Pharma

Pushing dual-antigen CAR-T into neuro-autoimmune indications, one of the fastest-moving frontiers in cell therapy

SNIPR Biome [BIOTECH][273] — SNIPR Biome · CARB-X supported

Extends CRISPR gene-editing tools beyond human genome correction into precision antimicrobial phage therapy, a distinct application vector for editing platforms; CARB-X backed.

Epicrispr Biotechnologies [BUSINESS][274] — Epicrispr Biotechnologies

First clinical evidence for an AAV-delivered epigenetic silencer (DUX4); could validate editing without DNA cuts.

Kyverna Therapeutics [BUSINESS][275] — Kyverna Therapeutics

Likely first FDA-approved CAR-T for autoimmune disease.

Flags / unverified

■ Updated Casgevy commercial adoption: new value = $43M Q1 2026 revenue; 500+ patients initiated

■ Updated lonvo-z regulatory status: new value = Rolling BLA initiated April 2026; submission completion targeted H2 2026; potential US launch H1 2027 (first-ever in vivo CRISPR product)

■ Updated Casgevy commercialization (Q1 2026): new value = $43.0M Q1 2026 revenue; >500 patients initiated globally

■ Updated VERVE-102 Phase 1b Heart-2 readout: new value = Single-dose in vivo base edit: -62% LDL-C, -88% PCSK9 at top dose; NEJM publication May 25, 2026; Phase 2 H2 2026

■ Updated casgevy_eligible_age: new value = ages 2 and older (July 1, 2026); ~5,500 more US children eligible

■ Updated PM359 (Prime Medicine) regulatory designation: new value = RMAT (Jun 22, 2026)

■ Updated ?: new value = ?

■ Updated Aurora Therapeutics program_status: new value = Abandoned lead PKU gene-editing program and cut staff, Aug 5, 2026, citing competitive pressure

■ Updated impact_so_far: new value = Casgevy Q2 2026 revenue $76M, +151% YoY and +78% QoQ, after FDA pediatric expansion to age 2+.

■ Updated First autoimmune CAR-T BLA: new value = Kyverna miv-cel rolling BLA by end-2026

Quantum error correction (logical qubits below threshold)

Impact so far. Crossed theoretical threshold for scalable quantum computing; demonstrated exponential error suppression on real hardware for the first time, moving from physics milestone to engineering milestone.

Watch for next. Fault-tolerant quantum computers with 10+ logical qubits demonstrating practical speedup (2025-2026); Riverlane's MegaQuOp (1 million error-free operations) target by 2026.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Hartmut Neven[276]	Founder and Lead, Google Quantum AI; VP Engineering Google	Biz	★ TIME's 100 Most Influential People in AI (2025)
John Preskill[277]	Director, Institute for Quantum Information & Matter; Professor of Theoretical Physics California Institute of Technology (Caltech)	Acad.	★ Leading figure in quantum information theory; multiple honors for QEC research
Ilyas Khan[278]	Founder & Chief Product Officer; Vice-Chairman of Board Quantinuum	Biz	★ Founding Chairman, The Stephen Hawking Foundation (2019); Founding Chairman, Topos Institute
Barbara Terhal[279]	Professor of Quantum Information; Research Scientist QuTech & DIAM, Delft University of Technology (TU Delft)	Acad.	★ Leading researcher in quantum error correction theory and surface codes
Steve Brierley[280]	Founder and CEO Riverlane	Biz	★ Built world's first QEC chip (DD1); $75M Series C funding lead (2024)
David Hayes[281]	Senior Quantum Physicist; Logical Qubit Research Lead Quantinuum	Biz	★ Co-author of Microsoft-Quantinuum 800x logical qubit achievement (2024)
John Martinis[282]	Co-founder & CTO Qolab	Cross	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Surface code below threshold[283]	Google Willow (Dec 2024, 105 qubits) was the first hardware to show that increasing the surface-code distance suppresses logical error exponentially — the long-promised threshold theorem realized in silicon.	Google Willow surface-code experiment
Logical qubits with virtualization[284]	Quantinuum + Microsoft demonstrated 4 logical qubits with 800x lower error than physical qubits using ion-trap hardware and run-time mid-circuit measurement, validating a different path to QEC.	Quantinuum H2 / Helios · Microsoft Azure Quantum
Bivariate bicycle / qLDPC codes[285]	IBM's bivariate bicycle codes (Nature 2024) need ~10x fewer physical qubits per logical qubit than the surface code, potentially shrinking the path to fault tolerance from millions of qubits to ~hundreds of thousands.	IBM Quantum (Heron, Loon) · Bivariate bicycle codes
Cat qubits & bosonic codes[286]	Encoding a logical qubit in the photons of a microwave cavity (Alice & Bob, AWS) intrinsically suppresses bit-flip errors, reducing the QEC overhead and offering an alternative to surface codes.	Alice & Bob cat qubits · AWS Ocelot · Yale circuit-QED
Real-time decoding hardware[287]	Riverlane's DD1 and Google's custom decoders move syndrome decoding into dedicated FPGA / ASIC, the missing piece for sustained logical operation — the 'CPU' for the quantum 'GPU'.	Riverlane DD1 · Google QEC decoder · QEC decoder competitions

Sub-idea	Description	Examples
Logical-qubit advantage on real applications (neutral atoms)[288]	First demonstration that logical qubits outperform physical qubits on an end-to-end real-world task (differential equations) on a neutral-atom processor at 99.4% gate fidelity across 1,000 equations.	Pasqal neutral-atom run, 1000 differential equations, logical beats physical by ~50% avg, up to 10x on hard cases
Barbell codes (qLDPC)[289]	A new family of quantum low-density parity-check error-correcting codes designed for IQM's superconducting Constellation processor, reducing logical error rates and physical-qubit overhead versus the surface code.	IQM barbell codes on the Constellation processor
Dual-rail superconducting qubits for fault tolerance[290]	Erasure-biased dual-rail encoding (photon-loss/leakage detected as erasures) in superconducting circuits, now pursued by D-Wave via its Quantum Circuits Inc. acquisition as a lower-overhead route to logical qubits, distinct from transmon surface-code approaches.	D-Wave gate-model roadmap: 17 physical qubits 2026 -> 181 by 2028 -> 10 logical 2030 -> 100 logical 2032 · Built on Quantum Circuits Inc. dual-rail tech (acquired Jan 2026)
Q-PLANET	Pasqal-coordinated EU Chips JU pilot manufacturing line for neutral-atom quantum chip components	
Crypto-Agility Readiness Score[291]	Benchmark of 43 repos vs NIST PQC standards; legacy libs avg 24.9/100	
Mitten codes (qLDPC)[292]	High-rate (~20%) qLDPC code family with fault-tolerant surgery gadgets and a fast decoder (Caltech, arXiv:2607.28795, Jul 30 2026).	mitten codes
Verifiable provable advantage[293]	Advantage demonstrations with a mathematical proof (not conjecture) and efficient classical verification of the output.	Quantinuum H2 complement-sampling, Nat. Commun. Sep 2026
Sandia/Quantinuum/NVIDIA launch QUOPS cross-platform quantum benchmark[294]	The QUOPS framework benchmarks full-system quantum performance: Quantinuum Helios-1 scored Q=1,504 physical and Q=40 on an 8-logical-qubit Steane [[7,1,3]] configuration vs Google Willow's Q=216 and IBM ibm_boston's Q=204; utility-scale problems need Q≈2.5–3.4×10^8.	
Scalable logical qubit definition[295]	Proposed grading standard for logical-qubit claims.	

b) Current research frontier (last 12–18 months)

01. Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first demonstration of exponential error suppression with increasing code size[283]

02. Microsoft-Quantinuum (April 2024): Demonstrated 4 logical qubits with error rates 800x lower than physical qubits; 14,000+ experiments without error using trapped-ion hardware with qubit virtualization[284]

03. IBM (2024): Surface code distance-5 implementation with logical error suppression, decoding in 135 microseconds on laptop; bivariate bicycle codes show 27x speedup at 0.3% circuit-level error rates[285]

04. Quantinuum Helios (2025): 3rd generation ion-trap quantum computer with 99.921% 2-qubit gate fidelity; first complete quantum chemistry simulation using QEC on real hardware[296]

05. Riverlane DD1 QEC Chip (2024): Dedicated quantum error correction ASIC/FPGA achieving logical error rate of 1 in a trillion; published in Nature Electronics with end-to-end hardware validation[287]

06. NVIDIA Ising open AI models accelerate QEC decoding/calibration with sub-microsecond latency, 2.5x speedup and 3x accuracy gains over classical baselines.[297]

07. US federal industrial policy is now explicitly steering quantum hardware build-out at the foundry level — a structural shift from research grants to manufacturing capacity formation, with $2B in CHIPS funding distributed across nine vendors on May 21, 2026.[298]

08. Flatiron Institute CCQ + Boston University demonstrated (Science, May 21 2026) that a tensor-network algorithm, running on a laptop, can classically simulate the same 100-qubit highly-entangled dynamics problem D-Wave used to claim 'beyond-classical' computation with its 5,000-qubit Advantage2 annealer in March 2025 — resetting where the quantum-advantage line sits.[299]

09. IBM disclosed $10B five-year quantum push (May 28, 2026) spanning R&D, manufacturing, partnerships and the Anderon foundry with DoC ($1B CHIPS proposed); targets Starling fault-tolerant machine by 2029.[300]

10. On a trapped-ion QCCD processor, two optimized QEC code constructions (12-qubit Knill-inspired and 16-qubit tesseract colour code) combined with scalable error detection achieve logical error rates 11x to 800x lower than physical-circuit baselines; now published

in Nature (vol 654, pp 349-355, 2026).[301]

11. Quantinuum 98-qubit Helios (Nature, June 17): 99.9975% 1q / 99.921% 2q fidelity, 99.967% SPAM; Sandia certified beyond-classical operation.[302]

12. HPE's hybrid quantum-HPC platform (HPE Discover, June 15-18) integrates Quantinuum + IQM QPUs with Cray supercomputing for enterprise quantum-classical workflows.[303]

13. IQM directional tile codes: up to 1,000x lower logical error rate vs surface code at ~30 physical qubits/logical, native iSWAP gates on existing hardware.[304]

14. University of Sydney + IBM (156-qubit Heron r2): mid-circuit measurement idling noise is a dominant QEC error source; addressing it lifts logical qubit survival from <90% to >96% per cycle (Nature Communications, July 6, 2026).[305]

15. At HPE Discover (June 15-18), HPE unveiled a hybrid classical-quantum platform integrating multiple qubit modalities with Cray supercomputing; Quantinuum announced a quantum-HPC collaboration and IQM (already tracked) contributed superconducting QPU tech live at Oak Ridge — pushing quantum-classical hybrid orchestration toward enterprise deployment.[306]

16. IBM Nighthawk r2 (120 qubits): tunable-coupler reset ~200µs→25ns, ~25x throughput, 11x lower init error; Qiskit v2.5.[307]

17. Quantinuum (Nature, Jul 15 2026): first universal gate set from anyon braiding+fusion on the 54-qubit H2 trapped-ion system (non-Abelian S3 order).[308]

18. 12 logical qubits (Steane code) running QFT on Quantinuum Helios; 98 physical qubits for the approximate variant.[309]

19. A Swap-Wait-Swap CZ gate between dual-rail cavity qubits reached ~99.9% two-qubit fidelity at ~500ns with ~0.5% erasure rate and post-selected residual Pauli errors below 0.1% - a hardware step toward efficient error-corrected gate-model quantum computing.[310]

20. Helix architecture on Helios: logical memory 9.3e-5/round, logical Clifford 2.8e-4 (4.28x below physical), 3-logical-qubit GHZ >99.925%, no post-selection.[311]

21. IBM Nighthawk r2 (120q): 100K+ circuits/sec, accurate observables at 7,500+ two-qubit gates — 2026 roadmap milestone early.[312]

22. [313]

23. [314]

24. Jülich benchmark: Quantinuum Helios-1 stands out on QEC-style mid-circuit measurement — Preprint arXiv 2610.05928 (Jülich Supercomputing Centre; covered Oct 8, 2026) compared 10 Quantinuum/IBM/IQM processors: IBM measurement-based circuits showed ~10x higher effective error, while Helios-1 stayed above random up to 81 (surface) and 91 (color) data qubits; the 30-qubit chain comparison was inconclusive. Not a QEC demonstration; not peer reviewed.[315]

c) Who could be next

PsiQuantum [STARTUP][316] — PsiQuantum (photonic quantum computing) · Significant DARPA backing; private funding round details proprietary

DARPA US2QC program finalist (Feb 2025) pursuing surface codes on photonic qubits; broke ground on largest US quantum computing project at Illinois facility; targeting utility-scale quantum computer by 2033

Atom Computing [STARTUP][317] — Atom Computing (neutral-atom quantum computing) · Series B/C rounds; corporate backing from aerospace partners

Scaling neutral-atom systems for QEC; announced 1,000+ qubit demonstration targets; competing with Quantinuum (trapped ions) and Google (superconducting) on QEC roadmaps

Alice & Bob [STARTUP][286] — Alice & Bob (bosonic quantum computing) · $50M+ Series B; European backing

Developing cat-qubits and bosonic codes as alternative to surface codes; co-hosted Fault Tolerance Summit (Les Houches 2024); raised $50M+ Series B

Anderon [STARTUP][300] — IBM (spinout) + U.S. Department of Commerce · $2B total ($1B IBM cash + IP/assets; $1B proposed CHIPS Act incentive)

First purpose-built US quantum foundry — provides shared, scaled 300mm fabrication for IBM, Quantinuum, Diraq, Microsoft and other modalities, addressing the wafer-supply bottleneck for fault-tolerant systems.

HRL Laboratories [LAB][318] — HRL Laboratories (Boeing/GM joint venture) · M&A by IBM (terms undisclosed)

Silicon-spin-qubit + quantum-sensing expertise folded into IBM's fault-tolerance roadmap (Starling 2029)

Quantum Circuits Inc (QCI) [STARTUP][319] — Quantum Circuits Inc (acquired by D-Wave, Jan 2026) · Acquired by D-Wave for $550M (Jan 2026)

Originated the dual-rail erasure-qubit modality now central to D-Wave's gate-model roadmap - the dependency entity behind this week's Nature result

Infleqtion (x Cisco quantum networking) [STARTUP][320] — Infleqtion / Cisco Quantum Labs · Public company (NYSE: INFQ via SPAC, 2026)

First neutral-atom vendor pairing with a networking incumbent on a full distributed-quantum stack; also strong in atomic clocks/sensing.

IonQ QCE26 research program [LAB][321] — IonQ

Four Best Paper Awards in one year signals maturing hybrid quantum-classical applications across chemistry, logistics and AI fine-tuning

Pasqal FTQC national programme [CROSS][322] — Pasqal FTQC national programme

Civil state funding for neutral-atom fault tolerance in Europe.

Flags / unverified

■ None. Primary sources verified: Google Nature paper (Dec 2024), Microsoft blog (Apr 2024), Quantinuum press releases (2024-2025), IBM Research (2024), Riverlane Nature Electronics publication (2024).

■ Updated Physical-to-logical qubit ratio (memory): new value = 2:1 (per-round error ~1.3e-13)

■ Updated largest_quantum_simulated_protein: new value = 12,635 atoms (trypsin); IBM Heron 156q

■ Updated ibm_cloud_quantum_milestone: new value = 10 years; 156q Heron at ~1.17e-3 two-qubit error

■ Updated IonQ–SkyWater acquisition status: new value = Shareholder-approved May 8, 2026; ~$1.8B; closing expected Q2-Q3 2026

■ Updated PsiQuantum US government support: new value = $100M CHIPS LOI signed May 21, 2026 for fault-tolerant photonic component development

■ Updated QuEra fault-tolerant roadmap / cloud availability: new value = Libra: 256+ logical qubits, 1e-6 logical error rate, ~1M logical ops, on AWS Braket by 2028

■ Updated ?: new value = ?

■ Updated current_frontier[0].claim: new value = Krypton-sputtered tantalum qubit films deposit at 200C instead of >400C while staying CMOS-fab compatible (Cornell, Nature Materials, Aug 2026).

Reusable rockets (Falcon 9 + Starship catch)

Impact so far. Demonstrated booster reusability at scale (Falcon 9) and tower-catch recovery capability for heavy-lift (Starship/Mechazilla), reducing per-flight costs by 50%+ and enabling rapid cadence launch operations.

Watch for next. Starship Ship catch on tower (targeted 2025-2026); suborbital relaunch within 24 hours; Blue Origin New Glenn cadence increases; Stoke Space and Rocket Lab first operational flights.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Elon Musk[323]	Founder, CEO, and Chief Engineer SpaceX	Biz	★ TIME Person of the Year (2023); multiple recognitions for SpaceX achievements
Gwynne Shotwell[324]	President and Chief Operating Officer SpaceX	Biz	★ National Academy of Engineering member; recognized for commercial spaceflight advancement
Tim Dodd[325]	Lead Starship Systems Engineer (public spokesperson for Starship updates) SpaceX	Biz	★ NASA commendations for rapid iteration and testing protocols
Lars Blackmore[326]	Director of Entry, Descent, and Landing (EDL); Falcon 9 Booster Landing Algorithm Lead SpaceX	Biz	★ JPL/NASA recognition for guidance and control algorithms
Josh Boehm[327]	Director of Launch & Recovery Operations; Mechazilla Tower Systems Lead SpaceX	Biz	★ Internal recognition for tower catch system design and execution
Mikael Dolsten[328]	Board director, Varda Space Industries (former Pfizer Chief Scientific Officer) Varda Space Industries	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Falcon 9 booster reuse[329]	SpaceX has flown more than 380 booster reflights with one booster reaching 22+ flights — turning rockets from single-use ammunition into commercial aircraft and pulling launch cost below $2,000/kg.	Falcon 9 Block 5 · B1062 (22+ flights)
Mechazilla booster catch (Starship)[327]	Starship IFT-5 (Oct 2024) caught the Super Heavy booster mid-air on the launch tower's chopstick arms — replacing landing legs with the tower itself, the architectural choice that makes 24-hour reflight plausible.	Starship IFT-5 / IFT-7 catch · Mechazilla launch tower
Fully reusable upper stage[330]	The much harder problem — bringing the upper stage back through orbital reentry and reusing it — is what Starship's Ship is designed for, and what Stoke Space's Nova attempts with active heat-shield cooling.	SpaceX Starship Ship · Stoke Space Nova · Rocket Lab Neutron
Fairing recovery[331]	Falcon 9 fairing halves are now recovered and reflown on essentially every mission (~100% rate over 300+ flights), removing what used to be a $6M throwaway part per launch.	Falcon 9 fairing recovery (Doug / Mr. Steven)
Heavy-lift competition opens up[332]	Blue Origin's New Glenn (first booster reuse Apr 2026), Rocket Lab Neutron (debut targeted 2026), and Stoke Nova mean Falcon 9 / Starship will not be the only reusable launch options for the first time.	Blue Origin New Glenn · Rocket Lab Neutron · Stoke Space Nova
Rocket Lab Neutron AVac engine	Second-stage engine full-duration qualification hot fire ahead of Neutron debut	

Sub-idea	Description	Examples
LOXSAT	NASA Tipping Point-funded, Eta Space-built cryogenic LOX fluid-management demo on a Rocket Lab bus	
Starship Flight 13[333]	First Starlink V3 deploy attempt (20 sats), in-space Raptor relight, both-stage recovery; July 16 pad abort, July 23 weather scrub, 2nd attempt July 24	
Stoke Space raises $1B Series E, unveils 15-ton Nova Block 2[334]	Stoke closed a $1B Series E (total raised $2.3B) to fund Nova Pathfinder's first flight, targeted early 2027, and develop the larger fully-reusable Nova Block 2, which delivers 15 metric tons to LEO in a 5m fairing. Backfill: Sep 8 announcement with sustained coverage through the window.	

b) Current research frontier (last 12–18 months)

01. Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 weeks between flights[329]

02. Starship IFT-5 (October 13, 2024): Historic first catch of Super Heavy booster with Mechazilla 'chopstick' arms at launch tower; booster rested safely on tower arms ~7 minutes after liftoff on first catch attempt[327]

03. Starship IFT-7 (January 16, 2025): Second successful Super Heavy booster catch achieved; booster 14 (B14) caught after 8+ minutes of flight; demonstrates repeatability of tower catch system[335]

04. SpaceX Launch Rate Dominance (2026): 55 launches (54 Falcon 9, 1 Falcon Heavy) through May 6, 2026; targeting 140-145 Falcon 9 launches by year-end; industry leading reusability infrastructure[336]

05. Falcon 9 Pricing & Reusability Economics (May 2026): Advertised launch cost $74 million; enables cost-leadership vs. competitors; 100% fairing recovery success on 307+ missions through February 2025[331]

06. Starship V3 to debut mid-May 2026 with ~3x prior payload capacity (100+ MT LEO); Pad 2 first flight at Starbase.[337]

07. NASA CLPS Moon Base awards: Astrobotic $297.9M, Firefly $144.2M, Intuitive Machines $148.3M for four lunar deliveries (~2028).[338]

08. SpaceX plans consumer Starlink mobile service (direct-to-device + possible terrestrial network), leveraging ~$17B EchoStar spectrum (disclosed at IPO roadshow, Jun 26).[339]

09. US Space Force SB-AMTI awarded Rocket Lab $397M to build/launch/operate a Neutron-launched tracking-satellite constellation — Neutron's first national-security customer, pre-2028 deadline.[340]

10. Starship Flight 14 FCC filing: first orbital-insertion profile, ~20 Starlink V3 deploy, ship splashdown; FAA approval pending as of Sep 2, 2026.[341]

c) Who could be next

Blue Origin New Glenn [STARTUP][332] — Blue Origin (heavy-lift reusable rocket) · Internal Blue Origin funding; Jeff Bezos backed

Successfully reused booster for first time (April 19, 2026; second and third New Glenn flights); competing with Starship for heavy-lift market; achieving operational reusability milestone

Rocket Lab Neutron [STARTUP][342] — Rocket Lab (partially reusable medium-lift rocket) · Public company (NASDAQ: RKLB); $500M+ operational runway

Targeting Q4 2026 debut after propellant tank failure setbacks; pursuing booster reusability for medium-lift class; complementary to Starship/New Glenn in launch market segmentation

Stoke Space Nova [STARTUP][330] — Stoke Space (fully reusable medium-lift rocket) · Venture-backed; Series C through development phase

100% reusable architecture; full-flow staged-combustion engines (Zenith, Andromeda); targeting early 2026 first flight; heat-shield reentry recovery approach distinct from SpaceX/Blue Origin

Rocket Lab Neutron [STARTUP][343] — Rocket Lab USA (RKLB) · Public (RKLB); $2.2B+ backlog

Most credible Falcon 9 alternative for medium-lift reusable launch (~13t reusable / 15t expendable, 9 Archimedes engines); growing backlog signals real demand pull, not just announcements.

EarthDaily Analytics [COMPANY][344] — EarthDaily Constellation

EDC-08 (launched July 7 on SpaceX Transporter-17) completes 'Launch III,' bringing its 16-imager, 22-spectral-band constellation to the satellite count needed for daily global calibrated Earth monitoring; commercial ops expected later 2026.

Eta Space [STARTUP][345] — Eta Space · NASA Tipping Point-funded

Cryogenic fluid-management / orbital propellant-depot startup; NASA-funded LOXSAT demo toward a planned 2030 Cryo-Dock depot.

Inversion Space [STARTUP][346] — Inversion Space · $350K NASA study contract (on top of a $71M SpaceWERX/Space Force reentry-demo contract)

Fills the reentry/hypersonics-as-a-service gap sub-theme: Inversion is racing Varda and Outpost to commercialize point-to-point cargo reentry, and this NASA study extends its Arc vehicle toward aerocapture for interplanetary use.

Blue Origin [BUSINESS][347] — Blue Origin

Capital and IPO signals for the main Starship competitor.

Flags / unverified

■ None. All data from official SpaceX statements, Blue Origin announcements, and third-party space tracking (Wikipedia/Spaceflight Now) verified through May 8, 2026.

■ Updated Falcon 9 cadence checkpoint: new value = 45 payloads deployed in single rideshare on May 2-3, 2026

■ Updated starship_v3_first_flight_status: new value = Flight 12 (first V3) scrubbed May 21, 2026 — hydraulic pin tower-arm issue; retargeted May 22, 2026 from Starbase Pad 2

■ Updated Starship V3 first flight: new value = Starship V3 + Raptor 3 + Pad 2 debut succeeded May 22, 2026; 20 mock Starlinks deployed, Ship reached Indian Ocean target

■ Updated Blue Origin New Glenn status: new value = Vehicle lost on pad during May 28, 2026 static fire; planned June 4 Kuiper launch delayed

■ Updated SpaceX total booster recoveries / single-drone-ship landings: new value = 619th total recovery; OCISLY reaches 200 catches (June 3, 2026)

■ Updated Blue Origin New Glenn return-to-flight target: new value = Before end of 2026 (per CEO Dave Limp, June 2026)

■ Updated Falcon 9 single-booster reuse record: new value = 35 flights (booster B1067), achieved 2026-06-07/08

■ Updated SpaceX corporate/financial status: new value = Public company (Nasdaq: SPCX) as of 2026-06-12; IPO priced $135/share, closed ~$161 (+19%), ~$2.1T valuation at first close (largest IPO ever)

■ Updated ?: new value = ?

■ Updated New Glenn return-to-flight: new value = Root cause = BE-4 main oxygen valve; retrofit ready end-August; RTF still targeted before year-end 2026

■ Updated Starship heat-shield status (company claim): new value = Musk (Aug 4, 2026) says tile problem 'solved' per Flight 13 data — company claim, not independently verified

■ Updated current_frontier[0].claim: new value = SpaceX set a 38-minute dual-coast Falcon 9 turnaround record on Aug 15, 2026 (prior record 65 minutes, 2024).

■ Updated starship_flight_14_status: new value = Ship 41 six-engine static fire complete Aug 21; Flight 14 NET Sept 15

■ Updated Starship Flight 14 target launch date: new value = NET September 22, 2026

■ Updated Starship Flight 14 target launch date / mission profile: new value = NET Sep 28, 2026 — wet dress rehearsal complete Sep 24; first orbital-insertion attempt with 26 V3 Starlink sats deployed

■ Updated Starship Flight 14 status: new value = Flew 2026-09-28: first orbit, 26 Starlink V3 deployed, ship splashdown intact; mission shortened; booster Gulf splashdown

■ Updated Neutron first launch: new value = 2027

■ Updated Starship Flight 15 plan: new value = Ship tower-catch attempt, ~late Nov 2026

■ Updated B1088 flights: new value = 20

■ Updated New Glenn return to flight: new value = Dec 2026 target

Renewables + battery cost collapse

Impact so far. Battery costs collapsed 80% since 2010 ($1,200 to $108/kWh); solar & wind LCOE below fossil gas; renewable capacity additions accelerating; grid storage emerging as competitive energy arbitrage asset class.

Watch for next. Sodium-ion parity with LFP cost (2026-2027); Form Energy iron-air deployment at utility scale (2025-2026); Tesla Megablock (20 MWh modular unit) launch late 2026; long-duration (12-100hr) storage cost-competitiveness crossing threshold.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jay Whitacre[348]	CEO, CTO, and Founder; Trustee Professor in Energy Stratus Materials (and Carnegie Mellon University)	Cross	★ Founder of Aquion Energy (2008); recognized for cathode innovations in battery chemistry
Tejas Patel[349]	Co-Founder and CEO Form Energy	Biz	★ MIT Technology Review Climate Tech Company to Watch (2023-2024)
Varun Sridhar[350]	Executive Lead, CATL Sodium-Ion Battery Division CATL (Contemporary Amperex Technology Co., Limited)	Biz	★ CATL leads global battery market; Chief Scientist announcement on Naxtra mass production (2026)
Wang Xiangming[351]	General Manager, BYD Battery Division BYD (Build Your Dreams)	Biz	★ BYD world leader in EV and battery production; global EV market leadership
Andrew Zito[352]	Executive Lead, Energy Business; Tesla Megapack Division Director Tesla Energy (Tesla, Inc.)	Biz	★ Leadership of fastest-growing grid battery deployment company; Q1 2026 record deployments
Albert Haché[353]	Chief Commercial Officer; Executive Leadership BloombergNEF (Bloomberg Finance L.P.)	Biz	★ Director of most-cited renewable cost benchmarking reports globally

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Solar PV cost collapse[354]	Utility-scale solar LCOE fell to $38–78/MWh ($20–45/MWh with PTC), making solar the cheapest source of new electricity in most of the world for 10+ years running per Lazard's annual benchmark.	Lazard LCOE+ · BloombergNEF Solar Module Index
Lithium-ion battery price below $100/kWh[353]	BloombergNEF's annual price survey logged $108/kWh for packs and $99/kWh for BEV packs in 2024 — the threshold at which a new EV reaches sticker parity with a comparable ICE vehicle without subsidies.	BNEF battery price survey
LFP chemistry takeover[351]	Lithium iron phosphate (LFP) has displaced NMC for most stationary storage and a growing share of EVs — cheaper, no cobalt, longer cycle life, the dominant chemistry in Tesla Megapacks and BYD vehicles.	CATL LFP · BYD Blade Battery · Tesla Megapack (LFP)
Sodium-ion mass production[350]	CATL Naxtra (~$40/kWh cell, 2026 mass production) and BYD's 30 GWh sodium-ion line are commercializing a chemistry that needs no lithium, no cobalt, no nickel — important for grid storage and emerging-market EVs.	CATL Naxtra · BYD sodium-ion · HiNa Battery
Long-duration storage (iron-air)[355]	Form Energy's iron-air batteries (~$250/kWh, 100-hour duration) target the multi-day grid-storage gap that lithium-ion is too expensive to fill — first utility deployments with Xcel and Georgia Power are live.	Form Energy iron-air · EnerVenue metal-hydrogen · ESS Inc iron flow
High-energy-density sodium-ion at GWh scale[356]	Sodium-ion cells engineered for 261 Wh/kg (high-energy), 162 Wh/kg with -50C discharge (power), and 180Ah/>20,000 cycle storage variant — closing the gap to LFP while using a non-lithium supply chain.	Gotion Gnascent (261 Wh/kg) · CATL Naxtra · BYD sodium-ion · HiNa Battery

Sub-idea	Description	Examples
ESS Bridge sodium-ion BESS[357]	ESS Inc.'s first non-iron-flow product: a Bridge BESS using Alsym Energy sodium-ion cells (1-16+ hour duration) for data-center/commercial storage; ~$1B in demand signals since the April 2026 Alsym partnership (report July 8, 2026).	
Eos Energy Z3	Non-flammable aqueous zinc-chemistry (Znyth) long-duration battery, ~91% domestic content, made in Pittsburgh	
Iron-air / metal-air multi-day storage[358]	Iron-air 'rust batteries' target 24-100h multi-day discharge at costs lithium-ion cannot match; Ore Energy just signed continental Europe's largest iron-air offtake (up to 1 GWh, 400-MWh phase 1) with utility Budget Thuis for 2028 delivery.	Ore Energy · Form Energy · Budget Thuis (offtake)
CATL opens direct-to-buyer sales of 587Ah LFP storage cells at ~$65/kWh[359]	CATL began listing its largest 587Ah LFP energy-storage cell (1.88kWh, 8,000-cycle life at 70% SoH) for direct online sale at 435 CNY (~$65)/kWh, bypassing distributors.	
ZrO2 nano-scaffold interfacial perovskite-silicon tandem cell[360]	New interfacial engineering approach (discrete monoclinic ZrO2 nanoparticle layer between TCO and SAM) that reduces recombination/leakage in perovskite-silicon tandems, setting a new efficiency and voltage record just weeks after HZB's 30.3% CsCl seed-layer result.	

b) Current research frontier (last 12–18 months)

01. BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of $108/kWh; stationary storage at $70/kWh (45% drop YoY); BEV packs at $99/kWh (2nd year below $100/kWh threshold)[353]

02. Solar & Wind LCOE Competitiveness (Lazard 2025): Utility-scale solar $38-78/MWh; onshore wind $37-86/MWh; with PTCs solar drops to $20-45/MWh; wind and solar least-expensive generation 10th consecutive year[354]

03. CATL Sodium-Ion Battery (April 2026): Naxtra technology at $40/kWh cell level (20% cheaper than LFP); mass production targeting 2026; operating range -40°C to 70°C; 10,000 cycle life with 90% retention at -40°C[350]

04. Form Energy Iron-Air Battery (2024-2025): Commercial-scale factory operational in West Virginia; utility deals with Xcel Energy, Southern Company, Georgia Power; target ~$250/kWh energy (reservoir); 100-hour duration; first deployments Q4 2025[355]

05. Tesla Megapack Deployment Surge (Q1 2026): Record 10.4 GWh deployed in Q1 2026 (156% YoY growth); full-year 2025 deployments 31.4 GWh (2.1x 2023); Shanghai Megafactory producing 1 GWh+ annually by Q2 2025[361]

06. JinkoSolar: 34.82% certified perovskite-silicon tandem cell (SIMIT-certified), up from 34.76%, near LONGi's 34.85% record.[362]

07. HZB/Humboldt ESTI-confirmed 25.5% perovskite-CIGS tandem cell (1 cm²), new world record.[363]

08. Fervo-Google 396 MW EGS PPA (Cape Station Phase 2, ~2028) with option for ~600 MW more by 2030 — largest next-gen geothermal PPA.[364]

09. MN8/Eos/Google Mammoth Solar: 86 MW PV + 10 MW/100 MWh Eos Z3 zinc LDES + 70 MW/280 MWh Li-ion on a reclaimed WV coal mine, PJM 2028-2030.[365]

10. SHARE VPP (PG&E + Rewiring America, Google-funded): ~21,000 enrolled home devices, grid support from fall 2026 through a 2027 proof-of-concept.[366]

c) Who could be next

BYD Sodium-Ion [STARTUP][351] — BYD (battery and EV manufacturer) · Internal BYD funding; $37B+ market cap; profitable EV/battery leader

30 GWh sodium-ion factory breakground (Jan 2024); Q3 2025 delivery timeline; targeting <70% of LFP cost long-term; competing with CATL for sodium-ion market leadership

Northvolt (now Lyten acquisition) [STARTUP][367] — Northvolt (Swedish battery manufacturer) / Lyten (US battery company acquirer) · Lyten-backed restructuring; previously $5B+ valuation

Bankruptcy restructuring (Nov 2024, Mar 2025); Lyten acquisition of Northvolt assets including 16 GWh capacity; Northvolt Ett resuming production Q4 2025; European manufacturing recovery watch

Envision AESC / HiNa Battery Technology [STARTUP][368] — Envision AESC (joint venture); HiNa Battery Tech (Chinese sodium-ion leader) · Corporate backing (Envision Group, HiNa venture funding)

Sodium-ion alternatives gaining traction; HiNa targeting mass production 2025-2026; diversifying battery chemistry supply chains away from LFP dominance

Ford Energy [STARTUP][369] — Ford Motor Company · Internal Ford-funded subsidiary, leveraging existing Glendale KY plant capex

US-assembled LFP BESS at 20 GWh/year scale targeting utilities, data centers, and industrial customers; repurposes the Kentucky EV-battery plant from the dissolved BlueOval SK JV; Lisa Drake named president.

QuantumScape [STARTUP][370] — QuantumScape Corporation · Public (NYSE: QS); backed by Volkswagen

Leading solid-state lithium-metal battery developer; anodeless ceramic-separator cells promise higher energy density and faster charging. Two major OEMs (VW/PowerCo, Honda) now validate its platform toward commercialization.

JinkoSolar [PUBLIC_COMPANY][362] — JinkoSolar Holding · Public (NYSE: JKS)

One of the world's largest solar manufacturers and a perovskite-silicon tandem record contender (34.82% certified, June 2026); key player pushing tandem cells past the single-junction limit toward mass production.

Plug Power / Orica Hunter Valley Hydrogen Hub [COMPANY][371] — Plug Power (electrolyzers); Orica (offtake)

50MW GenEco PEM electrolyzer order for Orica's Hunter Valley Hydrogen Hub (NSW), which reached final investment decision July 7, 2026 as Australia's largest renewable-hydrogen project (~4,700 tonnes/yr).

ComEd Scheduled Dispatch VPP [COMPANY][372] — Commonwealth Edison (ComEd)

Illinois' first virtual power plant, approved by the ICC June 30, 2026 under the Clean and Reliable Grid Affordability Act; aggregates customer battery storage for grid dispatch starting 2027.

Alsym Energy [STARTUP][357] — Alsym Energy (non-lithium/sodium-ion cells)

Entity reconciliation: Alsym's sodium-ion cells are the dependency powering ESS's new Bridge BESS; a notable non-lithium chemistry supplier targeting data-center/commercial storage.

Eos Energy Enterprises [PUBLIC_COMPANY][373] — Eos Energy Enterprises, Inc. · Nasdaq: EOSE

US aqueous zinc-chemistry long-duration battery maker (~91% domestic content); won a Golden Dome prototype contract.

Ore Energy [STARTUP][358] — Ore Energy (Delft, NL) · Venture-backed (Delft spinout)

Iron-air 'rust battery' startup targeting 24-100h multi-day discharge; signed continental Europe's largest iron-air offtake (up to 1 GWh) with utility Budget Thuis, a notable entrant in the long-duration storage gap.

Steady Energy [STARTUP][374] — Steady Energy (VTT spin-off, Finland) · Up to €40M EIB convertible loan (Sep 2026)

First EIB financing for SMR technology; heat-only design targets district heating, a segment distinct from electricity-focused SMRs

Flags / unverified

■ Northvolt status in flux (bankruptcy restructuring as of May 2026); actual Form Energy commercial deployment timing contingent on Q4 2025 utility project completion. All other figures from BloombergNEF, Lazard, Tesla official reports, and industry tracking verified.

■ Updated Firm 24/7 solar+storage LCOE (best regions): new value = $54-82/MWh

■ Updated CATL sodium-ion capacity: new value = 40 GWh/year sodium-ion expansion announced May 9, 2026 ($735M)

■ Updated Global energy storage 2025 installs (BNEF): new value = 112 GW / 307 GWh installed in 2025; 158 GW / 459 GWh forecast for 2026

■ Updated CATL sodium-ion (Naxtra) production status: new value = Manufacturing bottlenecks solved; mass production confirmed for 2026 (May 30, 2026)

■ Updated QuantumScape OEM partners: new value = Honda R&D added as second major OEM (multi-year joint development on solid-state lithium-metal cells & manufacturing), 2026-06-18

■ Updated standalone battery storage cost trend (LCOS): new value = rising in 2026, reversing prior-year decline, on Li-ion import tariffs/FEOC rules

■ Updated Si-perovskite tandem cell efficiency (LONGi, ESTI-certified): new value = 35.5% (Jul 14 2026)

■ Updated ?: new value = ?

■ Updated Samsung SDI US LFP production: new value = Mass-production validation underway; cell output begins October 2026, SBB 2.0 deliveries by year-end

■ Updated current_frontier[0].claim: new value = Form Energy raised a $750M Series G (T. Rowe Price) on Aug 12, 2026; iron-air backlog now ~80 GWh, up from ~20 GWh, incl. Google, Xcel, Crusoe, FuturEnergy Ireland.

■ Updated co_evaporated_tandem_efficiency: new value = 30.3%

Self-driving cars / robotaxis

Impact so far. Robotaxis crossed from pilot to scaled commercial reality between 2023–2025 — Waymo passing 4M paid driverless trips with peer-reviewed safety data, Tesla launching FSD-based robotaxis, and Aurora running fully driverless freight — finally validating a 15-year capital cycle.

Watch for next. Whether end-to-end neural-net autonomy (Tesla, Wayve) catches up with Waymo's HD-map / LiDAR stack on safety; expansion of Waymo One beyond 5–10 cities; Aurora's commercial trucking ramp; and whether Chinese AV scale (Apollo Go, Pony) fragments the global market.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Tekedra Mawakana[375]	Co-CEO Waymo (Alphabet)	Biz	—
Dmitri Dolgov[375]	Co-CEO and CTO Waymo (Alphabet)	Biz	—
Ashok Elluswamy[376]	Director of Autopilot / FSD Tesla	Biz	—
Sebastian Thrun[377]	Founder; Stanford Professor (emeritus from CS faculty) Kitty Hawk / Stanford University; founded Google self-driving project (2009)	Cross	★ DARPA Grand Challenge winner 2005
Chris Urmson[378]	Co-founder and CEO Aurora Innovation; previously CTO of Google Self-Driving Car (now Waymo)	Cross	—
Raquel Urtasun[379]	Founder and CEO; Professor Waabi; University of Toronto	Cross	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Waymo's mapped + LiDAR robotaxi stack[380]	HD maps + multi-sensor (LiDAR/radar/camera) + rules-and-learning hybrid is the only stack with peer-reviewed safety data at commercial scale and with paid driverless rides in 5+ US cities.	Waymo One (Phoenix, SF, LA, Austin, Atlanta) · Waymo Driver
End-to-end neural-net driving[381]	Replace hand-coded planners with a single learned model trained on video — Tesla FSD v12+, Wayve LINGO/GAIA, Comma.ai. Same scaling-laws logic as LLMs, applied to driving.	Tesla FSD v12+ · Wayve LINGO-2 / GAIA-2 · Comma.ai openpilot
Autonomous trucking[382]	Long-haul freight on highways is structurally easier than urban robotaxis; Aurora launched commercial driverless freight (Dallas–Houston) in May 2025, with Kodiak and Waabi following — a near-term economic prize.	Aurora Driver · Kodiak Robotics · Waabi Driver
Generative-AI simulation[383]	Neural-rendering simulators (Waabi World, NVIDIA Cosmos, Wayve GAIA-1) generate synthetic edge cases at scale — closing the rare-event data gap that pure on-road miles can't fill.	Waabi World · NVIDIA Cosmos · Wayve GAIA-1 · Applied Intuition
Chinese robotaxi at scale[384]	Baidu Apollo Go, Pony.ai, and WeRide run commercial driverless services across multiple Chinese cities; Apollo Go has reported 9M+ cumulative rides — a parallel deployment story to the US Waymo arc.	Baidu Apollo Go · Pony.ai · WeRide
NHTSA AV first-responder directive[385]	NHTSA's July 8, 2026 directive to AV developers labeled repeated interference with emergency scenes a 'functional insufficiency'; the agency will convene meetings with AV makers by July 31, 2026 to hear proposed solutions.	Waymo

Sub-idea	Description	Examples
SF AV regulation push[386]	Mayor Lurie push for tougher robotaxi rules after incidents	
Airport robotaxi operations[387]	Driverless curbside pickup/drop-off at major airports — the highest-value, most regulated robotaxi use case; Zoox at Las Vegas Harry Reid is an early US instance.	Zoox at Harry Reid International, Sep 2026
Waymo, Nihon Kotsu, GO to launch Tokyo robotaxi service in 2027[388]	Waymo announced on Sep 14, 2026 a partnership with Japanese taxi firms Nihon Kotsu and GO to open Japan's first fully autonomous commercial taxi service in Tokyo in 2027, scaling to roughly 100 vehicles pending regulatory approval.	

b) Current research frontier (last 12–18 months)

01. Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing 4 million paid driverless trips by early 2025 and on track for 10M+ in 2026.[380]

02. Tesla launched its initial robotaxi service in Austin in June 2025 using FSD-only / vision-only AI inference (no LiDAR / HD maps), the first major commercial alternative to Waymo's stack.[389]

03. Waymo's safety data (Aug 2024 paper, 7M+ rider-only miles) reported 81% fewer airbag-deploying crashes and 78% fewer injury-causing crashes vs. a human-driven benchmark, the first peer-reviewed at-scale evidence that robotaxis can be safer than humans.[390]

04. End-to-end neural-net driving (Tesla FSD v12+, Wayve, Comma.ai) replaced hand-coded planners with a single learned model trained on video, mirroring the LLM-style scaling-laws approach in the autonomy stack.[381]

05. Chinese robotaxi operators (Baidu Apollo Go, Pony.ai, WeRide) are running commercial driverless services across Beijing, Wuhan, and Guangzhou; Apollo Go reported 9M+ cumulative rides by mid-2025.[384]

06. Waymo's first purpose-built robotaxi (Zeekr-made Ojai, 6th-gen Driver with cheaper sensor stack) went live with public rides on May 28, 2026.[391]

07. DA-WAM reports PDMS 93.7 (NAVSIM-v1) and EPDMS 87.7 (NAVSIM-v2) by aligning world-model latents with decision scoring.[392]

08. Waymo public service live in Denver, San Diego, Tampa (Sep 1, 2026): 14 US metros, >4,000-vehicle fleet incl. Ojai.[393]

09. Tesla: 1M unsupervised robotaxi miles; paid public Cybercab rides in Austin from Sep 4, 2026 (~420 registered autonomous vehicles in Texas, ~45 Cybercabs).[394]

10. [395]

11. Uber and Pony.ai to start Gen-7 robotaxi testing in London — Oct 8, 2026 (backfill, one day before window): Uber and Pony.ai plan to begin road testing Pony.ai Gen-7 robotaxis in London within weeks, part of a 2,000+ European robotaxi push; no passenger-service date yet. Uber separately has a London service planned with Wayve.[396]

c) Who could be next

Wayve [STARTUP][397] — Wayve AI Ltd. · Series C, $1.05B led by SoftBank (May 2024); also NVIDIA, Microsoft

Pioneering end-to-end embodied-AI autonomy; raised $1.05B Series C in May 2024 led by SoftBank, with NVIDIA and Microsoft participating; partnerships with Uber and OEMs to ship as a software-only AV stack.

Waabi [STARTUP][398] — Waabi Innovation Inc. · Series B, $200M led by Uber & Khosla (June 2024)

Raquel Urtasun's startup is building generative-AI simulation-first autonomous trucking; raised $200M Series B (June 2024) led by Uber and Khosla, targeting fully driverless trucking deployment in 2026.

Aurora Innovation [STARTUP][382] — Aurora Innovation, Inc. (NASDAQ: AUR) · Public (NASDAQ: AUR); ~$2B in cash reserves

Public company; launched the first fully driverless freight runs between Dallas and Houston in May 2025 — a leading indicator that long-haul autonomous trucking ships before passenger autonomy generalizes.

Applied Intuition [STARTUP][399] — Applied Intuition Inc. · Series F-equivalent, $250M at $15B valuation (2025)

Simulation and ADAS infrastructure provider used by most Western OEMs; raised $250M at $15B valuation (2025), positioning as the AWS of autonomy software.

Stellantis x Wayve x Uber L4 robotaxi partnership [STARTUP][400] — Stellantis / Wayve / Uber · Non-binding MoU; no financial terms disclosed

Three-way ecosystem (OEM platform + mapless embodied-AI driver + global ride network) to mass-produce and deploy L4 robotaxis without city-by-city mapping; signals convergence on Wayve's end-to-end approach.

Pony.ai [BUSINESS][396] — Pony.ai

Chinese robotaxi operator expanding into Europe via Uber (Zagreb live, London Gen-7 testing).

Flags / unverified

■ Tesla robotaxi rollout used safety operators in Austin in 2025 — fully driverless status varies by report

■ Waymo cumulative ride numbers cited from company-published statistics; independent audit limited

■ Updated Waymo service area: new value = 1,400 sq mi across 11 cities (May 13, 2026)

■ Updated waymo_freeway_status: new value = All freeway rides paused; 4 Sun Belt cities paused (May 21, 2026)

■ Updated Pony.ai global robotaxi fleet size: new value = 1,700+ vehicles as of 2026-05-24 (+250 since 2025 annual report); WeRide ~1,000 domestic

■ Updated Tesla Robotaxi Austin status (geofence vs active fleet): new value = Entire Austin metro geofence as of June 3, 2026, but only ~20 active unsupervised vehicles (down from 25 late-April peak); Dallas 3, Houston 6; fleet ramp deferred to late 2026/early 2027 pending FSD v15

■ Updated Waymo consumer monetization model: new value = Waymo Premier, an invite-only $29.99/mo subscription (priority pickups, 10% cash back, free cancellations), launched in SF, LA, Phoenix on Jun 11, 2026

■ Updated Waymo geographic expansion: new value = Registered Waymo Germany GmbH for EU robotaxi entry (June 25 2026)

■ Updated operating_cities: new value = adds San Diego, Las Vegas, Tampa, Denver (employee driverless); 6th-gen Driver on Hyundai IONIQ 5

■ Updated Tesla Robotaxi paid miles (quarterly): new value = ~700,000 miles (Q2 2026)

■ Updated ?: new value = ?

■ Updated Zoox commercialization: new value = Begins charging for driverless rides in Las Vegas Aug 10, 2026 — first paid US service for a purpose-built no-wheel robotaxi

■ Updated current_frontier[0].claim: new value = CPUC approved paid fully driverless Waymo service across 18 California counties (Aug 14, 2026), adding Sacramento and San Diego.

■ Updated nevada_robotaxi_permits: new value = Tesla 5,000 / Waymo 1,000 / Uber 1,000 approved (Aug 20, 2026)

■ Updated Tesla FSD China regulatory status: new value = No commercial operating permit as of Sep 17, 2026; Q3 2026 approval deadline expires Sep 30, 2026

■ Updated Waymo financing: new value = $5B debt facility (Oct 2026); 15 markets

■ Updated Tesla Cybercabs registered in Texas: new value = 319

Humanoid robotics

Impact so far. Humanoid robotics moved from research demos to low-volume industrial production between late 2024 and May 2026 — Tesla and Boston Dynamics in factories, Figure on the BMW line, 1X taking consumer pre-orders, with VLA models (RT-2, π0, Helix) finally enabling task generalization beyond hand-coded behaviors.

Watch for next. Whether Helix-class full-body autonomy generalizes to unstructured homes (1X NEO, Figure 03), whether teleop data factories drop unit-economic costs below $50/hour, and whether Tesla Optimus V3 actually hits 1M units/year — the inflection from industrial pilot to mainstream tooling.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Sergey Levine[401]	Co-founder & Research Lead Physical Intelligence	Biz	—
Jitendra Malik[402]	Professor, Robotics Researcher UC Berkeley EECS	Acad.	—
Dieter Fox[403]	Senior Director, Robotics Research NVIDIA / Allen Institute for AI	Cross	—
Pieter Abbeel[404]	Robotics & RL researcher Amazon / UC Berkeley	Cross	—
Brett Adcock[405]	CEO & Founder Figure AI	Biz	—
Marc Raibert[406]	Founder Boston Dynamics; Boston Dynamics AI Institute	Biz	—
Rohit Prasad[407]	CEO Boston Dynamics	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Vision-Language-Action (VLA) foundation models[408]	End-to-end neural policies that combine VLM backbones with robot action tokens — RT-2, RT-X / Open-X-Embodiment, Physical Intelligence π0/π0.5, and Figure's Helix enable zero-shot generalization across embodiments and tasks.	Google DeepMind RT-2 · RT-X / Open X-Embodiment · Physical Intelligence π0 / π0.5 · Figure Helix 02
Tactile sensing & dexterous manipulation[409]	Multi-axis fingertip and palm sensors (3–5 mN sensitivity) fused with vision close the in-hand-manipulation gap — slip detection, contact-rich assembly, and delicate grasping become tractable.	Sanctuary AI Phoenix 75-DOF hands · Figure 03 fingertip + palm sensors · Toyota Punyo soft-skin · PaXini DexH13
Sim-to-real RL & whole-body control[410]	Domain-randomized RL trained in Isaac Lab/Sim plus hybrid imitation pipelines now transfer zero-shot to hardware, with whole-body controllers handling balance and locomotion across embodiments.	NVIDIA Isaac GR00T N1.6 · Boston Dynamics electric Atlas · Humanoid-Gym · Fourier GR-1
Data factories & teleoperation scaling[411]	Large-scale teleop pools (45–50 tutors per 100 robots) and at-home gig-economy recording have driven cost-per-tutor-hour down ~60% to ~$118 in 2026, accelerating data-hungry VLA training.	Tutor Intelligence DF1 · Hyundai Robotics Metaplant · Scale AI / Encord teleop · DoorDash chore data
Form-factor convergence (26–32 DOF bipedal)[412]	The industry has converged on ~26–32 DOF bipedal designs with 16–22 DOF tendon-driven hands, enabling vertical integration and shipment scaling toward 90K+ units in 2026.	Tesla Optimus V3 · Boston Dynamics electric Atlas · Figure 03 · Unitree G1
MolmoAct2 (open VLA)[413]	Open-weight Vision-Language-Action foundation model for bimanual humanoid manipulation, trained on 720 h of teleoperation, with public weights, dataset, and eval code.	87.1% real-world DROID success with unseen objects · ~37x faster inference vs MolmoAct1 · Open weights + 720 h dataset released

Sub-idea	Description	Examples
Ambi Robotics + Pickle Robot inbound-logistics automation[414]	Integrated trailer-unload (Pickle) to AI palletize (AmbiStack) with zero human handoff at the dock door.	Pickle Robot trailer-unloading arms · AmbiStack AI stacking
RCM constraint-consistent dynamic control for surgical robots (ICRA 2026 Best Paper, Medical Robotics)[415]	Remote-center-of-motion constraint-consistent dynamic control; TUM / Haddadin lab.	ICRA 2026 Best Paper in Medical Robotics
1X NEO 25-DoF tendon-driven tactile hand[416]	25-DoF (22 actuated + 3 wrist) tendon-driven hand with force-transparent joints, fingertip/palm tactile sensing for real-time slip detection, and IP68 sealing; 1X claims capacity to build 10,000 hands in 2026.	
XAG X Series autonomous farm-robotics system[417]	Spray drone + XA1 auto-docking/recharge station + LM1 mixing/refill unit + RM80 unmanned mower, launched July 6 in Guangzhou for fully unmanned crop-protection workflows.	
Masked Visual Actions[418]	One video model as forward+inverse dynamics for robotics (Stanford/Harvard; Fei-Fei Li, Jiajun Wu, Yilun Du et al.)	
Ordered Action Tokenization (OAT)[419]	Prefix-decodable causally-ordered action tokens let a VLA policy trade latency for precision at inference; RSS 2026 Outstanding Paper finalist, validated on 3 policy backbones and 60+ tasks.	OAT (RSS 2026)
BYD 'Xiao Di' service humanoid[420]	BYD's first humanoid robot for dealer showrooms (greeting, demos, sales), 2-3 units per showroom; signals automaker vertical entry into commercial humanoids.	Xiao Di
RSS 2026 Outstanding Paper: FlashSAC cuts sim-to-real humanoid RL to minutes[421]	FlashSAC, an off-policy SAC variant bounding weight/feature/gradient norms, beat PPO and other baselines across 60+ tasks in 10 simulators and cut sim-to-real humanoid locomotion training from hours to minutes.	
Rack-climbing warehouse robots at scale[422]	Climbing ACR robots that use existing shelving as the storage grid, competing with humanoids for logistics labor at far lower unit cost.	Hai Robotics HaiPick Climb, 1,500+ units, Europe, Sep 2026
"World-Action Models" survey formalizes world models + action policies for robots[423]	A Sep 13, 2026 arXiv survey proposes "world-action models" as a distinct category clarifying how world models, model-based RL, action-conditioned video generation and reactive VLA policies relate for robot learning and control.	

b) Current research frontier (last 12–18 months)

01. Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specific training.[424]

02. Tesla Optimus V3 entered internal factory deployment in Jan 2026 with ~1,000 robots in production at Fremont and Gigafactory Texas; Tesla targets mass production at 1M units/year by Q3 2026.[425]

03. 1X NEO took 10,000 pre-orders in 5 days (Oct 2025); the Hayward, CA factory began full-scale production in Q1 2026 with consumer delivery targeted for late 2026.[426]

04. Unitree G1 launched at $16K (April 2026) with 23 DOF; Unitree filed for a Shanghai IPO at ~$610M (March 2026), aiming for 20K shipments in 2026 (vs ~5.5K in 2025).[427]

05. Boston Dynamics' electric Atlas entered commercial production in Jan 2026 with all 2026 units committed to Hyundai and Google DeepMind, supported by autonomous battery-swapping for continuous operation.[428]

06. DreamZero (NVIDIA GEAR, 14B World Action Model on a video-diffusion backbone) achieves >2x zero-shot generalization vs state-of-the-art VLAs in real-robot experiments, real-time 7Hz control, and few-shot embodiment transfer in 30 min of play data.[429]

07. GAM (arXiv:2606.17046, ETH/KAIST) repurposes a 3D geometry foundation model for contact-rich manipulation, beating VLA/world-action baselines on accuracy, robustness, speed and size.[430]

08. NavWM (arXiv:2606.24101, CAS/Beihang) unifies latent world reasoning, action prediction and controllable generation via 'latent world tokens' for visual navigation.[431]

09. NVIDIA Halos for Robotics: full-stack physical-AI safety system (IGX Thor + Halos OS + certification lab); Agility Robotics first adopter.[432]

10. RSS 2026 Best Paper: FlashSAC (Holiday Robotics) cuts sim-to-real humanoid locomotion training from hours to minutes across 60+ tasks / 10 simulators.[421]

11. ICRA 2026 Best Paper (Robot Learning): conditioning ACT/Diffusion-Policy/SmolVLA on camera extrinsics improves generalization to unseen viewpoints.[433]

12. On July 28, 2026 the FCC added foreign-produced advanced robotic devices (humanoids and quadrupeds) and connected power inverters to its Covered List, blocking equipment authorization for new imports over supply-chain/cybersecurity risk; a Department of War national-security review can grant conditional approval.[434]

13. Beijing Humanoid Robot Innovation Center's Tiangong Ultra clocked 8.86s in the 100m semifinal on Aug 26 and 8.64s in the final, improving on its own 9.39s mark set Aug 22 - both faster than the 9.58s human world record.[435]

14. CMU researchers show world-action models trained to autoregressively predict point tracks, DINO features and depth before actions outperform those predicting future RGB, refining the world-action-model thesis for robot manipulation. Trending on alphaXiv.[436]

15. [437]

16. [438]

17. World-action models surge on alphaXiv: LeCun's H-JEPA and NVIDIA Long-WAM — Week of Oct 5, 2026: H-JEPA (NYU/AMI Labs/Inria/Brown, LeCun co-author; arXiv 2610.06805) learns hierarchical latent world models, lifting Visual AntMaze success from 18% to 73% with less planner compute (341 alphaXiv saves by Oct 13). NVIDIA/MIT Long-WAM (arXiv 2610.10528, Oct 7) scales world-action-model context (RoboCasa GR-1 63.3%→78.7%).[439]

c) Who could be next

Figure AI [STARTUP][405] — Figure AI · Series C ~$1.5B at $39.5B valuation (Sep 2025); ~$1.9B total raised

Now fully independent from OpenAI on its own Helix VLA stack; reported $39.5B valuation (Sep 2025) and BMW production deal — a likely IPO candidate.

Unitree Robotics [STARTUP][427] — Unitree Robotics · Series C ~$520M at $5.5B (Feb 2026); IPO ~$610M filing

Best price/performance leader globally (G1 at $16K); Shanghai IPO filing March 2026 unlocks ~$610M for 4× capacity expansion targeting 20K units in 2026.

Physical Intelligence [STARTUP][440] — Physical Intelligence (Pi) · Founded 2024 by Levine, Abbeel et al; venture-backed (specifics undisclosed)

The leading VLA foundation-model lab; π0.5 (April 2026) shows emergent open-world generalization, positioned to license to OEMs across the industry.

Sanctuary AI [STARTUP][441] — Sanctuary AI · Series B ~$50–100M (estimate); Magna strategic partnership

75-DOF hydraulic hands with industry-leading dexterity and a Magna automotive partnership for production deployment.

Figure 03 production scale-up [STARTUP][442] — Figure AI · Series C+; ~$39B reported valuation

First humanoid OEM to credibly hit hourly production cadence with autonomous (non-teleop) locomotion on uneven terrain - a leading indicator of commercial humanoid roll-out.

ROBOTERA [STARTUP][443] — ROBOTERA (Beijing) · >$200M (May 8, 2026) led by SF Group, HSG, IDG Capital

China's leading commercial humanoid; logistics product-market fit with China Post and SF Group; >95% of core components developed in-house

Unitree Robotics [STARTUP][444] — Unitree Robotics · Targeting ~$6.2B valuation, raising ~4.2B yuan (~$616M) via STAR Market IPO; backers include Meituan, Sequoia China, Tencent, Alibaba, Ant Group

First embodied-AI company cleared for a China A-share (STAR Market) IPO and chosen as the hardware base for NVIDIA's open Isaac GR00T Reference Humanoid; 2025 revenue 1.699B yuan at ~60% gross margins, but carries U.S. national-security/legislative risk.

NEURA Robotics [STARTUP][445] — NEURA Robotics GmbH · Series C of up to $1.4B (led by Tether; Qualcomm, Amazon, NVIDIA, Bosch, Schaeffler, EIB)

One of the largest robotics/physical-AI rounds on record; positions a European player against US/China humanoid leaders with a shared-learning 'Neuraverse' and 4NE1 humanoid.

Theker [STARTUP][446] — Theker · $85M Series A (led by CRV; Samsung, Aglaé Ventures)

Reconfigurable-form factory robots as an alternative to fixed humanoids; strategic backing from Samsung, LVMH's Aglaé, and Inditex signals manufacturing/retail pull.

XDOF [STARTUP][447] — XDOF (robot teleoperation data) · $70M launch round — Thrive Capital, Spark Capital, a16z, Lux Capital, WndrCo

Fills the teleoperation-data-collection gap; open-sourced ABC-130K (130K bimanual trajectories); serves frontier-lab customers via GELLO-derived teleop hardware.

Pickle Robot [STARTUP][414] — Pickle Robot Co. · Venture-backed (see coverage)

Trailer-unloading manipulation for inbound logistics; core warehouse-logistics-robots player surfaced via the Ambi integration.

Agility Robotics [COMPANY][448] — Agility Robotics (Digit humanoid) · ~$2.5B SPAC (Churchill Capital Corp XI), >$620M gross proceeds

Going public via Churchill Capital Corp XI SPAC at ~$2.5B, raising >$620M (largest humanoid raise to date); >$300M booked robots-as-a-service revenue (~1,000 Digit units) with Amazon, Toyota and GXO Logistics.

XAG [COMPANY][417] — XAG (agricultural robotics & drones)

Entity reconciliation: maker of this week's X Series drone+dock+refill system; a leading agricultural-robotics/drone player and the first item activating the ag-robotics gap sub-theme.

Holiday Robotics [STARTUP][449] — Holiday Robotics · undisclosed

RSS 2026 Best Paper (FlashSAC) — fast/stable off-policy RL for high-dimensional humanoid control; first Korean team to win RSS Best Paper

Beijing Humanoid Robot Innovation Center [LAB][450] — Beijing Humanoid Robot Innovation Center · n/a

Distinct from Unitree/ROBOTERA - a state-backed national humanoid platform racking up human-beating locomotion records

XPeng [PUBLIC_COMPANY][451] — XPeng · Public (NYSE: XPEV / HKEX: 9868)

Vertically integrated EV/autonomy stack applied to humanoids; one of few players with a dedicated automated humanoid line.

Flags / unverified

■ Tesla Optimus V3 specs and production timeline come largely from non-Tesla sources; treat with caution.

■ Figure's claims about full Helix capabilities are demo-tested, not peer-reviewed for unseen-task generalization.

■ Humanoid market projections (90K shipments 2026, 1.2M by 2030) are highly speculative.

■ Updated Figure Helix-02 longest autonomous run: new value = Full 8-hour autonomous factory shift at near-human throughput (~3 sec/package)

■ Updated Figure 03 production rate: new value = 1 robot/hour at BotQ; 350+ Gen-3 units; new System 0 perception-conditioned whole-body control

■ Updated Boston Dynamics ownership: new value = SoftBank exercised put on remaining ~9.65%; full ownership moving to Hyundai affiliates + Chairman Chung Euisun

■ Updated Unitree IPO status/financials: new value = CSRC-approved ~$619M STAR Market IPO; FY2025 gross margin 60.27%, revenue ~$250M, net profit RMB278M

■ Updated ?: new value = ?

■ Updated watch_list[Unitree Robotics].status: new value = Public (SSE STAR Market, listed Aug 19, 2026; $904M IPO; ~$47B day-one market cap)

■ Updated XPeng IRON commercial sales start: new value = H2 2027 (internal pilots from early 2027)

Fusion energy

Impact so far. Fusion went from public-science promise to active commercial track between 2022 and 2026: NIF demonstrated repeatable net energy gain, CFS broke ground on SPARC with HTS magnets, and Helion shipped a D-T plasma machine on the way to a 50 MW commercial PPA with Microsoft.

Watch for next. First private-fusion net-electricity-on-grid demonstrations (CFS SPARC Q>1 in 2027, Helion Orion 2028), AI plasma control becoming standard, and validation of tritium breeding fuel-cycle infrastructure at commercial scale.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Omar Hurricane[452]	Chief Scientist, Inertial Confinement Fusion Program Lawrence Livermore National Laboratory	Gov	—
Bob Mumgaard[453]	Co-founder & CEO Commonwealth Fusion Systems	Biz	—
David Kirtley[454]	Co-founder & CEO Helion Energy	Biz	—
Michel Laberge[455]	Founder & CTO General Fusion	Biz	—
Pietro Barabaschi[456]	Director-General ITER Organization	Gov	—
Sam Altman[457]	Chairman of the Board Helion Energy (also CEO, OpenAI)	Cross	—
Bernard Looney[458]	Independent director, Type One Energy (former CEO, BP) Type One Energy	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Tokamak (magnetic confinement)[459]	The most-developed fusion approach: a toroidal magnetic chamber confines a hot plasma. CFS SPARC, ITER, KSTAR/EAST, JT-60SA, and DIII-D are the major active programs.	CFS SPARC (HTS magnets) · ITER · KSTAR (100M°C, 48s record) · EAST (403s H-mode)
Inertial confinement (laser-driven)[460]	High-energy lasers compress a fusion fuel pellet to ignition. NIF reached repeatable net energy gain (8.6 MJ from 2.08 MJ in April 2025) — the first machine of any kind to do so.	NIF (LLNL) · Marvel Fusion (proton-boron) · Pacific Fusion
HTS (high-temperature superconductor) magnets[461]	Rare-earth barium copper oxide (REBCO) tape enables 20+ tesla magnetic fields, cutting tokamak size and cost dramatically — the enabling technology for compact commercial fusion machines.	CFS 20-tesla TF magnet · Tokamak Energy ST40 / Demo4 · MIT-CFS HTS collaboration
AI plasma control[462]	Deep RL controllers stabilize plasma in real time on DIII-D, KSTAR, and TCV; DeepMind's TORAX simulator (May 2024) is now part of CFS's planning loop and broadly adopted across the field.	DeepMind TORAX · DeepMind + EPFL TCV (Nature 2024) · DIII-D + KSTAR ELM-suppression AI
Alternative confinement concepts[457]	Field-reversed configuration (Helion, 150M°C D-T, Feb 2026), sheared-flow Z-pinch (Zap Energy, 1.6 GPa pressures), stellarators (Wendelstein 7-X, 43s record May 2025), and acoustically driven MTF (General Fusion).	Helion Polaris (FRC) · Zap Energy FuZE-3 (Z-pinch) · Wendelstein 7-X (stellarator) · General Fusion LM26 (MTF)

b) Current research frontier (last 12–18 months)

01. NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fusion.[463]

02. Commonwealth Fusion Systems installed the first of 18 toroidal-field HTS magnets on SPARC in Jan 2026; first plasma is targeted for 2026 with a Q>1 demonstration in 2027.[464]

03. Helion Polaris reached 150 million °C with deuterium-tritium fuel (Feb 2026) — the first privately-built D-T fusion machine; Orion commercial plant construction began Jul 2025 with a 50 MW Microsoft PPA targeting 2028.[457]

04. KSTAR set a 100 million °C, 48-second plasma record (2024) with a 300-second target by 2026; Wendelstein 7-X reached a 43-second high-fusion-product plasma in May 2025.[465]

05. AI/RL plasma control was demonstrated on DIII-D and KSTAR in 2024; DeepMind's TORAX simulator (May 2024) is now embedded in CFS's daily operational planning and adopted across the field.[462]

06. A desktop-scale (sub-5-inch) electrostatic/magnetic 'orbitrap' fusion device heated plasma to roughly 11 million degrees C (above 1 keV apparent ion temperature) on less than $50M of investment, demonstrating a compact path toward fusion conditions.[466]

07. Helion received world-first regulatory licenses (WA radioactive-materials + air-emissions) for its Orion commercial fusion plant; enabled by NRC byproduct-material framework (ADVANCE Act).[467]

08. CFS published 5 peer-reviewed papers (Journal of Plasma Physics) on the ARC plant's physics basis for ~400 MW net electricity.[468]

09. Realta Fusion demonstrated direct energy conversion — drawing multiple amps at ~100V from plasma kinetic energy on the WHAM magnetic-mirror machine at UW-Madison (June 29, 2026); projects DEC could cut plant costs 10-20% vs thermal cycles.[469]

10. [470]

c) Who could be next

Commonwealth Fusion Systems (SPARC) [STARTUP][471] — CFS (MIT spinout) · $3.0B total; Series B2 $863M (Aug 2025) led by Nvidia, Google, Bill Gates

Largest private fusion effort: $3B+ raised, magnet assembly underway, first plasma targeted 2026 and Q>1 in 2027 — closest to a credible commercial tokamak.

Helion Energy (Polaris / Orion) [STARTUP][472] — Helion Energy · Series E+ via Sam Altman, Khosla Ventures; Microsoft PPA May 2023

Only private firm with a firm grid PPA (Microsoft 50 MW by 2028); FRC achieved 150M°C D-T (2026); Orion commercial plant under construction.

TAE Technologies (Norm / Copernicus) [STARTUP][473] — TAE Technologies · $1.3B+ equity raised; $150M latest round (Chevron, Google, NEA)

Neutral-beam injection breakthrough (Apr 2025) cut machine complexity ~50%; Copernicus targeting net energy before 2030; planned TMTG-style merger at $6B valuation (mid-2026).

Tokamak Energy (ST40 / Demo4) [STARTUP][461] — Tokamak Energy · Multiple rounds; UK government and private capital

Spherical tokamak architecture with 11.8-tesla HTS magnets; claims 30–40× smaller plasma volume vs LTS designs and a clear path to ST-E1 commercial pilot.

Thea Energy [STARTUP][474] — Thea Energy · $100M (May 27, 2026); leads Prelude Ventures, US Innovative Technology, General Innovation Capital Partners

Planar-coil stellarator as a simpler-to-manufacture alternative to tokamaks; $100M to build a demonstration prototype amid AI-data-center power demand.

Focused Energy [STARTUP][475] — Focused Energy GmbH (Germany / Austin, TX) · $240M Series A (June 2026) led by RWE; ~$300M private + $200M grants total

Now one of the most heavily funded fusion startups; inertial-confinement laser fusion using direct-drive targets at high rep-rate, building a 'Lighthouse' demo at a former RWE nuclear site.

Type One Energy [STARTUP][458] — Type One Energy Group · Series B (closing 2026)

Leading stellarator-path fusion developer (Infinity One prototype at TVA Bull Run, grid power targeted ~2034); closing Series B, recruiting major-energy leadership (Bernard Looney) — a key non-tokamak competitor to track.

Realta Fusion [STARTUP][469] — Realta Fusion (magnetic-mirror fusion)

Entity reconciliation: UW-Madison spinout (WHAM machine) that just demonstrated first-of-kind direct plasma-to-electricity conversion; a distinct mirror-confinement approach vs tokamak/inertial peers.

General Fusion [PUBLIC_COMPANY][476] — General Fusion · ~$150M cash post-SPAC (2026)

First pure-play fusion company to go public (Nasdaq: GFUZ, via SPAC); magnetized target fusion (LM26) toward 2028 Lawson milestones.

Proxima Fusion [STARTUP][477] — Proxima Fusion · €411M ($468M) at €2.4B valuation (led by XTX Ventures & East X Ventures; RWE, Google, Lightspeed, Balderton, others)

Europe's largest private fusion raise; QI-HTS stellarator; 'Alpha' net-energy demonstrator near Munich

General Fusion (Nasdaq: GFUZ) [STARTUP][478] — General Fusion Group Ltd. · Public (Nasdaq: GFUZ); ~$150M cash reported Aug 2026

First fusion developer to trade publicly, forcing quarterly quantitative disclosure of magnetized-target-fusion milestones (0.72 keV on LM26) against a 1 keV / 10 keV / Lawson roadmap — a new source of auditable fusion progress data.

Inertia Enterprises [STARTUP][479] — Inertia Enterprises Inc. · n/a

Only private company with an exclusive integrated fusion-technology partnership with LLNL/NIF; ICF target manufacturing is central to laser-fusion economics

US-UK fusion partnerships (TVA/UK Infinity Fusion Consortium, UKAEA-PPPL, EPRI-Birmingham) [LAB][480] — UKAEA, PPPL, EPRI, TVA, Type One Energy, Tokamak Energy

Fusion moving from lab to market via bilateral supply-chain, computing and materials partnerships

Fujikura [BUSINESS][470] — Fujikura

HTS tape supply is a gating input for high-field fusion magnets.

Flags / unverified

■ Sam Altman is Chairman of Helion's board, not founder (David Kirtley founded the company).

■ TAE merger details and valuation are reports of plans, not finalized transactions.

■ Updated zap_energy_strategy: new value = Hybrid fission-fusion company strategy; Zabrina Johal named CEO (May 5)

■ Updated US federal fusion commercialization strategy: new value = Finalized DOE Fusion S&T Roadmap published 2026-06-09 targeting commercial fusion in the mid-2030s, implemented via a new Office of Fusion

■ Updated Helion Energy funding & valuation: new value = $465M Series G (2026-06-04), $15.5B post-money valuation, total raised ~$1.5B; led by Thrive Capital

■ Updated ?: new value = ?

■ Updated CFS total capital raised: new value = $4 billion (new $1B round; first pension/SWF direct fusion investment)

■ Updated ITER tokamak assembly: new value = 6 of 9 sector modules installed (two-thirds of core); final module retargeted to mid-2027

■ Updated dt_fuel_layer_production_time: new value = 2-3 hours

Brain-computer interfaces

Impact so far. BCIs crossed from lab to early commercial reality between 2024 and 2026: Neuralink put 5+ patients on the Telepathy implant, Precision Neuroscience took home the first wireless-cortical-BCI 510(k), BrainGate hit 110 cpm bimanual typing, UCSF restored speech at 78 wpm, and Meta shipped a 20.9 wpm sEMG wristband to consumers.

Watch for next. The first BCI Pre-Market Approval (likely Synchron), Neuralink's first Blindsight implant, Paradromics speech-restoration pivotal results, and whether the non-invasive consumer category (Meta Neural Band, Kernel) closes meaningful ground on invasive bandwidth.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Leigh Hochberg[481]	Director, Center for Neurotechnology and Neurorecovery; IDE sponsor-investigator, BrainGate Brown University / Massachusetts General Hospital	Cross	★ Sean M. Healey International Prize for Innovation in ALS (2024)
Edward Chang[482]	Director, Speech Neuroprosthesis Lab UCSF Department of Neurosurgery	Acad.	—
Jaimie Henderson[483]	Director, Stanford Neural Prosthetics Translational Lab; Co-PI, BrainGate2 Stanford School of Medicine	Acad.	—
Thomas Oxley[484]	Co-founder & CEO Synchron Inc.	Biz	—
Matt Angle[485]	Founder & CEO Paradromics	Biz	—
Benjamin Rapoport[486]	Co-founder & Chief Science Officer Precision Neuroscience	Biz	—
Mikhail Shapiro[487]	Co-founder, Merge Labs; Professor, Caltech Merge Labs / Caltech	Cross	—
Tyson Aflalo[487]	Co-founder, Merge Labs Merge Labs	Cross	—
Karl Deisseroth[488]	Professor; optogenetics pioneer; Nobel 2026 Stanford University	Acad.	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Intracortical microelectrode arrays (Utah arrays)[489]	High-density silicon arrays implanted directly into motor cortex (Blackrock NeuroPort) record from individual neurons — the gold standard for invasive BCIs since the early 2000s.	BrainGate clinical consortium · Blackrock NeuroPort · Stanford / Brown / MGH BrainGate2
Endovascular BCI (Stentrode)[484]	Synchron's stent-electrode is implanted via blood vessels — no open-brain surgery — trading some signal fidelity for dramatically lower surgical risk.	Synchron Stentrode · FDA COMMAND trial · Synchron + Apple Vision Pro integration
High-bandwidth thin-film cortical interfaces[486]	Flexible subdural electrode films (e.g. Precision Layer 7) put 1,024+ electrodes on a hair-thin sheet, with FDA 510(k) clearance and a Medtronic surgical-navigation partnership in 2025–26.	Precision Neuroscience Layer 7 · Mount Sinai 4,096-electrode demo · Medtronic + Precision partnership
Speech decoding & synthesis[490]	UCSF's Chang lab and Stanford have pushed motor-cortex speech decoders to 78 wpm with personalized voice synthesis; the BrainGate consortium hit 110 cpm bimanual typing.	UCSF Chang lab speech BCI · Stanford inner-speech decoder · BrainGate bimanual typing

Sub-idea	Description	Examples
Non-invasive consumer BCI (EMG / EEG / fNIRS)[491]	Wearable surface-EMG and EEG/fNIRS systems (Meta Neural Band, Kernel Flow, OpenBCI) are reaching consumer scale at meaningful — if lower — bandwidths than invasive devices.	Meta Neural Band sEMG wristband · Kernel Flow TD-fNIRS · OpenBCI
Intraoperative high-density micro-ECoG BCI[492]	Temporary high-density micro-electrocorticography arrays placed during routine awake craniotomy that yield same-day speech and cursor decoding, lowering the barrier to BCI use in surgical mapping and short-term clinical applications.	Johns Hopkins micro-ECoG study in 4 patients (May 2026) · Precision Neuroscience Layer 7 Cortical Interface
Neuralink transdural implant technique[493]	Neuralink's first transdural BCI implant threads electrode arrays through the intact dura (rather than removing it) with a thicker-needle robotic technique; performed in a Vancouver ALS patient, the first Canadian ALS recipient.	Neuralink
PRIMA (Science Corporation)[494]	Wireless subretinal vision-restoration implant + camera glasses; CE-marked, mean +25.5 ETDRS letters	

b) Current research frontier (last 12–18 months)

01. Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (visual prosthesis) received FDA breakthrough device designation (Sep 2024) with first-in-human implant planned for 2026.[495]

02. Paradromics received FDA IDE approval (Nov 2025) for the Connect-One speech-restoration trial with the Connexus BCI, claiming ~200+ bps information transfer (10–20× higher than BrainGate, 100–200× higher than Synchron) via thread arrays thinner than a human hair.[496]

03. INBRAIN Neuroelectronics completed enrollment for the world's first graphene-BCI clinical trial (April 2026) with zero device-related adverse events in the first cohort; 1,024-contact wireless rechargeable arrays.[497]

04. Precision Neuroscience's Layer 7 received FDA 510(k) clearance (April 2025) — the first wireless cortical BCI cleared for market — and is now in 30-day chronic trials at six U.S. medical centers; Medtronic partnership announced January 2026.[486]

05. BrainGate reported rapid bimanual typing (110 cpm, 1.6% WER) in two paralyzed patients (Nature Neuroscience, March 2026); the consortium raised $200M Series D in late 2025 for FDA PMA pivotal trials.[498]

06. Neuralink developing surgical robot for precision brain access with automated procedures and high-volume implant production targeted for 2026.[499]

07. Synchron's Stentrode is the first BCI integrated with Apple's BCI HID profile as a native input category, enabling hands-free Apple-device control for paralysis patients.[500]

08. Neuralink is developing a next-generation surgical robot capable of implanting threads in any brain region, paired with a 2026 push to high-volume implant manufacturing.[501]

09. A UC Davis-led intracortical BCI (Nature Medicine, 15 June 2026) enabled a man with ALS to communicate independently at home via speech-to-text (averaging 56 words/min) and cursor control without researcher support over 3,800+ hours, demonstrating long-term self-administered use.[502]

10. Stanford/Nature (June 17): every motor-cortex region encodes whole-body movement (8 BCI participants, 45 movements), upending the motor homunculus and implying fewer implants for versatile BCI control.[503]

11. China (NMPA) approved Neuracle's NEO epidural BCI for commercial use — first invasive BCI approved outside clinical trials.[504]

12. Perspective (Saez, Mount Sinai; Trends in Cognitive Sciences, Jul 17 2026) proposes closed-loop 'cognitive BCIs' for psychiatric disorders.[505]

13. Neuracle Technology implanted its NMPA-approved NEO epidural BCI in a paralyzed patient at Huashan Hospital on July 13, 2026, giving him robotic-glove control via an epidural electrode array.[506]

14. Merge Labs x Butterfly Network: multi-year exclusive license of Poseidon ultrasound-on-chip for ultrasound-based BCI (Sep 1, 2026).[487]

15. A single high-density ECoG implant supported simultaneous decoding of speech and upper-body gestures in paralyzed participants — the first system to produce both communication modes at once rather than switching between them (Nature Neuroscience, Sep 14, 2026).[507]

16. Nobel Medicine 2026 to Deisseroth, Hegemann, Nagel for optogenetics — Oct 5, 2026: the Nobel Assembly honored light-gated ion channels and optogenetics, the foundational tool for switching neurons on/off with light that underpins modern neurotech and BCI research.[508]

c) Who could be next

Neuralink Blindsight [STARTUP][509] — Neuralink · Reported $1B+ private equity, Musk-led

First visual-prosthesis implant planned in 2026 — direct visual-cortex stimulation that bypasses damaged optics. A major expansion from motor-only BCIs.

Synchron pivotal trial [STARTUP][484] — Synchron Inc. · $200M raise (Nov 2025) for pivotal trial execution

Preparing FDA Pre-Market Approval submission across 4+ U.S. sites; if approved, the first BCI PMA could define the regulatory gold standard for the field.

Paradromics Connexus [STARTUP][485] — Paradromics Inc. · Series-stage venture funding (specifics undisclosed)

Extreme bandwidth (~200+ bps), first FDA IDE for speech restoration as a primary endpoint, 3 clinical sites planned for early 2026.

INBRAIN Neuroelectronics [STARTUP][510] — INBRAIN (Barcelona) · Spanish/EU biotech funding; WEF recognition signals next round

Graphene-electrode material breakthrough with bi-directional 1,024-contact arrays; WEF Technology Pioneer 2025 and zero adverse events in first cohort.

Rice/Baylor BrainGate team [LAB][511] — Rice University & Baylor College of Medicine · Federal/clinical trial grants (BrainGate consortium)

First Texas-based BrainGate site; pairs Rice computational/engineering with Baylor clinical neuroscience to advance intracortical BCI for tetraplegia assistive robotics.

Neuracle Technology [STARTUP][504] — Neuracle Technology (with Tsinghua University) · Private (China)

Maker of NEO, the first invasive BCI approved for commercial use (China NMPA); direct competitor context for Neuralink/Synchron.

Science Corporation [STARTUP][494] — Science Corporation · Founded by Max Hodak (Neuralink co-founder); VC-backed

Leading non-Neuralink BCI/neural-interface company; PRIMA retinal implant CE-marked (Jul 2026), US FDA review underway

Merge Labs [STARTUP][487] — Merge Labs (Butterfly Network chip partner) · $252M seed (OpenAI/Altman-backed)

Best-funded bet on a non-surgical (ultrasound) BCI path; dependency on Butterfly's Poseidon ultrasound-on-chip.

CorTec (Brain Interchange) [STARTUP][512] — CorTec GmbH · n/a

Only fully implantable closed-loop BCI with two FDA Breakthrough designations (stroke rehab + communication); European counterweight to US implant makers.

Precision Neuroscience [STARTUP][513] — Precision Neuroscience · $250M Series D; $430M total raised to date

Large late-stage raise (broad investor base spanning sovereign wealth, VC and family offices) signals conviction in non-penetrating, surface-electrode BCI as a distinct category competing with Neuralink and Paradromics.

Flags / unverified

■ Paradromics' 200+ bps figure is a manufacturer claim; not yet independently verified in human use.

■ BCI regulatory pathway is fragmented across 510(k) / IDE / PMA tracks; no unified FDA BCI guidance exists yet.

■ Updated neuralink_surgical_robot_capability: new value = Next-gen R1 robot can implant in any brain region (motor + non-motor); threads through intact dura; real-time motion compensation

■ Updated paradromics_connexus_device_compatibility: new value = FDA-approved to interface with personal computing devices (Aug 26, 2026)

- Updated Paradromics longest preclinical stable recording: new value = >4 years (sheep)
- Updated Neuralink record cursor bandwidth: new value = 11.32 bps
- Updated Paradromics human implants: new value = First human Connexus implant (Jun 17, 2026; Connect-One EFS)

AI accelerators / advanced compute

Impact so far. GPU-dominated AI compute fractured into specialized lanes between 2024 and 2026 — wafer-scale (Cerebras), inference ASICs (Groq, Etched), photonic interconnect (Lightmatter), and per-hyperscaler custom silicon (TPU/Trainium/Maia/MTIA) — pulling 20–70% of opex out of training and inference.

Watch for next. Rubin (2H 2026) and the Cerebras OpenAI 750 MW deployment going live; Lightmatter Passage co-packaged optics in production; neuromorphic chips (Loihi 3, NorthPole) crossing the practical edge-AI threshold; whether NVIDIA-Groq integration cements inference dominance.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jensen Huang[514]	CEO & Co-founder NVIDIA	Biz	—
Andrew Feldman[515]	Co-founder & CEO Cerebras Systems	Biz	—
Jonathan Ross[516]	Founder Groq (acquired by NVIDIA Dec 2025)	Biz	—
Jim Keller[517]	CEO Tenstorrent	Biz	—
Norm Jouppi[518]	Engineering Fellow; lead architect, Google TPU Google	Biz	★ ACM-IEEE Eckert-Mauchly Award
Nick Harris[519]	Co-founder & CEO Lightmatter	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
GPU scaling (Hopper → Blackwell → Rubin)[520]	NVIDIA's flagship line keeps doubling effective AI throughput per generation: Blackwell (208B transistors, ~20 PFLOPS FP4 per B200) ships at scale in 2025, Rubin (3× Blackwell, 1.2 EFLOPS FP8 training) in 2H2026.	NVIDIA H100/H200 (Hopper) · NVIDIA B200 (Blackwell) · NVIDIA Rubin / Rubin Ultra
Wafer-scale processors[521]	Cerebras WSE-3 (5nm, 4 trillion transistors, 900K cores, 44 GB on-chip SRAM, 21 PB/s memory) eliminates chip-to-chip movement entirely — enabling single-chip training of multi-trillion-parameter models.	Cerebras WSE-3 · Cerebras CS-3 system · OpenAI 750 MW Cerebras deal
Inference-specialized ASICs[516]	Fixed-function silicon optimized for narrow inference workloads — Groq LPU's deterministic conveyor-belt architecture and Etched's transformer-only Sohu ASIC compete with GPUs on $/token.	Groq LPU (acquired by NVIDIA) · Etched Sohu · MatX
Hyperscaler custom silicon[522]	Cloud platforms have abandoned GPU monoculture: Google Trillium / TPU v6, AWS Trainium 3, Meta MTIA 300/500, Microsoft Maia 200 each deliver hyperscaler-specific compute economics.	Google TPU v6 Trillium · AWS Trainium 3 · Meta MTIA 300/500 · Microsoft Maia 200
Photonic & neuromorphic compute[523]	Lightmatter's photonic processor and IBM NorthPole / Intel Loihi 3 neuromorphic chips break with the GPU paradigm — using photons or spiking neurons for orders-of-magnitude better energy per inference for the right workloads.	Lightmatter Envise / Passage · IBM NorthPole · Intel Loihi 3
Google four-partner inference TPU supply chain[524]	Google diversifying custom AI silicon by adding Marvell (memory processing unit + next-gen inference TPU) alongside Broadcom and MediaTek, signaling hyperscaler push to commoditize inference accelerators and reduce NVIDIA dependence.	Marvell-designed MPU paired with TPU · Inference-only TPU separate from training-focused TPUs · ~2M MPU production target

Sub-idea	Description	Examples
NVIDIA RTX Spark (N1X) client AI superchip[525]	Arm-CPU + Blackwell-GPU SoC with 128GB unified memory bringing data-center-class local AI inference (120B-param LLMs, 1M-token context) to consumer Windows PCs.	RTX Spark Superchip / N1X · Windows-on-Arm agentic AI laptops from Dell/HP/Microsoft/ASUS/Lenovo/MSI
Cerebras Systems	Wafer-scale AI accelerator (CS-3/WSE-3) maker expanding European inference capacity	
Azure AMD Helios (UALink)[526]	72 MI455X GPUs, 31TB HBM4, ~260TB/s intra-rack, open UALink-over-Ethernet fabric	
Co-packaged optics / silicon photonics[527]	Silicon-photonics foundries hit a mass-production inflection for co-packaged optics — TrendForce (Jul 30, 2026) reports UMC/SILITH shipped first mass-produced 12-inch SiPh wafers and TSMC's COUPE platform entered production, relieving copper-interconnect bandwidth limits for AI accelerators.	TSMC COUPE · UMC/SILITH · GlobalFoundries · Tower Semiconductor
Samsung zHBM (3D-stacked memory)[528]	Concept memory that stacks HBM directly atop the AI accelerator via wafer-bonding; projected ~8x perf / >10x density vs HBM5, targeting the memory-bandwidth wall.	zHBM · zNAND-O
Memory-safe portable GPU kernels[529]	Compiler-native, cross-vendor GPU offload in a memory-safe language as an alternative to hand-written CUDA for AI kernels.	rustc + LLVM Offload GPU backend (arXiv 2608.13759), nightly-only
UC Berkeley Gemmelos dual-chip edge AI SoC - fills consumer-NPU gap[530]	UC Berkeley presented Gemmelos, a two-chip Intel 16nm platform for multimodal edge AI inference, one of several on-device accelerator disclosures at Hot Chips 2026 alongside KU Leuven's ETHEREAL and Harvard/Lockheed's Pistil.	
Tower Semiconductor/NewPhotonics ship laser-integrated CPO PICs at volume[531]	Tower Semiconductor and NewPhotonics began high-volume shipments of laser-integrated, serviceable optical-engine PICs on Tower's PH18DA silicon-photonics platform supporting 800G–1.6T for AI scale-out interconnect, with 6.4T planned next.	

b) Current research frontier (last 12–18 months)

01. NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027.[514]

02. Cerebras Systems signed a >$10B agreement with OpenAI (Jan 2026) for ~750 MW of inference compute through 2028 and filed for IPO at ~$23B valuation in April 2026.[532]

03. NVIDIA acquired Groq for ~$20B (Dec 2025) and integrated the LPU into its 2026 GTC stack; Groq 3 LPU achieves ~1,500 tokens/sec on agentic AI inference.[533]

04. Hyperscaler custom silicon scaled across all four majors: Google Trillium (4× LLM training perf vs v5e, 2× HBM bandwidth), AWS Trainium 3 (2.52 PFLOPS MXFP8, 144 GB HBM3e), Meta MTIA generations 4× HBM bandwidth, Microsoft Maia 200 (10 PFLOPS FP4).[522]

05. Intel Loihi 3 (Jan 2026) delivers 8M neurons / 64B synapses at 4nm running at ~1.2W peak vs 300W+ for GPU equivalents — neuromorphic compute crossing the threshold for practical edge AI.[534]

06. Cerebras' $3.5B IPO at ~$26.6B valuation, anchored by a $20B+ OpenAI compute deal, validates wafer-scale silicon as a credible alternative to GPU dominance for AI inference.[535]

07. OpenAI + Broadcom 'Jalapeño' reticle-sized inference ASIC, design-to-tape-out in ~9 months, perf/watt 'substantially better' than SoTA, gigawatt-scale with Microsoft from end-2026.[536]

08. SK hynix sampling 12-high HBM4E (16 Gbps/pin, 48 GB/stack, >20% better power, +17% heat resistance vs HBM4) for next-gen AI accelerators.[537]

09. NVIDIA fully liquid-cooled AI server (closed-loop, 45°C supply, near-zero water); est. >$4M/yr savings at 50 MW.[538]

10. AMD+Cerebras disaggregated inference (Helios prefill + WSE token-gen): up to 5x tokens/sec/watt vs WSE-only on a 1T-param model; Cerebras Cloud H2 2026.[539]

11. Peking Univ. 40nm phase-change memristor chip: full neural dynamical system in 2.12ms; up to 24.7x more power-efficient; claimed 478x vs A100 on 3D cortical reconstruction (Science 2026).[540]

12. Hot Chips 2026 advance program lists Google's 8th-gen TPU (split training/serving), OpenAI's first in-house silicon talk, AMD MI400, Meta custom silicon and Microsoft MAIA 200 — verify against program.[541]

13. Nvidia publicly detailed the Rubin GPU and Vera CPU for the first time, claiming Vera Rubin NVL72 delivers 10x-30x more tokens per megawatt than GB300 NVL72 on SemiAnalysis' AgentX benchmark at high interactivity.[542]

14. AMD disclosed MI455X with 432GB of HBM4 at 23.3TB/s and 40.26 PFLOPS MXFP4; the 72-GPU Helios rack delivers 2.9 exaflops, 31TB HBM4 and 1.7PB/s of bandwidth, shipping Q3 2026.[543]

15. Startup Opticore presented a large-scale homodyne photonic crossbar for AI tensor processing, claiming up to 100x the energy efficiency and 25x the compute density of leading GPUs on standard foundry processes, with a scaled system demo targeted for 2026.[544]

16. [545]

17. [546]

18. Microsoft unveils Maia 300 as its primary in-house inference chip — Oct 12, 2026 (OCP Global Summit): Microsoft's third-gen Maia 300 (TSMC N2P, 288 GB HBM4) becomes its main inference platform, deploying from 2027 at multi-GW scale; Microsoft claims 40% more tokens per MW than leading GPU systems (vendor figures).[547]

c) Who could be next

Tenstorrent [STARTUP][548] — Tenstorrent · Series B (2024) led by Khosla Ventures and others

Jim Keller's open-source RISC-V + Metalium ASIC pursuing GPU alternatives for hyperscale clusters via 12×400 Gbps Ethernet scale-out; Blackhole successor scaling.

Rebellions [STARTUP][549] — Rebellions Inc. · Series C 2025, $1.4B valuation; Arm + Samsung Ventures + Synopsys

Korean accelerator with 4nm UCIe-Advanced quad-chiplet architecture (REBEL-Quad), 144 GB HBM3e, claiming ~3.2× tokens/W vs H200; Arm + Samsung backed.

FuriosaAI (RNGD next-gen) [STARTUP][550] — FuriosaAI · Rejected $800M Meta acquisition; raised capital via LG and OpenAI partnerships

Korean TSMC 5nm AI ASIC; founder declined Meta's $800M acquisition (March 2025) and is pursuing a 2027 IPO with LG and OpenAI partnerships.

Lightmatter (photonic compute) [STARTUP][523] — Lightmatter · Series D ~$400M (2024); SoftBank Vision Fund, GV

Shipping the Envise photonic processor and Passage L200/L20 co-packaged optics — the leading commercial bet on photonic interconnect for AI datacenters.

SambaNova Systems [COMPANY][551] — SambaNova (SN50 inference accelerators) · $1B Series F at $11B valuation, led by General Atlantic

$1B Series F (first tranche) led by General Atlantic at an $11B valuation (July 8, 2026); won JPMorgan Chase as an on-prem inference customer — a notable enterprise inference-ASIC win vs NVIDIA.

Taalas [STARTUP][552] — Taalas (being acquired by AMD) · Being acquired by AMD (Aug 2026)

Model-specific silicon — etches a model into the chip for extreme inference efficiency; AMD acquisition validates the inference-ASIC thesis.

Opticore [STARTUP][553] — Opticore Inc. · Seed/Series A (Origin Ventures, Jetha Global, Neotribe)

First disclosed photonic AI-accelerator startup at Hot Chips 2026; fills the long-open photonic/optical-compute gap

NewPhotonics [STARTUP][531] — NewPhotonics

Serviceable, laser-integrated co-packaged optics is the bottleneck technology for AI scale-up fabrics; 6.4T parts are next

Gimlet Labs [BUSINESS][546] — Gimlet Labs

Tests whether multi-silicon inference clouds can break NVIDIA lock-in at scale.

Flags / unverified

■ Etched Sohu's 500K tokens/sec figure is a manufacturer claim and the part remained unshipped at the time of writing.

■ FuriosaAI declined-acquisition story is widely reported but specifics are second-hand.

■ Updated nvidia_quarterly_data_center_revenue: new value = $75.2B Q1 FY27 (May 20, 2026 print); Vera Rubin initial shipments Q3 2026; Q2 guide $91B +/- 2%

■ Updated nvidia_capital_return_policy: new value = $80B additional buyback authorization; quarterly dividend raised 25x to $0.25/share; $20B returned in Q1 FY27

■ Updated Cerebras post-IPO status: new value = CBRS at $266.90 on May 27, 2026 (+10.4%); $66B market cap; $20B OpenAI capacity deal active

■ Updated TSMC Arizona total committed investment: new value = $265B across 12 facilities (adds CoWoS packaging)

■ Updated watch_list: new value = Etched: $700M at $21B valuation (Aug 18, 2026), first production rack delivered to Jane Street, >$1B signed contracts — no longer pre-shipment.

AI dev ecosystem & agent primitives (MCP, skills, agents, orchestration)

Impact so far. In roughly 18 months the AI-agent stack produced a set of durable, fast-standardizing developer primitives — MCP for connectivity, Agent Skills for portable expertise, subagents and orchestration apps for parallel work, and SDKs plus evals for governance — now backed by both major labs and a wave of well-funded startups.

Watch for next. Watch for consolidation around open standards (MCP and Skills now under foundation/shared governance), multi-agent 'post-IDE' orchestration workspaces becoming the default interface, and evals/agent governance maturing into a required production discipline.

a) People behind it

Name	Role / Affiliation	Type	Prizes
David Soria Parra[554]	Member of Technical Staff; co-creator of the Model Context Protocol (MCP) Anthropic	Biz	—
Justin Spahr-Summers[554]	Engineer; co-creator of the Model Context Protocol (MCP) Anthropic	Biz	—
Charlie Holtz[555]	Co-founder & CEO of Conductor (parallel coding-agent orchestration app); previously growth at Replicate Conductor (Melty Labs), YC S24	Biz	—
Jackson de Campos[555]	Co-founder of Conductor; 'the human interface to AI organizations' Conductor (Melty Labs), YC S24	Biz	—
Harrison Chase[556]	Founder & CEO of LangChain; creator of LangChain framework and the LangGraph agent-orchestration runtime LangChain	Biz	—
Amelia Wattenberger[557]	Product Lead for Intent (Augment Code agent-orchestration workspace); Partner, Sutter Hill Ventures; ex-GitHub Next Augment Code / Sutter Hill Ventures	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Model Context Protocol (MCP)[14]	An open standard giving AI assistants a universal two-way way to connect to external data and tools, replacing per-integration connectors with one protocol.	Pre-built MCP servers (Drive, Slack, GitHub, Postgres) · Early adopters Block and Apollo · Donated to the Linux Foundation's Agentic AI Foundation (Dec 2025)
Agent Skills (Claude Skills / portable skills)[558]	Modular folders (a SKILL.md plus scripts and resources) that agents discover and load on demand via progressive disclosure to specialize at tasks.	PDF/Excel/PowerPoint/Word document skills · skill-creator skill · Open cross-platform standard at agentskills.io (Dec 2025)
Subagents / multi-agent orchestration in coding agents[559]	Specialized sub-agents that each run in their own context window with custom prompts, tools, and permissions, letting a primary agent delegate and preserve its main context.	Claude Code Explore/Plan/general-purpose subagents · Custom subagents in ~/.claude · Subagents in the Claude Agent SDK
Agent-orchestration desktop/cloud apps[557]	Standalone workspaces to run and coordinate multiple coding agents in parallel across isolated git worktrees, then review and merge their work from one interface.	Conductor (run a team of coding agents on your Mac) · Augment Code's Intent (coordinator/implementor/verifier teams) · Zed / Codex parallel agents
Agent SDKs / frameworks[560]	Libraries and runtimes for building, orchestrating, and deploying multi-agent workflows programmatically.	Claude Agent SDK · OpenAI Agents SDK (Responses API) · LangChain LangGraph

Sub-idea	Description	Examples
Context engineering & evals / agent governance[561]	An emerging discipline focused on curating what enters an agent's context window and on systematically evaluating, grading, and governing agent behavior for production reliability.	OpenAI Evals for Agents (trace grading, prompt optimization) · OpenAI Guardrails + Connector Registry · LangChain context-engineering framing
Agentic commerce via MCP + x402[562]	Exposing financial accounts and pay-per-use data/compute to LLM agents through an MCP server plus the x402 micropayment standard, letting agents transact autonomously within user-set limits.	Coinbase for Agents MCP server usable in ChatGPT and Claude · x402 used to pay for premium research APIs and on-demand compute without login · x402 standard backed by AWS, Anthropic, Circle, Near
Agentforce Help Agent	Prepackaged autonomous customer-service agent with outcome-based ($2/resolution) pricing	
Alibaba Agent-Native Cloud[563]	AgentLoop + AgentTeams on AgentRun for multi-agent orchestration/governance	
Spec-driven development (GitHub Spec Kit)[564]	A structured spec — not ad-hoc prompts — becomes the durable source of truth; agents compile/verify code from it. GitHub Spec Kit is the open-source reference toolkit, compatible with 30+ coding agents.	GitHub Spec Kit · Claude Code Skills as spec artifacts · Martin Fowler / Thoughtworks SDD writeups
Pluggable agent harnesses[565]	Agent runtimes where model, tools, sandbox, session and UI are all swappable plugins rather than a fixed loop — the harness itself becomes the product surface.	DeepSeek Harness v0.1 (MIT, Aug 2026) · Prime Intellect Prime Agent (MIT, Aug 2026)
Recursive / self-rewriting agent harnesses[566]	Agents that treat their own prompts, skills and sub-agent structure as mutable state edited through a persistent code interpreter, rather than as fixed scaffolding.	Prime Intellect Prime Agent (MIT, Aug 2026)
Coding-agent supply-chain compromise[567]	Attacks that ride AI-generated or AI-suggested code through CI/CD into internal systems — attribution to the agent is itself contested, which is part of the problem.	Wiz: Copilot Autofix -> Snowflake Jira compromise (Aug 2026, disputed by GitHub)
ToolHazard scales adversarial testing for LLM agent tool-use security[568]	ToolHazard synthesizes adversarial tool-use environments to evaluate and align LLM agents against indirect prompt injection; trending on arXiv and Hugging Face papers this week as agentic-security benchmarking scales. (arXiv submission Aug 12, 2026; logged on this week's traction surge.)	
Sakana AI ships Fugu Max/Ultra v2, an orchestrator-as-a-model system[569]	Sakana AI released Fugu Max and Fugu Ultra v2 on Sep 11, 2026 — not single models but orchestrators that route one API request across a pool of models; Fugu Ultra v2 scored 74.3 on the DeepSWE repair benchmark.	
Agentic AI Foundation launches first vendor-neutral MCP certification (MCPA)[570]	The Agentic AI Foundation and Linux Foundation Education launched the Model Context Protocol Associate (MCPA) exam on Sep 14, 2026, covering MCP fundamentals, architecture, execution, security/governance and ecosystem use cases — the first official MCP certification.	
Google previews persistent Gemini agent across Workspace and Cloud[571]	Oct 8, 2026 (Gemini at Work): Google previewed a persistent enterprise Gemini agent with memory and skills spanning Gmail, Drive, Docs, Sheets, Calendar and the CLI, for multi-hour/day tasks; private preview.	
Claude Code 'mods': TypeScript middleware that rewrites the agent loop[572]	Oct 1, 2026: Anthropic launched Claude Code mods — plugin-shipped TypeScript functions that rewrite prompts, gate/retry tool calls and redact secrets; unsandboxed, with a built-in 'sec-default' mod on Team/Enterprise. Backfill: launched one day before the window; widely covered within 24h.	

b) Current research frontier (last 12–18 months)

01. MCP went from an Anthropic experiment (announced Nov 25, 2024) to a cross-industry standard; in Dec 2025 Anthropic donated MCP to the Agentic AI Foundation, a Linux Foundation directed fund co-founded with Block and OpenAI.[573]

02. Anthropic launched Agent Skills on Oct 16, 2025 across Claude apps, Claude Code, the Agent SDK and the API, then on Dec 18, 2025 published Skills as an open cross-platform standard (agentskills.io).[558]

03. OpenAI released the Responses API and open-source Agents SDK in March 2025, then at DevDay (Oct 6, 2025) launched AgentKit — Agent Builder, ChatKit, Connector Registry and expanded Evals — to move agents from prototype to production.[561]

04. Conductor, the parallel coding-agent desktop app (YC S24), reports use at companies including Linear, Vercel, Notion and Stripe and announced a Series A on Mar 30, 2026.[555]

05. LangChain, maker of the LangGraph agent-orchestration runtime, reached unicorn status with a $125M Series B at a $1.25B valuation in Oct 2025, led by IVP with CapitalG and Sapphire Ventures.[556]

06. OpenAI Daybreak releases GPT-5.5-Cyber (85.6% CyberGym) + Codex Security (30M+ commits / 30K repos scanned), with government Trusted Access partnerships.[574]

07. Agon (arXiv:2606.24177) autonomous research orchestrator ran 444 loops across domains with no human-written experimental code.[575]

08. Agents' Last Exam (ALE), built with 250+ industry experts over 1,000+ real economically-valuable tasks across 55 industries, finds frontier agents average only 2.6% on its hardest tier despite Terminal-Bench scores up to 82% — a large benchmark-vs-real-work gap.[576]

09. [577]

10. [578]

11. [579]

12. [580]

13. Anthropic replaces Git with in-house Scion/Antique as merges hit 30k/day — Oct 13, 2026: Anthropic's engineering blog says merged changes rose from ~2,500/day (May) to 30,000+/day (Sep), forcing a custom server-centric source-control system (Scion) with an AI-assisted review tool (Antique), migrated in ~five months; Scion reportedly uses 200x less CPU/memory than the git mirror fleet (vendor-reported).[581]

c) Who could be next

Conductor [STARTUP][555] — Conductor (Melty Labs) · YC Summer 2024; Series A announced Mar 30, 2026 (amount not confirmed in primary source)

Pioneer of the 'run a team of coding agents on your Mac' category; parallel agents in isolated worktrees, adopted at Linear, Vercel, Notion, Stripe.

Augment Code (Intent) [STARTUP][557] — Augment Code · Augment backed by ~$472M total (incl. Sutter Hill Ventures); Intent in public beta

Intent is a 'post-IDE' agent-orchestration workspace using coordinator/implementor/verifier agent teams over a shared living spec; bring-your-own-agent (Claude Code, Codex, OpenCode).

LangChain [STARTUP][556] — LangChain · $125M Series B at $1.25B valuation (Oct 2025), led by IVP

Most widely adopted open-source agent framework; LangGraph is a leading multi-agent orchestration/memory runtime and LangSmith leads agent evals/observability.

Transformer Lab [STARTUP][582] — Transformer Lab (lab.cloud) · Not disclosed (open-source platform; for-teams commercial tier)

Open-source MLOps/experimentation layer (compute orchestration, training, eval, quantization) with a teams tier; a credible primary-source research shop on efficiency/quantization to track for the edge/efficiency lane.

Virtue AI [STARTUP][583] — Virtue AI · VC-backed (Walden Catalyst and others); founders Bo Li, Dawn Song, Sanmi Koyejo, Carlos Guestrin

AI-native security platform (red-teaming + runtime guardrails) for LLMs and autonomous agents — the tooling layer for the agentic-AI-security surge; MCP-tool injection testing + agent action guards

Tessl [STARTUP][584] — Tessl · Founded by Guy Podjarny (founder of Snyk). Prior funding ~$125M ($25M seed + $100M Series A, Nov 2024; Index Ventures, Accel, GV, boldstart), reported ~$500-750M valuation; no 2026 round found.

Commercial spec-centric / 'AI-native' dev platform and spec registry; a lead bet on the prompt-less/spec-driven paradigm from a proven founder.

Amazon Kiro [STARTUP][585] — Amazon Web Services · Amazon (AWS) product

Spec-driven agentic IDE from AWS; generates requirements/design/task specs before code — a hyperscaler entrant in spec-first tooling.

Prime Intellect [STARTUP][566] — Prime Intellect · Venture-backed (round details not verified this run)

One of the few labs shipping permissively licensed frontier-grade agent harnesses and decentralized training stacks; entity added after its Prime Agent release surfaced in the weekly scan without the company being tracked.

Wiz Research [LAB][567] — Wiz (Google Cloud) · Subsidiary of Google Cloud

Emerging as the main source of primary incident reporting on coding-agent and CI/CD supply-chain compromise; entity added after its Snowflake write-up surfaced without Wiz being tracked.

Model Context Protocol [LAB][586] — Agentic AI Foundation / Anthropic · n/a

Defines the trajectory of the tool-calling infrastructure underlying most agent products

Agentic AI Foundation (AAIF) [LAB][587] — Linux Foundation · n/a

Now hosts both A2A and MCP, the two leading agent/tool interoperability standards

Sierra [BUSINESS][588] — Sierra

Agentic CX company co-leading the Personal Agent Protocol with Meta for agent-to-business access.

Flags / unverified

■ Conductor Series A dollar amount unverified against a primary source (YC page confirms the round exists, dated Mar 30, 2026, but not the amount).

■ MCP co-creator attributions sourced from the a16z 'MCP co-creator' podcast; the official Anthropic MCP announcement does not name individuals.

■ Updated MCP protocol + enterprise agent primitives status (mid-2026): new value = MCP statelessness in the 2026-07-28 spec RC (published May 2026); Anthropic adds enterprise-managed MCP connectors (Okta first) and self-hosted sandboxes + MCP tunnels for private MCP server access from managed agents

■ Updated ?: new value = ?

■ Updated Microsoft Agent Framework Harness: new value = GA (Aug 2026): supported runtime with planning/history/approvals/OpenTelemetry, plus GitHub Copilot SDK & Claude Agent SDK connectors

Five cross-cutting themes

1. Scaling laws are the through-line.

Every one of the AI breakthroughs (transformers, reasoning models, diffusion) explicitly relies on a scaling-law thesis. Battery and solar cost-down curves (Wright's law) are the same shape: roughly 20% cost reduction per doubling of cumulative volume. Reusable rocketry is the same curve in cost-per-kg-to-orbit.

2. Verifiable rewards close the loop.

Reasoning models train on tasks with checkable answers. AlphaFold is verified against crystal structures. QEC has a hard threshold. CRISPR and obesity Phase-3 trials hit pre-registered endpoints. Where the reward is dense and verifiable, RL — and capital — pour in.

3. Delivery is the new bottleneck in biology.

mRNA vaccines, in vivo CAR-T, in vivo CRISPR, and personalized cancer vaccines share one stack: a payload wrapped in a lipid nanoparticle. Whoever solves non-hepatocyte targeted delivery unlocks the rest of the genome.

4. Reusability and amortization beat raw novelty.

Falcon 9's moat is the same booster flying 22+ times. MoE LLMs reuse a small fraction of parameters per token. Surface-code logical qubits trade more physical qubits for arbitrarily long lifetime. Battery cycle life, not raw energy density, is what brought $/kWh-cycle below grid parity.

5. The lab → spinout → IPO path is now standard.

AlphaFold launched Isomorphic Labs. Baker's IPD seeded Xaira / Generate / Outpace. Karikó & Weissman seeded BioNTech. Doudna & Charpentier seeded CRISPR Tx, Editas, Intellia. Liu's lab spawned Beam, Prime, and Editas. Sutton & Barto's RL textbook seeded an entire research lineage now valued at hundreds of billions.

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